

Project Data Page:





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LP Condensate Flash Vessel

DESIGN CALCULATION

In Accordance with ASME Section VIII Division 1

ASME Code Version : 2025

Analysis Performed by : SL User

Job File : ...\\LP CONDENSATE FLASH DRUM (V8301C) -7 Percent.pvdb

Date of Analysis : Sep 16,2026 3:41pm

PV Elite 28 U1, March 2026

**Vessel Design Summary:****ASME Code, Section VIII Division 1, 2025**

Diameter Spec : 1900.000 mm. ID	
Vessel Design Length, Tangent to Tangent	5100.00 mm.
Distance of Bottom Tangent above Grade	3300.00 mm.
Distance of Base above Grade	300.00 mm.
Specified Datum Line Distance	0.00 mm.
Internal Design Temperature	230 °C
Internal Design Pressure	3.500 bars
External Design Temperature	198 °C
External Design Pressure	1.034 bars
Maximum Allowable Working Pressure	7 bars
External Max. Allowable Working Pressure	1 bars
Shop Test Pressure	9.620 bars
Field Test Pressure	4.550 bars
Required Minimum Design Metal Temperature	-1.0 °C
Warmest Computed Minimum Design Metal Temperature	-29.0 °C
Wind Design Code	ASCE/SEI 7-16
Earthquake Design Code	ASCE/SEI 7-16

Materials of Construction:

Component Type	Material	Class	Thickness	UNS #	Material Treatment Options
Shell	SA-516 70	...	...	K02700	
Head	SA-516 70	...	...	K02700	
Skirt	SA-516 70	...	...	K02700	
Nozzle	SA-106 B	...	...	K03006	
Nozzle	SA-516 70	...	...	K02700	
Nozzle	SA-105	...	...	K03504	
Re-Pad	SA-516 70	...	...	K02700	
Nozzle Flg	SA-105	...	...	K03504	
Nozzle Flg	SA-106 B	...	...	K03006	
Rings	SA-516 70	...	...	K02700	
Basing	SA-516 70	...	...	K02700	
Base Bolting	SA-307 A	...	6 <= t <= 100		

+FG - Fine Grained, +R - rolled, +T - Tempered, +N - Normalized

+Q - quenched, +CBV - Cb & V added, +IT - Impact tested

Element Pressures and MAWP (bars & mm.):

Element Description or Type	Design Pressure + Stat. head	Ext. Press.	Element M.A.W.P	Total Corrosion Allowance	Str. Flg. Gov.	In Creep Range
Skirt-2	...	...	...	3.0000	N/A	No
Skirt-1	...	...	...	3.0000	N/A	No
Bottom Dish	3.758	1.03	7.000	3.0000	No	No
shell	3.711	1.03	11.000	3.0000	N/A	No
shell	3.500	1.03	13.000	3.0000	N/A	No
Top dish	3.500	1.03	11.000	3.0000	No	No

Liquid Level: 2750.00 mm. Dens.: 0.001 kg./cm3 Sp. Gr.: 0.958

Stiffener Ring Specifications:

Elevation mm. Selected Type Ring Description



5250.00

Bar 150.0 x 12.0

Ring R1 Fr40

Element Types and Properties:

Element Type	"To" Elev mm.	Element Length mm.	Nominal Thickness mm.	Finished Thickness mm.	Reqd Thk Internal mm.	Reqd Thk External mm.	Long Eff	Circ Eff
Skirt	2000.0	2000.0	10.0	10.0	...	...	0.70	0.70
Skirt	3000.0	1000.0	10.0	10.0	...	...	0.70	0.50
Ellipse	3050.0	50.0	12.0	9.0	6.0	...	0.85	0.85
Cylinder	6050.0	3000.0	12.0	12.0	6.0	...	0.85	0.85
Cylinder	8050.0	2000.0	14.0	14.0	5.8	...	0.85	0.85
Ellipse	8100.0	50.0	14.0	12.0	5.8	...	0.85	0.85

External Pressure Calculations:

From	To	External Actual T. mm.	External Required T. mm.	External Design Pressure bars	External M.A.W.P. bars
10	20	...	No Calc	...	No Calc
20	30	...	No Calc	...	No Calc
30	40	9	8.04358	1.03421	1.46375
40	Ring	12	9.85799	1.03421	2.04687
Ring	50	12	10.5323	1.03421	1.62233
50	60	14	10.5412	1.03421	2.69915
60	70	12	8.05931	1.03421	3.27293

External Pressure Calculations:

From	To	Actual Length Bet. Stiffeners mm.	Allowable Length Bet. Stiffeners mm.	Ring Inertia Required cm ⁴	Ring Inertia Available cm ⁴
10	20	3000	No Calc	No Calc	No Calc
20	30	3000	No Calc	No Calc	No Calc
30	40	No Calc	No Calc	No Calc	No Calc
40	Ring	2408.33	4624.14	No Calc	No Calc
Ring	50	3008.33	4625.46	141.34	816.04
50	60	3008.33	7491.71	No Calc	No Calc
60	70	No Calc	No Calc	No Calc	No Calc

Local Stress Analysis Results:

Description	Analysis Type	Max Stress Ratio	High Stress Location	Pass Fail
Lift Lug	WRC-107/537	0.727	n/a	Passed
N2-6"	WRC-107/537	0.845	n/a	Passed
D1-3"	WRC-107/537	0.767	n/a	Passed
N5-2"	WRC-297	0.726	Nozzle	Passed
N1-16"	WRC-297	0.896	Shell	Passed
N6-6"	WRC-297	0.705	Shell	Passed
N4-3"	WRC-297	0.705	Shell	Passed
N3-20"	WRC-107/537	0.929	n/a	Passed

Wind/Earthquake Shear, Bending:

From	To	Distance to Support mm.	Cumulative Wind Shear Kgf	Earthquake Shear Kgf	Wind Bending Kg-m.	Earthquake Bending Kg-m.
10	20	1000	2089.5	4307.08	11502.9	26417.3
20	30	2500	1867.28	4250.39	7546.02	17859.7



30	40	3025	1755.85	4204.56	5734.42	13632.1
40	50	4550	1390.02	3816.82	5655.77	13431.6
50	60	7050	1029.52	2232.66	2026.39	4357.2
60	70	8075	788.734	1680.97	208.093	443.495

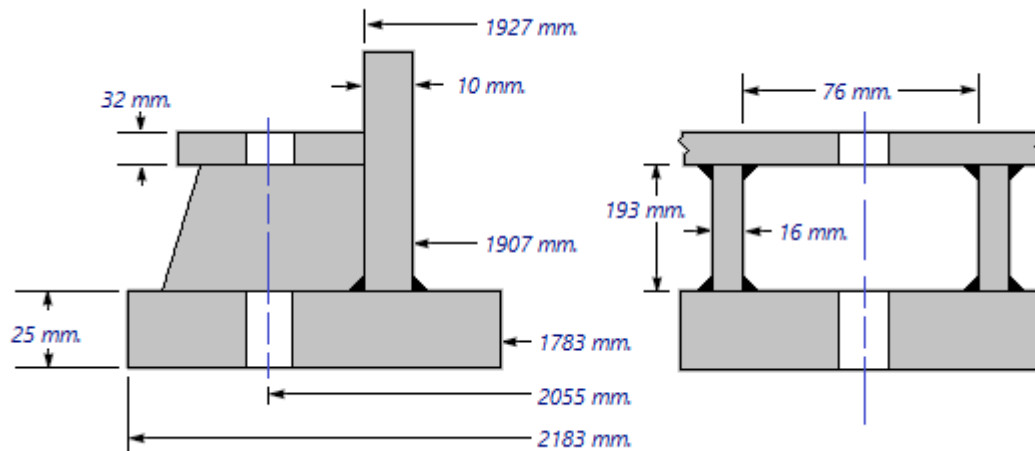
Abs Max of the all of the Stress Ratio's : 0.2471

Basering Data : Continuous Top Ring W/Gussets

Thickness of Basering	25.0000	mm.
Inside Diameter of Basering	1783.0000	mm.
Outside Diameter of Basering	2183.0000	mm.
Nominal Diameter of Bolts	36.0000	mm.
Diameter of Bolt Circle	2055.0000	mm.
Number of Bolts	12	

Thickness of Gusset Plates	16.0000	mm.
Average Width of Gusset Plates	118.0000	mm.
Height of Gussets	193.0000	mm.
Distance between Gussets	76.0000	mm.
Thickness of Top Plate or Ring	32.0000	mm.
Radial Width of the Top Plate	128.0000	mm.
Number of Gussets	24	

Basering Sketch for Cylindrical Skirts:



Basering Cross Section View

Loads for Foundation/Support Design:

Total Wind Shear on Support	2090.	Kgf
Total Earthquake Shear on Support	4307.	Kgf
Wind Moment on Support	11578.	Kg-m.
Earthquake Moment on Support	26492.	Kg-m.
Maximum Initial Bolt Load	2449.	Kgf

Wind and Earthquake moments include the effects of user defined forces and moments if any exist in the job and were specified to act (compute loads and stresses) during these cases. Also included are moment effects due to eccentric weights if any are present in the input.

Weights:

Fabricated - Bare w/o removable internals	8031.7	kg.
Shop Test - Fabricated + water (full)	26080.3	kg.
Shipping - Fab. + rem. intls.+ shipping app.	8031.7	kg.
Erected - Fab. + rem. intls.+ insul. (etc)	15789.7	kg.
Empty - Fab. + intls. + details + wgths.	15789.7	kg.
Operating - Empty + operating liquid (no ca)	22890.5	kg.

LP Condensate Flash Vessel

PV Elite 28 U1 Licensee: SL User

FileName : LP CONDENSATE FLASH DRUM (V8301C) -7 Percent Page 5 of 171

Vessel Design Summary: Step: 41 3:41pm Sep 16,2026



HEXAGON

Field Test - Empty weight + test medium (full)

32038.3 kg.

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**Nozzle Calculation Summary (bars & mm.):**

Description	MAWP	Ext	MAPNC	UG-45	[tr]	Weld Path	Areas or Stresses
N2-6"	7	OK	...	OK	7.82	OK	Passed
D1-3"	7	...	...	OK	6.40	OK	No Calc[*]
M1-24"	11	OK	...	...		OK	Passed
K1-2"	11	...	...	OK	7.80	OK	No Calc[*]
K2-2"	11	...	...	OK	7.80	OK	No Calc[*]
N5-2"	11	...	...	OK	6.42	OK	No Calc[*]
N1-16"	13	OK	...	OK	7.96	OK	Passed
N6-6"	13	OK	...	OK	7.96	OK	Passed
N4-3"	13	...	...	OK	7.96	OK	No Calc[*]
N3-20"	10	OK	...	OK	7.92	OK	Passed
K1A-2"	11	...	...	OK	7.80	OK	No Calc[*]

Nozzle MAWP Summary:

Minimum MAWP Nozzles : 7 Nozzle : D1-3"
 Minimum MAWP Shells/Flanges : 7 Element : Bottom Dish
 Minimum MAPnc Shells/Flanges : 11 Element : Bottom Dish

Computed Vessel M.A.W.P. : 7 bars

[*] - This was a small opening and the areas were not computed or the MAWP of this connection could not be computed because the longitudinal bending stress was greater than the hoop stress.

Multiple output lines for the same nozzle indicates required Code calculations in both the longitudinal and circumferential planes of reinforcement where applicable.

Check the spatial relationship between the nozzles:

From Node	Nozzle Description	Y Coordinate mm.	Layout Angle deg	Dia. Limit mm.
30	N2-6"	0.000	270.000	279.120
30	D1-3"	0.000	90.000	145.262
40	M1-24"	1550.000	90.000	1183.200
40	K1-2"	1600.000	270.000	113.600
40	K2-2"	300.000	270.000	113.600
40	N5-2"	300.000	90.000	93.890
50	N1-16"	3650.000	270.000	956.000
50	N6-6"	4450.000	270.000	284.720
50	N4-3"	4050.000	90.000	164.400
60	N3-20"	0.000	0.000	980.000
60	K1A-2"	0.000	90.000	113.600

The nozzle spacing is computed by the following:

$$= \sqrt{ll^2 + lc^2} \text{ where}$$

ll - Arc length along the inside vessel surface in the long. direction.
 lc - Arc length along the inside vessel surface in the circ. direction

If any interferences/violations are found, they will be noted below.

No interference violations have been detected!

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**Nozzle Schedule:**

Description	Nominal or Actual Size	Schd or FVC Type	Flg Type	Nozzle O/Dia in	Wall Thk mm	Reinforcing Pad Diameter	Pad Thk mm	Cut Length mm	Flg Class
K1-2"	2.000 in	Actual	LWN	3.320	16.764	...	...	219.31	300
K2-2"	2.000 in	Actual	LWN	3.320	16.764	...	...	219.31	300
N5-2"	2.000 in	XXS	WNF	2.375	11.074	...	...	212.48	600
K1A-2"	2.000 in	Actual	LWN	3.320	16.764	...	...	235.51	300
D1-3"	3.000 in	160	WNF	3.500	11.125	190.00	10.00	214.86	300
N4-3"	3.000 in	Actual	LWN	4.620	20.574	...	...	215.84	300
N2-6"	6.000 in	160	WNF	6.625	18.237	318.27	10.00	226.25	300
N6-6"	6.000 in	160	WNF	6.625	18.237	318.27	12.00	222.52	300
N1-16"	20.000 in	Actual	WNF	20.000	18.000	950.00	12.00	302.59	300
N3-20"	20.000 in	Actual	WNF	20.000	12.000	622.00	12.00	180.86	150
M1-24"	24.000 in	Actual	WNF	24.000	12.000	909.60	12.00	362.22	150

General Notes for the above table:

The cut length is the outside projection + inside projection + drop + in-plane shell thickness. This value does not include weld gaps, nor does it account for shrinkage.

In the case of oblique nozzles, the outside diameter must be increased. The re-pad width around the nozzle is calculated as follows:

Width of Pad = (Pad Outside Dia. (per above) - Nozzle Outside Dia.)/2

For hub nozzles, the thickness and diameter shown are those of the smaller and thinner section.

Nozzle Material and Weld Fillet Leg Size Details (mm.):

Description	Material	Shl Grve Weld	Noz Shl/Pad Weld	Pad OD Weld	Pad Grve Weld	Inside Weld
K1-2"	SA-105	9.000	10.000	...	...	...
K2-2"	SA-105	9.000	10.000	...	...	...
N5-2"	SA-106 B	9.000	10.000	...	...	...
K1A-2"	SA-105	9.000	10.000	...	...	...
D1-3"	SA-106 B	9.000	10.000	8.000	10.000	...
N4-3"	SA-106 B	9.000	10.000	...	...	...
N2-6"	SA-106 B	9.000	10.000	8.000	10.000	...
N6-6"	SA-106 B	12.000	10.000	10.000	12.000	...
N1-16"	SA-516 70	10.000	10.000	8.000	12.000	...
N3-20"	SA-516 70	9.000	10.000	8.000	12.000	...
M1-24"	SA-516 70	9.000	10.000	8.000	12.000	...

Note: The Outside projections below do not include the flange thickness.

Nozzle Miscellaneous Data:

Description	Elev/Distance From Datum mm.	Layout Angle deg	Proj Outside mm.	Proj Inside mm.	Installed in Component
K1-2"	4600.000	270.0	206.38	0.00	shell
K2-2"	3300.000	270.0	206.38	0.00	shell
N5-2"	3300.000	90.0	200.00	0.00	shell
K1A-2"	...	90.0	206.38	0.00	Top dish
D1-3"	...	90.0	200.00	0.00	Bottom Dish
N4-3"	7050.000	90.0	200.02	0.00	shell
N2-6"	...	270.0	200.00	0.00	Bottom Dish
N6-6"	7450.000	270.0	204.79	0.00	shell
N1-16"	6650.000	270.0	250.00	0.00	shell
N3-20"	...	0.0	150.00	0.00	Top dish
M1-24"	4550.000	90.0	300.00	0.00	shell


Bill of Materials:

QTY	DESCRIPTION	MATERIAL
1	SKIRT: 10.0mm. THK X 1907.0mm. ID X 2000.0mm.	SA-516 70
1	SKIRT: 10.0mm. THK X 1900.0mm. ID X 1000.0mm.	SA-516 70
1	ELLIPTICAL HEAD: 2.0 X 1, 12.0mm. THK X 1900.0mm. ID X 50.0mm.	SA-516 70
1	CYLINDER: 12.0mm. THK X 1900.0mm. ID X 3000.0mm.	SA-516 70
1	ELLIPTICAL HEAD: 2.0 X 1, 14.0mm. THK X 1900.0mm. ID X 50.0mm.	SA-516 70
1	PLATFORM: 1000mm X 1500mm WIDE	
1	LADDER: 3000.0mm	...
2	INSULATION: 500mm X 80mm THK	
1	INSULATION: 3000mm X 80mm THK	
1	BAR RING STIFFENER: 1924.0mm ID X 2224.0mm OD X 12.0mm	SA-516 70
1	INSULATION: 2000mm X 80mm THK	
2	LIFTING LUGS:	SA-516 70
2	LIFTING LUG PADS: 420.0mm X 420.0mm X 12.0mm THK	
1	PLATFORM: 1000mm X 3928mm WIDE	
1	LADDER: 5820.0mm	...
1	CLASS 150 GR 1.1, 24.0" BLIND FLANGE(S)	SA-105
1	BASERING: 2183.0mm. OD X 1783.0 ID X 25.0mm.	SA-516 70
12	BASERING BOLTS: 36.0mm. DIA	SA-307 A
12	BASERING BOLT NUTS: 36.0mm. DIA	...
1	BASERING TOP RING: 2183.0mm. OD X 1927.0 ID X 32.0mm.	...
24	BASERING GUSSET PLATES: 118.0mm. X 118.0mm. X 193.0mm. X 16.0mm.	...
1	CYLINDRICAL SEGMENT 14.0mm. THK X 3217.3mm. X 2000.0mm.	SA-516 70
1	CYLINDRICAL SEGMENT 14.0mm. THK X 2751.6mm. X 2000.0mm.	SA-516 70
1	NAMEPLATE	...

**Minimum Design Metal Temperature Results Summary (°C):**

Description	Notes	Curve	Basic MDMT	Reduced MDMT	UG-20(f) MDMT	Thickness ratio	Gov Thk mm.	E*	PWHT reqd
Bottom Dish	[10]	B	-29	-37	-29	0.850	9.000	0.85	No
Bottom Dish	[7]	B	-23	-48	-29	0.571	12.000	0.85	No
shell	[8]	B	-23	-48	-29	0.567	12.000	0.85	No
shell	[8]	B	-18	-48	-29	0.451	14.000	0.85	No
Top dish	[10]	B	-23	-48	-29	0.547	12.000	0.85	No
Top dish	[7]	B	-18	-48	-29	0.451	14.000	0.85	No
Ring R1 Fr40	[23]	B	-23	-48	-29	0.567	12.000	0.85	No
N2-6"	[1]	B	-29	-42	-29	0.766	10.000	1.00	No
Nozzle Flg	[5]	A	-18	-104					
D1-3"	[1]	B	-29	-42	-29	0.766	9.000	1.00	No
Nozzle Flg	[5]	A	-18	-104					
M1-24"	[1]	B	-23	-48	-29	0.556	12.000	1.00	No
Nozzle Flg	[5]	A	-18	-48					
K1-2"	[1]	A	-3	-29	-29	0.556	12.000	1.00	No
Nozzle Flg	[5]	A	-18	-104					
K2-2"	[1]	A	-3	-29	-29	0.565	12.000	1.00	No
Nozzle Flg	[5]	A	-18	-104					
N5-2"	[1]	B	-8	-104	-29	0.028	9.690	1.00	No
Nozzle Flg	[5]	A	-18	-104					
N1-16"	[1]	B	9	-104		0.088	18.000	1.00	No
Nozzle Flg	[5]	A	-18	-104					
N6-6"	[1]	B	-23	-48	-29	0.450	12.000	1.00	No
Nozzle Flg	[5]	A	-18	-104					
N4-3"	[1]	B	-18	-48	-29	0.450	14.000	1.00	No
Nozzle Flg	[5]	A	-18	-104					
N3-20"	[1]	B	-23	-48	-29	0.493	12.000	1.00	No
Nozzle Flg	[5]	A	-18	-48					
K1A-2"	[1]	A	-3	-36	-29	0.493	12.000	1.00	No
Nozzle Flg	[5]	A	-18	-104					
Nozl. Bolting[22]			-48			1.000	64.000	1.00	No

Warmest MDMT: 9 -29

Required Minimum Design Metal Temperature -1.0 °C

Warmest Computed Minimum Design Metal Temperature -29.0 °C

Notes:

[!] - This was an impact tested material.

[1] - Governing Nozzle Weld.

[4] - ASME Flange MDMT Calcs; Thickness ratio per UCS-66(b)(1)(-c).

[5] - ASME Flange MDMT Calcs; Thickness ratio per UCS-66(b)(1)(-b).

[6] - MDMT Calculations at the Shell/Head Joint.

[7] - MDMT Calculations for the Straight Flange.

[8] - Cylinder/Cone/Flange Junction MDMT.

[9] - Calculations in the Spherical Portion of the Head.

[10] - Calculations in the Knuckle Portion of the Head.

[11] - Calculated (Body Flange) Flange MDMT.

[12] - Calculated Flat Head MDMT per UCS-66.3

[13] - Tubesheet MDMT, shell side, if applicable

[14] - Tubesheet MDMT, tube side, if applicable

[15] - Nozzle Material

[16] - Shell or Head Material

[17] - Impact Testing required

[18] - Impact Testing not required, see UCS-66(b)(3)

[20] - Cylinder/Cone Junction MDMT based on Longitudinal Stress considerations

[21] - Body Flange Bolting Material

[22] - Nozzle Flange Bolting Material

[23] - Stiffening Ring to Shell Weld

[24] - Saddle to Shell Weld

UG-84(b)(2) was not considered.

UCS-66(g) was not considered.



UCS-66(i) was not considered.

Notes:

Impact test temps were not entered in and not considered in the analysis.

UCS-66(i) applies to impact tested materials not by specification and

UCS-66(g) applies to materials impact tested per UG-84.1 General Note (c).

The Basic MDMT includes the (30F) PWHT credit if applicable.

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Class From To : Basic Element Checks.

Class From To: Check of Additional Element Data

There were no geometry errors or warnings.

PV Elite performs all calculations internally in Imperial Units to remain compliant with the ASME Code and any built in assumptions in the ASME Code formulas. The finalized results are reflected to show the set of selected units for this analysis.

Information Regarding Nozzle Loads

Please note that nozzle loadings, if included, are assumed to be local in nature and will not contribute to or create a net section bending moment. Therefore, the addition of nozzle loads will not affect the support load calculation. If you wish to create a load on the support(s) from nozzle loads, you can enable that feature in tools->configuration->Job Specific Settings.



One or more nozzles in this model was specified using the actual thickness basis and a standard flange is also specified. In many cases a matching nominal diameter cannot be determined and the weight of the flange and optional blind flange cannot be determined. When actual thickness basis is specified, you must calculate the weight of the assembly (by hand) and enter it into the 'Overriding Weight' field in the nozzle dialog when there is a standard flange specified. However, for FVC nozzles that have their nominal diameter specified in the FVC dialog, entering the overriding weight for that nozzle is not required.



The level of precision is set to a value of less than 3. At this level you may not obtain proper resolution on some results. This value can be changed in Tools->Set Configuration Parameters->Default Values->Level of Precision.



The ASME Code Section VIII Division 1 does not explicitly require stresses on elements to be computed under test conditions. Please see interpretation VIII-1-10-12 for more information. PV Elite also computes weight due to test fluid and uses the weight in various calculations such as saddle and leg loads. PV Elite also computes static pressure due to operating and test liquids. The static pressure is used in conjunction with the design and test pressures to determine a total pressure in the determination of required thickness, MAWP etc. This ensures compliance with UG-22 requirements.

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**Units used in this Analysis (barsUniCode):**

Name	System Unit	Constant	User Unit
Length	Feet	304.8000	mm.
Force	Pounds	0.4536	Kgf
Mass	Pounds	0.4536	kg.
Area	sq. inches	6.4516	cm ²
Moment	ft. lbs.	0.1383	Kg-m.
Stress	lbs./sq.in.	0.0069	N./mm ²
Temperature	Degrees F	0.5556	°C
Pressure	psig	0.0689	bars
Elast. Modulus	lbs./sq.in.	6.8946	KPa.
Pipe Density	lbs./cu.in.	0.0277	kg./cm ³
Ins. Density	lbs./cu.ft.	0.1600E-04	kg./cm ³
Fluid Density	lbs./cu.ft.	0.1600E-04	kg./cm ³
Wind Speed	miles/hr	1.6093	Km/hr
Tray Weight	lbs./sq.ft.	0.0005	Kgs/cm ²
Inertia	in.4	41.6231	cm ⁴
G Load	G's	1.0000	g's
Wind Load	lbs./sq.ft.	4.8825	Kgs/m ²
Elevation	Feet	304.8000	mm.
Volume	in.3	16.3900	cm ³
Diameter	inches	25.4000	mm.
Thickness	inches	25.4000	mm.

PV Elite Vessel Analysis Program: Input Data

LP Condensate Flash Vessel

Design Internal Pressure (for Hydrotest) 3.5 bars
 Design Internal Temperature 230.0 °C
 Projection of Nozzle from Vessel Top 500 mm.
 Projection of Nozzle from Vessel Bottom 500 mm.
 Minimum Design Metal Temperature -1.0 °C
 Type of Construction Welded
 Special Service Air/Water/Steam
 Degree of Radiography RT-3
 Use Higher Longitudinal Stresses (Flag) Y
 Select t for Internal Pressure (Flag) N
 Select t for External Pressure (Flag) N
 Select t for Axial Stress (Flag) N
 Select Location for Stiff. Rings (Flag) N
 Consider Vortex Shedding N

Shop Pressure Test:

Type of Pressure Test User Entered Pressure
 Pressure Test Position Horizontal
 Test Performed in Corroded Condition No

Field Pressure Test:

Type of Pressure Test UG-99(b) Note [35]
 Pressure Test Position Vertical
 Test Performed in Corroded Condition Yes

Load Case 1 NP+EW+1.1WI+FW+BW
 Load Case 2 NP+EW+EE+FS+BS
 Load Case 3 NP+OW+1.1WI+FW+BW
 Load Case 4 NP+OW+EQ+FS+BS
 Load Case 5 NP+HW+HI
 Load Case 6 NP+HW+HE
 Load Case 7 IP+OW+1.1WI+FW+BW
 Load Case 8 IP+OW+EQ+FS+BS
 Load Case 9 EP+OW+1.1WI+FW+BW
 Load Case 10 EP+OW+EQ+FS+BS
 Load Case 11 HP+HW+HI
 Load Case 12 HP+HW+HE



Load Case 13	IP+WE+EW
Load Case 14	IP+WF+CW
Load Case 15	IP+VO+OW
Load Case 16	IP+VE+EW
Load Case 17	NP+VO+OW
Load Case 18	FS+BS+IP+OW
Load Case 19	FS+BS+EP+OW
Load Case 20	BL+IP+OW

Wind Design Code	ASCE/SEI 7-16
Wind Load Reduction Scale Factor	1.100
Basic Wind Speed [V]	97.2 Km/hr
Surface Roughness Category	C: Open Terrain
Type of Surface	Moderately Smooth
Base Elevation	300 mm.
Percent Wind for Hydrotest	50.0
Using User defined Wind Press. Vs Elev.	N
Height of Hill or Escarpment H or Hh	0 mm.
Distance Upwind of Crest Lh	0 mm.
Distance from Crest to the Vessel x	0 mm.
Type of Terrain (Hill, Escarpment)	Flat
Damping Factor (Beta) for Wind (Ope)	0.0100
Damping Factor (Beta) for Wind (Empty)	0.0000
Damping Factor (Beta) for Wind (Filled)	0.0000
Site Elevation above Sea Level	0 mm.

Seismic Design Code	ASCE/SEI 7-16
Seismic Load Reduction Scale Factor	1.000
Importance Factor	1.250
Table Value Fa	1.600
Table Value Fv	2.400
Max. Mapped Res. Acceleration [Ss]	0.147
Max. Eff. Ground Acceleration [S]	0.065
Force Modification Factor R	1.250
Site Class	D
Component Elevation Ratio z/h	0.100
Amplification Factor Ap	2.500
Force Factor	0.000
Consider Vertical Acceleration	Yes
Minimum Acceleration Multiplier	0.000
User Value of Sds (used if > 0)	0.000
User Value of Sd1 (used if > 0)	0.000
Moment Reduction Factor Tau	1.000

M.A.W.P + Static Head to Nozzle	Y
Consider MAP New and Cold in Noz. Design	N
Consider External Loads for Nozzle Des.	Y
Use ASME VIII-1 Appendix 1-9	N

Perform Blast Load Analysis	No
Material Database Year	Current w/Addenda or Code Year

Configuration Directives:

g0 is taken as min(g0, attached shell thk)	Yes
Do not use Nozzle MDMT Interpretation VIII-1 01-37	No
Use Table G instead of exact equation for "A"	Yes
Shell Head Joints are Tapered	Yes
Compute "K" in corroded condition	Yes
Use Code Case 2286	No
Use the MAWP to compute the MDMT	Yes
For thickness ratios <= 0.35, MDMT will be -155F (-104C)	Yes
For PWHT & P1 Materials the MDMT can be < -55F (-48C)	No
Using Metric Material Databases, ASME II D	Yes
Calculate B31.3 type stress for Nozzles with Loads	Yes
Reduce the MDMT due to lower membrane stress	Yes
Consider Longitudinal Stress in MDMT Calculations	Yes

**Complete Listing of Vessel Elements and Details:**

Element#	1/6
Element From Node	10
Element To Node	20
Element Type	Skirt Support
Description	Skirt-2
Distance "FROM" to "TO"	2000 mm.
Skirt Outside Diameter	1927 mm.
Diameter of Skirt at Base	1927 mm.
Skirt Thickness	10 mm.
Internal Corrosion Allowance	3 mm.
Nominal Thickness	10 mm.
External Corrosion Allowance	0 mm.
Design Temperature Internal Pressure	230 °C
Design Temperature External Pressure	198 °C
Effective Diameter Multiplier	1.42
Material Name	SA-516 70
Allowable Stress, Ambient	138 N./mm ²
Allowable Stress, Operating	138 N./mm ²
Allowable Stress, Hydrotest	234 N./mm ²
Material Density	0.00775 kg./cm ³
P Number Thickness	32.004 mm.
Yield Stress, Operating	220 N./mm ²
UCS-66 Chart Curve Designation	B
External Pressure Chart Name	CS-2
UNS Number	K02700
Product Form	Plate
Efficiency, Longitudinal Seam	0.7
Efficiency, Head-to-Skirt or Circ. Seam	0.7
Weld is pre-Heated	No

Element#	2/6
Element From Node	20
Element To Node	30
Element Type	Skirt Support
Description	Skirt-1
Distance "FROM" to "TO"	1000 mm.
Skirt Inside Diameter	1900 mm.
Diameter of Skirt at Base	1925 mm.
Skirt Thickness	10 mm.
Internal Corrosion Allowance	3 mm.
Nominal Thickness	10 mm.
External Corrosion Allowance	0 mm.
Design Temperature Internal Pressure	230 °C
Design Temperature External Pressure	198 °C
Effective Diameter Multiplier	1.42
Material Name	SA-516 70
Efficiency, Longitudinal Seam	0.7
Efficiency, Head-to-Skirt or Circ. Seam	0.5
Weld is pre-Heated	No

Element#	3/6
Element From Node	30
Element To Node	40
Element Type	Elliptical
Description	Bottom Dish
Distance "FROM" to "TO"	50 mm.
Inside Diameter	1900 mm.
Element Thickness	9 mm.
Internal Corrosion Allowance	3 mm.
Nominal Thickness	12 mm.
External Corrosion Allowance	0 mm.
Design Internal Pressure	3.5 bars
Design Temperature Internal Pressure	230 °C



Design External Pressure	1.0342	bars
Design Temperature External Pressure	198	°C
Effective Diameter Multiplier	1.42	
Material Name	SA-516 70	
Efficiency, Longitudinal Seam	0.85	
Efficiency, Circumferential Seam	0.85	
Elliptical Head Factor	2.0	
Weld is pre-Heated	No	

Element From Node	30	
Detail Type	Platform	
Detail ID	PLAT:[1 OF 1]	
Dist. from "FROM" Node / Offset dist	0	mm.
Platform Start Angle (degrees)	180.0	
Platform End Angle (degrees)	0.0	
Platform Wind Area	42192	cm2
Platform Weight	1577	Kgf
Platform Railing Weight	0	Kgf/mm.
Platform Grating Weight	0.017	Kgs/cm2
Platform Width	1500	mm.
Platform Height	1000	mm.
Platform Clearance or End Offset	120	mm.
Platform Force Coefficient	1.0	
Ladder Layout Angle	90.0	
Ladder Start Elevation	0	mm.
Ladder End Elevation	3000	mm.
Unit Weight of Ladder	0.037	Kgf/mm.
Platform Length (top head platform)	0	mm.

Element From Node	30	
Detail Type	Liquid	
Detail ID	Liquid: 10	
Dist. from "FROM" Node / Offset dist	-450	mm.
Height/Length of Liquid	500	mm.
Liquid Density	0.0009565	kg./cm3

Element From Node	30	
Detail Type	Insulation	
Detail ID	Ins: 10	
Dist. from "FROM" Node / Offset dist	-450	mm.
Height/Length of Insulation	500	mm.
Thickness of Insulation	80	mm.
Density	0.0003516	kg./cm3

Element From Node	30	
Detail Type	Nozzle	
Detail ID	N2-6"	
Dist. from "FROM" Node / Offset dist	300	mm.
Nozzle Diameter	6	in.
Nozzle Schedule	160	
Nozzle Class	300	
Layout Angle	270.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	122.97	Kgf
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-106 B	

Element From Node	30	
Detail Type	Nozzle	
Detail ID	D1-3"	
Dist. from "FROM" Node / Offset dist	200	mm.
Nozzle Diameter	3	in.
Nozzle Schedule	160	
Nozzle Class	300	
Layout Angle	90.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	39.315	Kgf
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-106 B	



Element#	4/6
Element From Node	40
Element To Node	50
Element Type	Cylinder
Description	shell
Distance "FROM" to "TO"	3000 mm.
Inside Diameter	1900 mm.
Element Thickness	12 mm.
Internal Corrosion Allowance	3 mm.
Nominal Thickness	12 mm.
External Corrosion Allowance	0 mm.
Design Internal Pressure	3.5 bars
Design Temperature Internal Pressure	230 °C
Design External Pressure	1.0342 bars
Design Temperature External Pressure	198 °C
Effective Diameter Multiplier	1.42
Material Name	SA-516 70
Efficiency, Longitudinal Seam	0.85
Efficiency, Circumferential Seam	0.85
Weld is pre-Heated	No
Element From Node	40
Detail Type	Liquid
Detail ID	Liquid: 20
Dist. from "FROM" Node / Offset dist	0 mm.
Height/Length of Liquid	2250 mm.
Liquid Density	0.0009565 kg./cm3
Element From Node	40
Detail Type	Insulation
Detail ID	Ins: 20
Dist. from "FROM" Node / Offset dist	0 mm.
Height/Length of Insulation	3000 mm.
Thickness of Insulation	80 mm.
Density	0.0003516 kg./cm3
Element From Node	40
Detail Type	Ring
Detail ID	Ring R1 Fr40
Dist. from "FROM" Node / Offset dist	2200 mm.
Inside Diameter of Ring	1924 mm.
Thickness of Ring	12 mm.
Outside Diameter of Ring	2224 mm.
Material Name	SA-516 70
Height of Section Ring	0 mm.
Using Custom Stiffener Section	No
Element From Node	40
Detail Type	Nozzle
Detail ID	M1-24"
Dist. from "FROM" Node / Offset dist	1500 mm.
Nozzle Diameter	24 in.
Nozzle Schedule	None
Nozzle Class	150
Layout Angle	90.0
Blind Flange (Y/N)	Y
Weight of Nozzle (Used if > 0)	397.3 Kg
Grade of Attached Flange	GR 1.1
Nozzle Matl	SA-516 70
Element From Node	40
Detail Type	Nozzle
Detail ID	K1-2"
Dist. from "FROM" Node / Offset dist	1550 mm.
Nozzle Diameter	2 in.
Nozzle Schedule	None



Nozzle Class	300	
Layout Angle	270.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	0	Kgf
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-105	

Element From Node	40	
Detail Type	Nozzle	
Detail ID	K2-2"	
Dist. from "FROM" Node / Offset dist	250	mm.
Nozzle Diameter	2	in.
Nozzle Schedule	None	
Nozzle Class	300	
Layout Angle	270.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	0	Kgf
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-105	

Element From Node	40	
Detail Type	Nozzle	
Detail ID	N5-2"	
Dist. from "FROM" Node / Offset dist	250	mm.
Nozzle Diameter	2	in.
Nozzle Schedule	XXS	
Nozzle Class	600	
Layout Angle	90.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	8.7136	Kgf
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-106 B	

Element From Node	40	
Detail Type	Weight	
Detail ID	clip weight	
Dist. from "FROM" Node / Offset dist	2500	mm.
Miscellaneous Weight	475	Kgf
Offset from Element Centerline	0	mm.

Element From Node	40	
Detail Type	Weight	
Detail ID	davit	
Dist. from "FROM" Node / Offset dist	1800	mm.
Miscellaneous Weight	50	Kgf
Offset from Element Centerline	1500	mm.

Element#	5/6	
Element From Node	50	
Element To Node	60	
Element Type	Cylinder	
Description	shell	
Distance "FROM" to "TO"	2000	mm.
Inside Diameter	1900	mm.
Element Thickness	14	mm.
Internal Corrosion Allowance	3	mm.
Nominal Thickness	14	mm.
External Corrosion Allowance	0	mm.
Design Internal Pressure	3.5	bars
Design Temperature Internal Pressure	230	°C
Design External Pressure	1.0342	bars
Design Temperature External Pressure	198	°C
Effective Diameter Multiplier	1.42	
Material Name	SA-516 70	
Efficiency, Longitudinal Seam	0.85	
Efficiency, Circumferential Seam	0.85	
Weld is pre-Heated	No	



Element From Node	50
Detail Type	Insulation
Detail ID	insulation
Dist. from "FROM" Node / Offset dist	0 mm.
Height/Length of Insulation	2000 mm.
Thickness of Insulation	80 mm.
Density	0.0003516 kg./cm3

Element From Node	50
Detail Type	Nozzle
Detail ID	N1-16"
Dist. from "FROM" Node / Offset dist	600 mm.
Nozzle Diameter	20 in.
Nozzle Schedule	None
Nozzle Class	300
Layout Angle	270.0
Blind Flange (Y/N)	N
Weight of Nozzle (Used if > 0)	288.61 Kg
Grade of Attached Flange	GR 1.1
Nozzle Matl	SA-516 70

Element From Node	50
Detail Type	Nozzle
Detail ID	N6-6"
Dist. from "FROM" Node / Offset dist	1400 mm.
Nozzle Diameter	6 in.
Nozzle Schedule	160
Nozzle Class	300
Layout Angle	270.0
Blind Flange (Y/N)	N
Weight of Nozzle (Used if > 0)	54.089 Kg
Grade of Attached Flange	GR 1.1
Nozzle Matl	SA-106 B

Element From Node	50
Detail Type	Nozzle
Detail ID	N4-3"
Dist. from "FROM" Node / Offset dist	1000 mm.
Nozzle Diameter	3 in.
Nozzle Schedule	None
Nozzle Class	300
Layout Angle	90.0
Blind Flange (Y/N)	N
Weight of Nozzle (Used if > 0)	0 Kg
Grade of Attached Flange	GR 1.1
Nozzle Matl	SA-106 B

Element#	6/6
Element From Node	60
Element To Node	70
Element Type	Elliptical
Description	Top dish
Distance "FROM" to "TO"	50 mm.
Inside Diameter	1900 mm.
Element Thickness	12 mm.
Internal Corrosion Allowance	3 mm.
Nominal Thickness	14 mm.
External Corrosion Allowance	0 mm.
Design Internal Pressure	3.5 bars
Design Temperature Internal Pressure	230 °C
Design External Pressure	1.0342 bars
Design Temperature External Pressure	198 °C
Effective Diameter Multiplier	1.42
Material Name	SA-516 70
Efficiency, Longitudinal Seam	0.85
Efficiency, Circumferential Seam	0.85



Elliptical Head Factor	2.0	
Weld is pre-Heated	No	
Element From Node	60	
Detail Type	Platform	
Detail ID	PLAT:[1 OF 1]	
Dist. from "FROM" Node / Offset dist	700	mm.
Platform Start Angle (degrees)	0.0	
Platform End Angle (degrees)	0.0	
Platform Wind Area	62933	cm2
Platform Weight	2839.7	Kgf
Platform Railing Weight	0	Kgf/mm.
Platform Grating Weight	0.017	Kgs/cm2
Platform Width	3928	mm.
Platform Height	1000	mm.
Platform Clearance or End Offset	1964	mm.
Platform Force Coefficient	1.0	
Ladder Layout Angle	0.0	
Ladder Start Elevation	180	mm.
Ladder End Elevation	6000	mm.
Unit Weight of Ladder	0.037	Kgf/mm.
Platform Length (top head platform)	3928	mm.
Element From Node	60	
Detail Type	Insulation	
Detail ID	Ins: 30	
Dist. from "FROM" Node / Offset dist	0	mm.
Height/Length of Insulation	500	mm.
Thickness of Insulation	80	mm.
Density	0.0003516	kg./cm3
Element From Node	60	
Detail Type	Nozzle	
Detail ID	N3-20"	
Dist. from "FROM" Node / Offset dist	0	mm.
Nozzle Diameter	20	in.
Nozzle Schedule	None	
Nozzle Class	150	
Layout Angle	0.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	124.25	Kgf
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-516 70	
Element From Node	60	
Detail Type	Nozzle	
Detail ID	K1A-2"	
Dist. from "FROM" Node / Offset dist	600	mm.
Nozzle Diameter	2	in.
Nozzle Schedule	None	
Nozzle Class	300	
Layout Angle	90.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	0	Kgf
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-105	
Element From Node	60	
Detail Type	Weight	
Detail ID	Exchanger	
Dist. from "FROM" Node / Offset dist	500	mm.
Miscellaneous Weight	1800	Kgf
Offset from Element Centerline	0	mm.



XY Coordinate Calculations:

From	To	X (Horiz.) mm.	Y (Vert.) mm.	DX (Horiz.) mm.	DY (Vert.) mm.
Skirt-2		...	2000	...	2000
Skirt-1		...	3000	...	1000
Bottom Dish		...	3050	...	50
shell		...	6050	...	3000
shell		...	8050	...	2000
Top dish		...	8100	...	50

The calculated dimensions are based on the given element lengths. Due to variability in manufacturing (weld gaps etc.), the Tangent to Tangent length may not be exact. The length includes gasket thicknesses and raised face heights.

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Internal Pressure Results Summary:

Element Thickness, Pressure, Diameter and Allowable Stress :

From	To	Int. Press + Liq. Hd bars	Nominal Thickness mm.	Total Corr Allowance mm.	Element Diameter mm.	Allowable Stress(SE) N./mm ²	Element#
Skirt-2		...	10	3	1927	...	1
Skirt-1		...	10	3	1900	...	2
Bottom Dish		3.7582	12	3	1900	117.3	3
shell		3.7113	12	3	1900	117.3	4
shell		3.5	14	3	1900	117.3	5
Top dish		3.5	14	3	1900	117.3	6

Element Required Thickness and MAWP :

From	To	Design Pressure bars	M.A.W.P. Corroded bars	M.A.P. New & Cold bars	Minimum Thickness mm.	Required Thickness mm.	Element#
Skirt-2		...	No Calc	No Calc	10	No Calc	1
Skirt-1		...	No Calc	No Calc	10	No Calc	2
Bottom Dish		3.5	7.1526	11.1013	9	6.04182	3
shell		3.5	10.8031	14.7043	12	6.02119	4
shell		3.5	13.4452	17.1335	14	5.84888	5
Top dish		3.5	11.1127	14.797	12	5.83275	6

Minimum 7 11

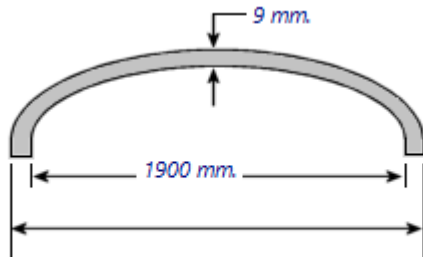
MAWP: 7 bars, limited by: Bottom Dish.

Elements Suitable for Design Internal Pressure.

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Internal Pressure Calculation Results:

ASME Code, Section VIII Division 1, 2025



Internal Pressure Results for: Bottom Dish

Elliptical Head Design Information:			
Design Pressure P	3.758 bars	Int. Design Temperature	230 °C
Material	SA-516 70	Ext. Design Temperature	198 °C
Length	50.000 mm.	External Pressure Chart	CS-2
		UNS Number	K02700
Int. Corr. All. c	3.0000 mm.	Allowable Stress (ope) S	138.0 N./mm ²
Ext. Corr. All. ce	0.0000 mm.	Allowable Stress (amb) Sa	138.0 N./mm ²
Inside Diameter Di	1900.000 mm.	Actual Stress	70.0 N./mm ²
Spec. Min. Thk t _e , or sa	9.0000 mm.	Spec. Nominal Thk.	12.000 mm.
Surface Area	43148.9 cm ²	SG of Contents	0.958
Element Volume	1039791.7 cm ³	Weight of Contents	992.0 Kg
Empty Weight	434.9 Kg	Operating Weight	1426.9 Kg
		Comp. All. Ope. Stress	82.0 N./mm ²
		Aspect Ratio	2.00

Radiography Information:			
User Defined		User Defined	
Circ. Joint Efficiency	85 %	Long. Joint Eff. E or v	85 %

Tolerance for Formed Heads per UG-81(a):

Head inner surface maximum deviation outside the specified shape, 1.25% of D: 23.750 mm.
 Head inner surface maximum deviation inside the specified shape, 0.625% of D: 11.875 mm.

Required Thickness due to Internal Pressure [tr]:

$$= (P \cdot D \cdot K_{cor}) / (2 \cdot S \cdot E - 0.2 \cdot P) \text{ Appendix 1-4(c)}$$

$$= (3.758 \cdot 1906.0 \cdot 0.996) / (2 \cdot 138.0 \cdot 0.85 - 0.2 \cdot 3.758)$$

$$= 3.0418 + 3.0000 = 6.0418 \text{ mm.}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

Less Operating Hydrostatic Head Pressure of 0.258 bars

$$= (2 \cdot S \cdot E \cdot t) / (K_{cor} \cdot D + 0.2 \cdot t) \text{ per Appendix 1-4 (c)}$$

$$= (2 \cdot 138.0 \cdot 0.85 \cdot 6.0) / (0.996 \cdot 1906.0 + 0.2 \cdot 6.0)$$

$$= 7 - 0 = 7 \text{ bars}$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$= (2 \cdot S \cdot E \cdot t) / (K \cdot D + 0.2 \cdot t) \text{ per Appendix 1-4 (c)}$$

$$= (2 \cdot 138.0 \cdot 0.85 \cdot 9.0) / (1.0 \cdot 1900.0 + 0.2 \cdot 9.0)$$

$$= 11 \text{ bars}$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$= (P \cdot (K_{cor} \cdot D + 0.2 \cdot t)) / (2 \cdot E \cdot t)$$

$$= (3.758 \cdot (0.996 \cdot 1906.0 + 0.2 \cdot 6.0)) / (2 \cdot 0.85 \cdot 6.0)$$

$$= 69.982 \text{ N./mm}^2$$



Straight Flange Required Thickness:

$$= (P \cdot R) / (S \cdot E - 0.6 \cdot P) + c \quad \text{per UG-27 (c)(1)}$$

$$= (3.758 \cdot 953.0) / (138.0 \cdot 0.85 - 0.6 \cdot 3.758) + 3.0$$

$$= 6.059 \text{ mm.}$$

Straight Flange Maximum Allowable Working Pressure:

Less Operating Hydrostatic Head Pressure of 0.258 bars

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \quad \text{per UG-27 (c)(1)}$$

$$= (138.0 \cdot 0.85 \cdot 9.0) / (953.0 + 0.6 \cdot 9.0)$$

$$= 11 - 0 = 11 \text{ bars}$$

Factor K, corroded condition [Kcor]:

$$= (2 + (\text{Inside Diameter} / (2 \cdot \text{Inside Head Depth}))^2) / 6$$

$$= (2 + (1906.0 / (2 \cdot 478.0))^2) / 6$$

$$= 0.995822$$

Percent Elong. per UCS-79, VIII-1-01-57 $(75 \cdot t_{\text{nom}} / R_f) \cdot (1 - R_f / R_o)$ 2.736 %**MDMT Calculations in the Knuckle Portion:**

Govrn. thk, tg = 9.0, tr = 6.0, c = 3.0 mm., E* = 0.85

Thickness Ratio = $tr \cdot E^* / (tg - c) = 0.85$, Temp. Reduction = 8 °C
 Min Metal Temp. w/o impact per UCS-66, Curve B -29 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -37 °C
MDMT Calculations in the Head Straight Flange:

Govrn. thk, tg = 12.0, tr = 6.044, c = 3.0 mm., E* = 0.85

Thickness Ratio = $tr \cdot E^* / (tg - c) = 0.571$, Temp. Reduction = 25 °C
 Min Metal Temp. w/o impact per UCS-66, Curve B -23 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -48 °C
 Min Metal Temp. w/o impact per UG-20(f) -29 °C

MDMT Design Information:			
MDMT Curve	B	Press. at MDMT	7.411 bars
UCS-66(b) red.	Yes	Min. Des. Mtl. Temp.	-1 °C
Governing Thk.	9.000 mm.	Basic MDMT	-29 °C
Stress/Thk. Ratio	0.850	Temperature Difference	8 °C
UCS-66(c) red.	No	Computed Min. Temp.	-37 °C
		MDMT per UG-20(f)	-29 °C
		Joint Efficiency E*	85 %

Note: Temperature values displayed above are for the location with the highest basic MDMT

Tolerance for Formed Heads per UG-81(a):

Head inner surface maximum deviation outside the specified shape, 1.25% of D: 23.750 mm.

Head inner surface maximum deviation inside the specified shape, 0.625% of D: 11.875 mm.

EXTERNAL PRESSURE RESULTS for: Bottom Dish**Elliptical Head From 30 to 40 Ext. Chart: CS-2 at 198 °C**

Elastic Modulus from Chart: CS-2 at 198 °C: 193850704 KPa.

Results for Maximum Allowable Ext. Pressure				MAEP
Tca	Outer Dia	Do/t	Factor A	Factor B
6.000	1918.00	319.67	0.0004345	42.11

$$\text{MAEP} = B / (K_0 \cdot D_o / t) = 42.1146 / (0.9 \cdot 319.6667) = 1.4638 \text{ bars}$$

LP Condensate Flash Vessel

PV Elite 28 U1 Licensee: SL User

FileName : LP CONDENSATE FLASH DRUM (V8301C) -7 Percent Page 24 of 171

Element: Bottom Dish Step: 4 3:41pm Sep 16,2026



HEXAGON

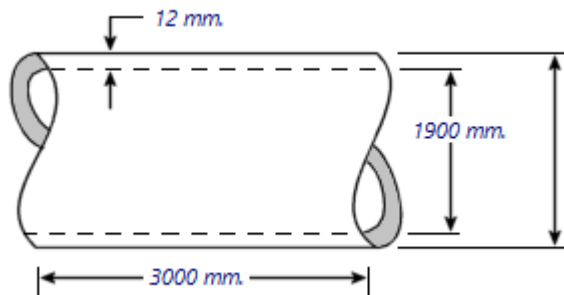
Results for Required Thickness				Tca
Tca	Outer Dia	Do/t	Factor A	Factor B
5.044	1918.00	380.29	0.0003652	35.40

MAEP = $B/(K0 \cdot Do/t)$ = $35.4014/(0.9 \cdot 380.2855)$ = 1.0343 bars

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Internal Pressure Calculation Results:

ASME Code, Section VIII Division 1, 2025



Internal Pressure Results for: shell

Cylindrical Shell Design Information:			
Design Pressure P	3.711 bars	Int. Design Temperature	230 °C
Material	SA-516 70	Ext. Design Temperature	198 °C
Length	3000.000 mm.	External Pressure Chart	CS-2
		UNS Number	K02700
Int. Corr. All. c	3.0000 mm.	Allowable Stress (ope) S	138.0 N./mm ²
Ext. Corr. All. ce	0.0000 mm.	Allowable Stress (amb) Sa	138.0 N./mm ²
Inside Diameter Di	1900.000 mm.	Actual Stress	46.5 N./mm ²
Spec. Min. Thk t, e, or sa	12.0000 mm.	Spec. Nominal Thk.	12.000 mm.
Surface Area	181332.8 cm ²	SG of Contents	0.958
Element Volume	8507386.0 cm ³	Weight of Contents	6108.8 Kg
Empty Weight	1676.0 Kg	Operating Weight	7784.8 Kg
		Comp. All. Ope. Stress	91.4 N./mm ²

Radiography Information:			
User Defined		User Defined	
Circ. Joint Efficiency	85 %	Long. Joint Eff. E or v	85 %

Required Thickness due to Internal Pressure [tr]:

$$\begin{aligned}
 &= (P \cdot R) / (S \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)} \\
 &= (3.711 \cdot 953.0) / (138.0 \cdot 0.85 - 0.6 \cdot 3.711) \\
 &= 3.0212 + 3.0000 = 6.0212 \text{ mm.}
 \end{aligned}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

Less Operating Hydrostatic Head Pressure of 0.211 bars

$$\begin{aligned}
 &= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)} \\
 &= (138.0 \cdot 0.85 \cdot 9.0) / (953.0 + 0.6 \cdot 9.0) \\
 &= 11 - 0 = 11 \text{ bars}
 \end{aligned}$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$\begin{aligned}
 &= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)} \\
 &= (138.0 \cdot 0.85 \cdot 12.0) / (950.0 + 0.6 \cdot 12.0) \\
 &= 15 \text{ bars}
 \end{aligned}$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$\begin{aligned}
 &= (P \cdot (R + 0.6 \cdot t)) / (E \cdot t) \\
 &= (3.711 \cdot (953.0 + 0.6 \cdot 9.0)) / (0.85 \cdot 9.0) \\
 &= 46.498 \text{ N./mm}^2
 \end{aligned}$$

% Elongation per Table UG-79-1 (50 * tnom / Rf * (1 - Rf / Ro)) 0.628 %

Minimum Design Metal Temperature Results:



Govrn. thk, tg = 12.0, tr = 6.006, c = 3.0 mm., E* = 0.85
 Thickness Ratio = tr * E*/(tg - c) = 0.567, Temp. Reduction = 25 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -23 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -48 °C
 Min Metal Temp. w/o impact per UG-20(f) -29 °C

MDMT Design Information:			
MDMT Curve	B	Press. at MDMT	7.364 bars
UCS-66(b) red.	Yes	Min. Des. Mtl. Temp.	-1 °C
Governing Thk.	12.000 mm.	Basic MDMT	-23 °C
Stress/Thk. Ratio	0.567	Temperature Difference	25 °C
UCS-66(c) red.	No	Computed Min. Temp.	-48 °C
		MDMT per UG-20(f)	-29 °C
		Joint Efficiency E*	85 %

EXTERNAL PRESSURE RESULTS for: shell

Cylindrical Shell From 40 to Ring R1 Fr40 Ext. Chart: CS-2 at 198 °C

Elastic Modulus from Chart: CS-2 at 198 °C: 193850704 KPa.

Results for Maximum Allowable Ext. Pressure					MAEP	
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
9.000	1924.00	2408.33	213.78	1.2517	0.0003386	32.82

$$MAEP = (4*B)/(3*(Do/t)) = (4*32.8201)/(3*213.7778) = 2.0469 \text{ bars}$$

Results for Required Thickness					Tca	
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
6.858	1924.00	2408.33	280.55	1.2517	0.0002245	21.76

$$MAEP = (4*B)/(3*(Do/t)) = (4*21.7638)/(3*280.5488) = 1.0343 \text{ bars}$$

Results for Maximum Stiffened Length					Slen	
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
9.000	1924.00	4624.14	213.78	2.4034	0.0001712	16.60

$$MAEP = (4*B)/(3*(Do/t)) = (4*16.5973)/(3*213.7778) = 1.0351 \text{ bars}$$

Cylindrical Shell From Ring R1 Fr40 to 50 Ext. Chart: CS-2 at 198 °C

Elastic Modulus from Chart: CS-2 at 198 °C: 193850704 KPa.

Results for Maximum Allowable Ext. Pressure					MAEP	
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
9.000	1924.00	3008.33	213.78	1.5636	0.0002684	26.01

$$MAEP = (4*B)/(3*(Do/t)) = (4*26.0128)/(3*213.7778) = 1.6223 \text{ bars}$$

Results for Required Thickness					Tca	
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
7.532	1924.00	3008.33	255.43	1.5636	0.0002044	19.81



$$MAEP = (4*B)/(3*(Do/t)) = (4*19.8147)/(3*255.4328) = 1.0342 \text{ bars}$$

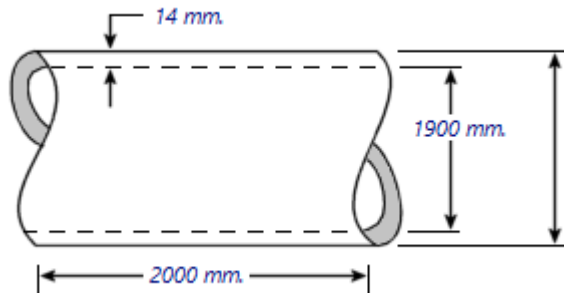
Results for Maximum Stiffened Length					Slen	
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
9.000	1924.00	4625.46	213.78	2.4041	0.0001712	16.59

$$MAEP = (4*B)/(3*(Do/t)) = (4*16.5924)/(3*213.7778) = 1.0348 \text{ bars}$$

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Internal Pressure Calculation Results:

ASME Code, Section VIII Division 1, 2025



Internal Pressure Results for: shell

Cylindrical Shell Design Information:					
Design Pressure	P	3.500 bars	Int. Design Temperature	230 °C	
Material		SA-516 70	Ext. Design Temperature	198 °C	
Length		2000.000 mm.	External Pressure Chart	CS-2	
			UNS Number	K02700	
Int. Corr. All.	c	3.0000 mm.	Allowable Stress (ope)	S	138.0 N./mm^2
Ext. Corr. All.	ce	0.0000 mm.	Allowable Stress (amb)	Sa	138.0 N./mm^2
Inside Diameter	Di	1900.000 mm.	Actual Stress	35.9 N./mm^2	
Spec. Min. Thk t _e , or s _a		14.0000 mm.	Spec. Nominal Thk.	14.000 mm.	
Surface Area		121139.8 cm^2	SG of Contents	0.000	
Element Volume		5671591.0 cm3	Weight of Contents	0.0 Kg	
Empty Weight		1304.9 Kg	Operating Weight	1304.9 Kg	
			Comp. All. Ope. Stress	95.8 N./mm^2	

Radiography Information:			
User Defined		User Defined	
Circ. Joint Efficiency	85 %	Long. Joint Eff.	E or v 85 %

Required Thickness due to Internal Pressure [tr]:

$$\begin{aligned}
 &= (P \cdot R) / (S \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)} \\
 &= (3.5 \cdot 953.0) / (138.0 \cdot 0.85 - 0.6 \cdot 3.5) \\
 &= 2.8489 + 3.0000 = 5.8489 \text{ mm.}
 \end{aligned}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

$$\begin{aligned}
 &= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)} \\
 &= (138.0 \cdot 0.85 \cdot 11.0) / (953.0 + 0.6 \cdot 11.0) \\
 &= 13 \text{ bars}
 \end{aligned}$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$\begin{aligned}
 &= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)} \\
 &= (138.0 \cdot 0.85 \cdot 14.0) / (950.0 + 0.6 \cdot 14.0) \\
 &= 17 \text{ bars}
 \end{aligned}$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$\begin{aligned}
 &= (P \cdot (R + 0.6 \cdot t)) / (E \cdot t) \\
 &= (3.5 \cdot (953.0 + 0.6 \cdot 11.0)) / (0.85 \cdot 11.0) \\
 &= 35.923 \text{ N./mm}^2
 \end{aligned}$$

% Elongation per Table UG-79-1 (50 * t_{nom} / R_f * (1 - R_f / R_o)) 0.731 %

Minimum Design Metal Temperature Results:

Govrn. thk, t_g = 14.0, t_r = 5.833, c = 3.0 mm., E* = 0.85



Thickness Ratio = $tr * E / (tg - c) = 0.451$, Temp. Reduction = 40 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -18 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -48 °C
 Min Metal Temp. w/o impact per UG-20(f) -29 °C

MDMT Design Information:			
MDMT Curve	B	Press. at MDMT	7.153 bars
UCS-66(b) red.	Yes	Min. Des. Mtl. Temp.	-1 °C
Governing Thk.	14.000 mm.	Basic MDMT	-18 °C
Stress/Thk. Ratio	0.451	Temperature Difference	30 °C
UCS-66(c) red.	No	Computed Min. Temp.	-48 °C
		MDMT per UG-20(f)	-29 °C
		Joint Efficiency E*	85 %

EXTERNAL PRESSURE RESULTS for: shell

Cylindrical Shell From 50 to 60 Ext. Chart: CS-2 at 198 °C

Elastic Modulus from Chart: CS-2 at 198 °C: 193850704 KPa.

Results for Maximum Allowable Ext. Pressure					MAEP	
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
11.000	1928.00	3008.33	175.27	1.5603	0.0003661	35.48

$$MAEP = (4*B)/(3*(Do/t)) = (4*35.4836)/(3*175.2727) = 2.6991 \text{ bars}$$

Results for Required Thickness					Tca	
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
7.541	1928.00	3008.33	255.66	1.5603	0.0002046	19.83

$$MAEP = (4*B)/(3*(Do/t)) = (4*19.833)/(3*255.6611) = 1.0343 \text{ bars}$$

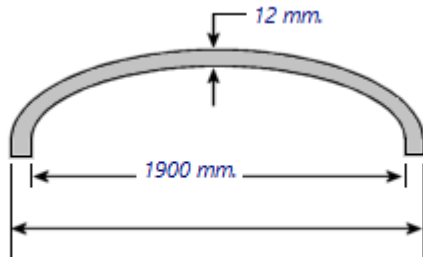
Results for Maximum Stiffened Length					Slen	
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
11.000	1928.00	7491.71	175.27	3.8857	0.0001403	13.60

$$MAEP = (4*B)/(3*(Do/t)) = (4*13.6035)/(3*175.2727) = 1.0348 \text{ bars}$$

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Internal Pressure Calculation Results:

ASME Code, Section VIII Division 1, 2025



Internal Pressure Results for: Top dish

Elliptical Head Design Information:			
Design Pressure P	3.500 bars	Int. Design Temperature	230 °C
Material	SA-516 70	Ext. Design Temperature	198 °C
Length	50.000 mm.	External Pressure Chart	CS-2
		UNS Number	K02700
Int. Corr. All. c	3.0000 mm.	Allowable Stress (ope) S	138.0 N./mm ²
Ext. Corr. All. ce	0.0000 mm.	Allowable Stress (amb) Sa	138.0 N./mm ²
Inside Diameter Di	1900.000 mm.	Actual Stress	43.5 N./mm ²
Spec. Min. Thk t _e , or sa	12.0000 mm.	Spec. Nominal Thk.	14.000 mm.
Surface Area	43322.2 cm ²	SG of Contents	0.000
Element Volume	1039791.7 cm ³	Weight of Contents	0.0 Kg
Empty Weight	508.4 Kg	Operating Weight	508.4 Kg
		Comp. All. Ope. Stress	91.4 N./mm ²
		Aspect Ratio	2.00

Radiography Information:			
User Defined		User Defined	
Circ. Joint Efficiency	85 %	Long. Joint Eff. E or v	85 %

Tolerance for Formed Heads per UG-81(a):

Head inner surface maximum deviation outside the specified shape, 1.25% of D: 23.750 mm.
Head inner surface maximum deviation inside the specified shape, 0.625% of D: 11.875 mm.

Required Thickness due to Internal Pressure [tr]:

$$= (P \cdot D \cdot K_{cor}) / (2 \cdot S \cdot E - 0.2 \cdot P) \text{ Appendix 1-4(c)}$$

$$= (3.5 \cdot 1906.0 \cdot 0.996) / (2 \cdot 138.0 \cdot 0.85 - 0.2 \cdot 3.5)$$

$$= 2.8327 + 3.0000 = 5.8327 \text{ mm.}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

$$= (2 \cdot S \cdot E \cdot t) / (K_{cor} \cdot D + 0.2 \cdot t) \text{ per Appendix 1-4 (c)}$$

$$= (2 \cdot 138.0 \cdot 0.85 \cdot 9.0) / (0.996 \cdot 1906.0 + 0.2 \cdot 9.0)$$

$$= 11 \text{ bars}$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$= (2 \cdot S \cdot E \cdot t) / (K \cdot D + 0.2 \cdot t) \text{ per Appendix 1-4 (c)}$$

$$= (2 \cdot 138.0 \cdot 0.85 \cdot 12.0) / (1.0 \cdot 1900.0 + 0.2 \cdot 12.0)$$

$$= 15 \text{ bars}$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$= (P \cdot (K_{cor} \cdot D + 0.2 \cdot t)) / (2 \cdot E \cdot t)$$

$$= (3.5 \cdot (0.996 \cdot 1906.0 + 0.2 \cdot 9.0)) / (2 \cdot 0.85 \cdot 9.0)$$

$$= 43.463 \text{ N./mm}^2$$



Straight Flange Required Thickness:

$$= (P \cdot R) / (S \cdot E - 0.6 \cdot P) + c \quad \text{per UG-27 (c)(1)}$$

$$= (3.5 \cdot 953.0) / (138.0 \cdot 0.85 - 0.6 \cdot 3.5) + 3.0$$

$$= 5.849 \text{ mm.}$$

Straight Flange Maximum Allowable Working Pressure:

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \quad \text{per UG-27 (c)(1)}$$

$$= (138.0 \cdot 0.85 \cdot 11.0) / (953.0 + 0.6 \cdot 11.0)$$

$$= 13 \text{ bars}$$

Factor K, corroded condition [Kcor]:

$$= (2 + (\text{Inside Diameter} / (2 \cdot \text{Inside Head Depth}))^2) / 6$$

$$= (2 + (1906.0 / (2 \cdot 478.0))^2) / 6$$

$$= 0.995822$$

Percent Elong. per UCS-79, VIII-1-01-57 $(75 \cdot t_{nom} / R_f) \cdot (1 - R_f / R_o)$ 3.182 %

MDMT Calculations in the Knuckle Portion:

Govrn. thk, tg = 12.0, tr = 5.791, c = 3.0 mm., E* = 0.85

Thickness Ratio = $tr \cdot E^* / (tg - c) = 0.547$, Temp. Reduction = 27 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -23 °C

Min Metal Temp. at Required thickness (UCS 66.1) -48 °C

Min Metal Temp. w/o impact per UG-20(f) -29 °C

MDMT Calculations in the Head Straight Flange:

Govrn. thk, tg = 14.0, tr = 5.833, c = 3.0 mm., E* = 0.85

Thickness Ratio = $tr \cdot E^* / (tg - c) = 0.451$, Temp. Reduction = 40 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -18 °C

Min Metal Temp. at Required thickness (UCS 66.1) -48 °C

Min Metal Temp. w/o impact per UG-20(f) -29 °C

MDMT Design Information:			
MDMT Curve	B	Press. at MDMT	7.153 bars
UCS-66(b) red.	Yes	Min. Des. Mtl. Temp.	-1 °C
Governing Thk.	12.000 mm.	Basic MDMT	-23 °C
Stress/Thk. Ratio	0.547	Temperature Difference	25 °C
UCS-66(c) red.	No	Computed Min. Temp.	-48 °C
		MDMT per UG-20(f)	-29 °C
		Joint Efficiency E*	85 %

Note: Temperature values displayed above are for the location with the highest basic MDMT

Tolerance for Formed Heads per UG-81(a):

Head inner surface maximum deviation outside the specified shape, 1.25% of D: 23.750 mm.

Head inner surface maximum deviation inside the specified shape, 0.625% of D: 11.875 mm.

EXTERNAL PRESSURE RESULTS for: Top dish

Elliptical Head From 60 to 70 Ext. Chart: CS-2 at 198 °C

Elastic Modulus from Chart: CS-2 at 198 °C: 193850704 KPa.

Results for Maximum Allowable Ext. Pressure				MAEP
Tca	Outer Dia	Do/t	Factor A	Factor B
9.000	1924.00	213.78	0.0006497	62.97

$$MAEP = B / (K \cdot Do / t) = 62.9749 / (0.9 \cdot 213.7778) = 3.2729 \text{ bars}$$

LP Condensate Flash Vessel

PV Elite 28 U1 Licensee: SL User

FileName : LP CONDENSATE FLASH DRUM (V8301C) -7 Percent Page 32 of 171

Element: Top dish Step: 7 3:41pm Sep 16,2026



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Results for Required Thickness				Tca
Tca	Outer Dia	Do/t	Factor A	Factor B
5.059	1924.00	380.29	0.0003652	35.40

$$MAEP = B / (K0 * Do / t) = 35.4011 / (0.9 * 380.289) = 1.0343 \text{ bars}$$

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**Hydrostatic Test Pressure Results:**

UG99(b)	= 1.30 * M.A.W.P. * Sa/S	9.298 bars
UG99b(b)(35)	= 1.30 * Design Pres * Sa/S	4.550 bars
UG99(c)	= 1.30 * M.A.P. - Head(Hyd)	14.147 bars
UG100	= 1.10 * M.A.W.P. * Sa/S	7.868 bars
PED	= max(1.43*DP, 1.25*DP*ratio)	5.005 bars
PED&UG99(b)(35)	= max(1.43*DP, 1.3*DP*ratio)	5.005 bars
App 27-4	= M.A.W.P.	7.153 bars

User Defined Hydrostatic Test Pressure at High Point 9.620 bars

Horizontal Test performed per: User Hydro Pressure

Please note that Nozzle, Shell, Head, Flange, etc MAWPs are all considered when determining the hydrotest pressure for those test types that are based on the MAWP of the vessel.

Stresses on Elements due to Test Pressure (N./mm² & bars):

From To	Stress	Allowable	Ratio	Pressure
Bottom Dish	121.9	234.0	0.521	9.81
shell	92.0	234.0	0.393	9.81
shell	79.0	234.0	0.338	9.81
Top dish	91.5	234.0	0.391	9.81

Stress ratios for Nozzle and Pad Materials (N./mm²):

Description	Pad/Nozzle	Ambient	Operating	Ratio
N2-6"	Nozzle	118.00	118.00	1.000
N2-6"	Pad	138.00	138.00	1.000
D1-3"	Nozzle	118.00	118.00	1.000
D1-3"	Pad	138.00	138.00	1.000
M1-24"	Nozzle	138.00	138.00	1.000
M1-24"	Pad	138.00	138.00	1.000
K1-2"	Nozzle	138.00	136.80	1.009
K2-2"	Nozzle	138.00	136.80	1.009
N5-2"	Nozzle	118.00	118.00	1.000
N1-16"	Nozzle	138.00	138.00	1.000
N1-16"	Pad	138.00	138.00	1.000
N6-6"	Nozzle	118.00	118.00	1.000
N6-6"	Pad	138.00	138.00	1.000
N4-3"	Nozzle	118.00	118.00	1.000
N3-20"	Nozzle	138.00	138.00	1.000
N3-20"	Pad	138.00	138.00	1.000
K1A-2"	Nozzle	138.00	136.80	1.009

Minimum 1.000

Stress ratios for Stiffening Ring Materials (N./mm²):

Description	Ambient	Operating	Ratio
Ring R1 Fr40	138.00	138.00	1.000

Minimum 1.000

Stress ratios for Pressurized Vessel Elements (N./mm²):

Description	Ambient	Operating	Ratio
Bottom Dish	138.00	138.00	1.000
shell	138.00	138.00	1.000
shell	138.00	138.00	1.000
Top dish	138.00	138.00	1.000



Minimum 1.000

Hoop Stress in Nozzle Wall during Pressure Test (N./mm2):

Description	Ambient	Operating	Ratio
N2-6"	4.78	216.00	0.022
D1-3"	4.09	216.00	0.019
M1-24"	24.52	234.00	0.105
K1-2"	2.07	225.00	0.009
K2-2"	2.07	225.00	0.009
N5-2"	2.66	216.00	0.012
N1-16"	13.45	234.00	0.057
N6-6"	4.78	216.00	0.022
N4-3"	2.40	216.00	0.011
N3-20"	20.37	234.00	0.087
K1A-2"	2.07	225.00	0.009

Elements Suitable for Test Pressure.

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**Hydrostatic Test Pressure Results:**

UG99(b)	= 1.30 * M.A.W.P. * Sa/S	9.298 bars
UG99b(b)(35)	= 1.30 * Design Pres * Sa/S	4.550 bars
UG99(c)	= 1.30 * M.A.P. - Head(Hyd)	13.790 bars
UG100	= 1.10 * M.A.W.P. * Sa/S	7.868 bars
PED	= max(1.43*DP, 1.25*DP*ratio)	5.005 bars
PED&UG99(b)(35)	= max(1.43*DP, 1.3*DP*ratio)	5.005 bars
App 27-4	= M.A.W.P.	7.153 bars

UG-99(b) Note 35, Test Pressure Calculation:

= Test Factor • Design Pressure • Stress Ratio

= 1.3 • 3.5 • 1.0

= 4.550 bars

Vertical Test performed per: UG-99b (Note 35)

Please note that Nozzle, Shell, Head, Flange, etc MAWPs are all considered when determining the hydrotest pressure for those test types that are based on the MAWP of the vessel.

Stresses on Elements due to Test Pressure (N./mm² & bars):

From To	Stress	Allowable	Ratio	Pressure
Bottom Dish	96.7	234.0	0.413	5.19
shell	64.4	234.0	0.275	5.14
shell	49.7	234.0	0.213	4.85
Top dish	57.7	234.0	0.247	4.65

Stress ratios for Nozzle and Pad Materials (N./mm²):

Description	Pad/Nozzle	Ambient	Operating	Ratio
N2-6"	Nozzle	118.00	118.00	1.000
N2-6"	Pad	138.00	138.00	1.000
D1-3"	Nozzle	118.00	118.00	1.000
D1-3"	Pad	138.00	138.00	1.000
M1-24"	Nozzle	138.00	138.00	1.000
M1-24"	Pad	138.00	138.00	1.000
K1-2"	Nozzle	138.00	136.80	1.009
K2-2"	Nozzle	138.00	136.80	1.009
N5-2"	Nozzle	118.00	118.00	1.000
N1-16"	Nozzle	138.00	138.00	1.000
N1-16"	Pad	138.00	138.00	1.000
N6-6"	Nozzle	118.00	118.00	1.000
N6-6"	Pad	138.00	138.00	1.000
N4-3"	Nozzle	118.00	118.00	1.000
N3-20"	Nozzle	138.00	138.00	1.000
N3-20"	Pad	138.00	138.00	1.000
K1A-2"	Nozzle	138.00	136.80	1.009

Minimum

1.000

Stress ratios for Stiffening Ring Materials (N./mm²):

Description	Ambient	Operating	Ratio
Ring R1 Fr40	138.00	138.00	1.000

Minimum

1.000

Stress ratios for Pressurized Vessel Elements (N./mm²):

Description	Ambient	Operating	Ratio
Bottom Dish	138.00	138.00	1.000



shell	138.00	138.00	1.000
shell	138.00	138.00	1.000
Top dish	138.00	138.00	1.000

Minimum 1.000

Hoop Stress in Nozzle Wall during Pressure Test (N./mm²):

Description	Ambient	Operating	Ratio
N2-6"	2.53	216.00	0.012
D1-3"	2.16	216.00	0.010
M1-24"	12.85	234.00	0.055
K1-2"	1.09	225.00	0.005
K2-2"	1.09	225.00	0.005
N5-2"	1.39	216.00	0.006
N1-16"	6.65	234.00	0.028
N6-6"	2.36	216.00	0.011
N4-3"	1.19	216.00	0.006
N3-20"	9.66	234.00	0.041
K1A-2"	0.98	225.00	0.004

Elements Suitable for Test Pressure.

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External Pressure Calculation Results Summary:

ASME Code, Section VIII Division 1, 2025

MDMT Calculations for: Ring R1 Fr40, Shell to Ring Junction, Curve: B

Govrn. thk, $t_g = 12.0$, $t_r = 6.006$, $c = 3.0$ mm., $E^* = 0.85$

Thickness Ratio = $t_r \cdot E^* / (t_g - c) = 0.567$, Temp. Reduction = 25 °C

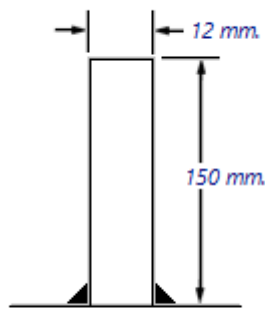
Min Metal Temp. w/o impact per UCS-66, Curve B -23 °C

Min Metal Temp. at Required thickness (UCS 66.1) -48 °C

Min Metal Temp. w/o impact per UG-20(f) -29 °C

Stiffening Ring Calculations for: Ring R1 Fr40, SA-516 70 , Bar Ring: 150.0 x 12.0 mm.

Material treatment options:



Effective Length of Shell: 145 mm.

	Area (cm ²)	Distance (mm.)	Area*Dist
Shell:	13.027	4.5000	58.623
Ring :	18.000	84.0000	1512.001
Total:	31.027		1570.625

Centroid of Ring plus Shell: 51 mm.

	Inertia	Distance	A*Dist ²
Shell:	0.879	46.1205	27710.656
Ring :	337.500	-33.3795	20055.494
Total:	338.379		47766.148

Available Moment of Inertia, Ring + Shell: 0.816E+03 cm⁴

Required Stress in Ring plus Shell Breq 15.44 N./mm²

Required Strain in Ring plus Shell Areq 0.0001590

Required Moment of Inertia, Ring plus Shell:

$$= (D_o^2 \cdot L_s(t + A_s/L_s)A) / 10.9$$

$$= (1924.0^2 \cdot 2708.333(9.0 + 18.0/2708.333)0.000159) / 10.9$$

$$= 141 \text{ cm}^4$$

External Pressure Calculations:

From	To	Section Length	Outside Diameter	Corroded Thickness	Factor A	Factor B
------	----	----------------	------------------	--------------------	----------	----------



		mm.	mm.	mm.		N./mm ²
10	20	3000	...	...	No Calc	No Calc
20	30	3000	...	...	No Calc	No Calc
30	40	No Calc	1918	6	0.00043448	42.1146
40	Ring	2408.33	1924	9	0.00033859	32.8201
Ring	50	3008.33	1924	9	0.00026836	26.0128
50	60	3008.33	1928	11	0.00036607	35.4836
60	70	No Calc	1924	9	0.00064969	62.9749

External Pressure Calculations:

From	To	External Actual T. mm.	External Required T. mm.	External Design Pressure bars	External M.A.W.P. bars
10	20	...	No Calc	...	No Calc
20	30	...	No Calc	...	No Calc
30	40	9	8.04358	1.03421	1.46375
40	Ring	12	9.85799	1.03421	2.04687
Ring	50	12	10.5323	1.03421	1.62233
50	60	14	10.5412	1.03421	2.69915
60	70	12	8.05931	1.03421	3.27293

Minimum

1

External Pressure Calculations:

From	To	Actual Length Bet. Stiffeners mm.	Allowable Length Bet. Stiffeners mm.	Ring Inertia Required cm ⁴	Ring Inertia Available cm ⁴
10	20	3000	No Calc	No Calc	No Calc
20	30	3000	No Calc	No Calc	No Calc
30	40	No Calc	No Calc	No Calc	No Calc
40	Ring	2408.33	4624.14	No Calc	No Calc
Ring	50	3008.33	4625.46	141.34	816.04
50	60	3008.33	7491.71	No Calc	No Calc
60	70	No Calc	No Calc	No Calc	No Calc

Elements Suitable for External Pressure.

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**Element and Detail Weights:**

From	To	Element Metal Wgt. kg.	Element ID Volume cm3	Corroded Metal Wgt. kg.	Corroded ID Volume cm3	Extra due Misc % kg.
10	20	1447.68	...	1168.64	...	101.338
20	30	468.144	...	398.063	...	32.7701
30	40	434.871	1039791	326.153	1049223	30.4409
40	50	1676	8507386	1258.97	8561203	117.32
50	60	1304.91	5671591	1026.89	5707468	91.3438
60	70	508.377	1039791	399.439	1049223	35.5864
Total		5839	16258560.00	4578	16367118.00	408

Weight of Details:

From	Type	Weight of Detail kg.	X Offset, Dtl. Cent. mm.	Y Offset, Dtl. Cent. mm.	Z Offset Dtl. Cent. mm.	Description
30	Plat	1687.36	1224.23	...	...	PLAT:[1 OF 1]
30	Liqd	992.003	...	-225	...	Liquid: 10
30	Insl	183.748	...	-200	...	Ins: 10
30	Nozl	131.579	-300	-596.194	...	N2-6"
30	Nozl	42.0675	200	-1182.79	...	D1-3"
40	Liqd	6108.83	...	1125	...	Liquid: 20
40	Insl	531.88	...	1500	...	Ins: 20
40	Ring	97.2624	...	2200	...	Ring R1 Fr40
40	Nozl	425.113	1100	1500	...	M1-24"
40	Nozl	11.0224	-1053.19	1550	...	K1-2"
40	Nozl	11.0224	-1053.19	250	...	K2-2"
40	Nozl	9.3236	1050	250	...	N5-2"
40	Wght	475	...	2500	...	clip weight
40	Wght	50	1500	1800	...	davit
50	Insl	355.294	...	1000	...	insulation
50	Nozl	308.815	-1075.01	600	...	N1-16"
50	Nozl	57.8752	-1052.39	1400	...	N6-6"
50	Nozl	19.8286	1050.01	1000	...	N4-3"
60	Plat	3038.43	...	700	...	PLAT:[1 OF 1]
60	Insl	161.335	...	250	...	Ins: 30
60	Nozl	132.951	...	594.006	...	N3-20"
60	Nozl	11.0224	600	515.461	...	K1A-2"
60	Wght	1800	...	500	...	Exchanger

Total Weight of Each Detail Type:

Platforms	4725.8
Liquid	7100.8
Insulation	1232.3
Stiffeners	97.3
Nozzles	1160.6
Weights	2325.0

Sum of the Detail Weights 16641.8 kg.

Weight Summation Results: (kg.)

	Fabricated	Shop Test	Shipping	Erected	Empty	Operating
Main Elements	6248.8	6248.8	6248.8	6248.8	6248.8	6248.8
Stif. Rings	97.3	97.3	97.3	97.3	97.3	97.3
Nozzles	1160.6	1160.6	1160.6	1160.6	1160.6	1160.6
Wld Weights	525.0	525.0	525.0	525.0	525.0	525.0
Platforms	...	...	...	4725.8	4725.8	4725.8
Insulation	...	...	...	1232.3	1232.3	1232.3
Empty Weights	...	...	...	1800.0	1800.0	...



Test Weights	...	1800.0	...	...	...	...
Ope Weights	...	...	...	...	...	1800.0
Ope. Liquid	...	...	...	...	...	7100.8
Test Liquid	...	16248.6	...	...	...	...
Totals	8031.7	26080.3	8031.7	15789.7	15789.7	22890.5

Field Installation Options:

Platforms installed before lifting.

Insulation installed before lifting.

Miscellaneous Weight Percent: 7.0 %

Note that the above value for the miscellaneous weight percent has been applied to the shells/heads/flange/tubesheets/tubes etc. in the weight calculations for metallic components.

Weight Summary:

Fabricated Wt.	- Bare weight without removable internals	8031.7 kg.
Shop Test Wt.	- Fabricated weight + water (full)	26080.3 kg.
Shipping Wt.	- Fab. weight + removable intls.+ shipping app.	8031.7 kg.
Erected Wt.	- Fab. wt + or - loose items (trays,platforms etc.)	15789.7 kg.
Ope. Wt. no Liq	- Fab. weight + internals. + details + weights	15789.7 kg.
Operating Wt.	- Empty weight + operating liq. uncorroded	22890.5 kg.
Field Test Wt.	- Empty weight + test medium (full)	32038.3 kg.

Weights used in Vortex Shedding Calculations:

Mass of the Upper 1/3 of the Vertical Vessel (operating)	8467.5 kg.
Mass of the Upper 1/3 of the Vertical Vessel (filled w/water)	17150.5 kg.
Mass of the Upper 1/3 of the Vertical Vessel (empty)	8978.2 kg.

Surface Areas of Elements:

From	To	Outside Surface Area cm^2	Inside Surface Area cm^2
10	20	121077	119820
20	30	60318.6	59690.3
30	40	43148.9	42116.4
40	50	181333	179071
50	60	121140	119381
60	70	43322.2	42116.4

Total	570339.3	562194.7 cm2
Total	613.9	605.1 ft ²

Element and Detail Weights:

From	To	Total Ele. Empty Wgt. kg.	Total. Ele. Oper. Wgt. kg.	Total. Ele. Hydro. Wgt. kg.	Total Dtl. Offset Mom. Kg-m.	Oper. Wgt. No Liquid kg.
10	20	1549.02	1549.02	1549.02	...	1549.02
20	30	500.914	500.914	500.914	...	500.914
30	40	638.958	3502.07	3549.22	2034.71	2510.07
40	50	2872.07	9512.77	11906.1	529.207	3403.95
50	60	1782.77	2138.07	7806.2	372.072	2138.07
60	70	2487.94	5687.7	6726.85	6.61357	5687.7

Cumulative Vessel Weight

From	To	Cumulative Ope Wgt. No Liquid kg.	Cumulative Oper. Wgt. kg.	Cumulative Hydro. Wgt. kg.
------	----	---	---------------------------------	----------------------------------



10	20	15789.7	22890.5	32038.3
20	30	14240.7	21341.5	30489.3
30	40	13739.8	20840.6	29988.4
40	50	11229.7	17338.5	26439.2
50	60	7825.77	7825.77	14533.1
60	70	5687.7	5687.7	6726.85

The cumulative operating weights no liquid in the column above are the cumulative operating weights minus the operating liquid weight minus any weights absent in the empty condition.

Cumulative Vessel Moment

From	To	Cumulative Empty Mom. Kg-m.	Cumulative Oper. Mom. Kg-m.	Cumulative Hydro. Mom. Kg-m.
10	20	2942.6	2942.6	2942.6
20	30	2942.6	2942.6	2942.6
30	40	2942.6	2942.6	2942.6
40	50	907.893	907.893	907.893
50	60	378.686	378.686	378.686
60	70	6.61357	6.61357	6.61357

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**Nozzle Flange MAWP Results: (bars & °C)**

Nozzle Description	Flange Rating Ope.	Ambient	Des. Temp	Class	Grade/Group	Equiv. Press	- - - UG-44	Max Pressure 50%	DNV
N2-6"	42.72	51.00	230	300	GR 1.1	...	...	...	...
D1-3"	42.72	51.00	230	300	GR 1.1	53.197	10.884	42.720	10.884
M1-24"	12.78	19.60	230	150	GR 1.1	...	...	...	...
K1-2"	42.72	51.00	230	300	GR 1.1	...	...	...	...
K2-2"	42.72	51.00	230	300	GR 1.1	...	...	...	...
N5-2"	85.38	102.00	230	600	GR 1.1	...	...	...	...
N1-16"	42.72	51.00	230	300	GR 1.1	...	...	...	...
N6-6"	42.72	51.00	230	300	GR 1.1	...	...	...	...
N4-3"	42.72	51.00	230	300	GR 1.1	...	...	...	...
N3-20"	12.78	19.60	230	150	GR 1.1	...	...	...	...
K1A-2"	42.72	51.00	230	300	GR 1.1	...	...	...	...
Min Rating	10.884	19.600, [for Core Elements]					...	...	...

Selected Method for Derating ASME B16.5 Flange MAWP: ASME UG-44(b)

The UG-44(b) Method is based on the paper PVP 2013-97814. PV Elite uses the maximum loads from each load category to determine ME and FE. In many cases, the computed maximum allowable pressure will be greater than the flange rating. In these cases, the minimum of the rating from the table and the UG-44(b) method will be used. SA-193 B8 Cl. 2 bolts or ones with higher allowable stresses at the specified bolt size shall be used.

The axial force FE must put the nozzle in tension (-P for 107/297; +Fr for PD 5500 AnnexG) to be considered.

Nozzle Description	Gasket Dimension G mm.	Fm Value	Resolved Moment ME Kg-m.	Axial Force FE Kgf	MAWP PD bars	Maximum Pressure Pmax bars	Pass Fail
D1-3"	111.125	0.50	129	612	7.153	10.884	Pass

Evaluating Flange on Nozzle [D1-3"]:

$$= 16 \cdot ME + 4 \cdot FE \cdot G \leq \pi \cdot G^3 \cdot [(PR - PD) + FM \cdot PR]$$

$$= 16(129) + 4 \cdot \text{abs}(612) \cdot 111.125 \leq \pi \cdot 111.125^3 \cdot [(42.720 - 7.153) + 0.500 \cdot 42.720]$$

$$= 0.23387E+04 \leq 0.25027E+04 \text{ Check Passes}$$

Maximum Pressure for Flange on Nozzle D1-3" [Pmax]:

$$= \min(PR, FM \cdot PR + PR - (16 \cdot ME + 4 \cdot FE \cdot G) / (\pi \cdot G^3))$$

$$= \min(42.720, 0.500 \cdot 42.720 + 42.720 - 2338.7 / (\pi \cdot 111.125^3))$$

$$= \min(42.720, 10.884)$$

$$= 10.884 \text{ bars}$$

Flange Pressure Ratings are per ASME B16.5 2025 Metric Edition

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The Natural Frequencies for the vessel have been computed iteratively by solving a system of matrices. These matrices describe the mass and the stiffness of the vessel. This is the generalized eigenvalue/eigenvector problem and is referenced in some mathematical texts.

The Natural Frequency for the Vessel (Empty.) is 7.93252 Hz.

The Natural Frequency for the Vessel (Ope...) is 7.25829 Hz.

The Natural Frequency for the Vessel (Filled) is 5.93868 Hz.

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Wind Load Calculation Results:

Input Values:

Wind Design Code	ASCE/SEI 7-16
Wind Load Reduction Scale Factor	1.100
Basic Wind Speed	[V] 97.2 Km/hr
Surface Roughness Category	C: Open Terrain
Type of Surface	Moderately Smooth
Base Elevation	300 mm.
Percent Wind for Hydrotest	50.0
Using User defined Wind Press. Vs Elev.	N
Height of Hill or Escarpment H or Hh	0 mm.
Distance Upwind of Crest Lh	0 mm.
Distance from Crest to the Vessel x	0 mm.
Type of Terrain (Hill, Escarpment)	Flat
Damping Factor (Beta) for Wind (Ope)	0.0100
Damping Factor (Beta) for Wind (Empty)	0.0000
Damping Factor (Beta) for Wind (Filled)	0.0000
Site Elevation above Sea Level	0 mm.

Wind Analysis Results

The importance factor is not used in the calculation of wind pressure vs. elevation for the selected wind code.

Static Gust-Effect Factor, Operating Case [G]:
 $= \min(0.85, 0.925((1 + 1.7 \cdot gQ \cdot I_{zbar} \cdot Q) / (1 + 1.7 \cdot gV \cdot I_{zbar})))$
 $= \min(0.85, 0.925((1 + 1.7 \cdot 3.4 \cdot 0.226 \cdot 0.943) / (1 + 1.7 \cdot 3.4 \cdot 0.226)))$
 $= \min(0.85, 0.895)$
 $= 0.850$

Natural Frequency of Vessel (Operating)	7.258 Hz
Natural Frequency of Vessel (Empty)	7.933 Hz
Natural Frequency of Vessel (Test)	5.939 Hz

User Entered Importance Factor is	1.000
Force Coefficient	[Cf] 0.556
Structure Height to Diameter ratio	4.355
Height to top of Structure	8667.000 mm.

This is classified as a rigid structure. Static analysis performed.

Sample Calculation for the First Element:

The ASCE code performs calculations in Imperial Units. The wind pressure is therefore computed in these units.

Value of $[\alpha]$ and $[Zg]$:
 Exposure Category: C from Table 26.11-1
 $\alpha = 9.5$; $Zg = 274320.0$ mm.

Effective Height [z]:
 $= \text{Centroid Height} + \text{Vessel Base Elevation}$
 $= 1000.0 + 300.0 = 1300.0$ mm.
 $= 4.265$ ft. Imperial Units

Velocity Pressure coefficient evaluated at height z [Kz]:
 Because z (4.265 ft.) < 15 ft.
 $= 2.01(15 / Zg)^{2/\alpha}$
 $= 2.01(15/900.0)^{2/9.5}$
 $= 0.849$

Type of Hill: No Hill

Wind Directionality Factor [Kd]:
 $= 1$, per Table 26.6-1



As there is No Hill Present: [Kzt]:

$$K1 = 0, K2 = 0, K3 = 0$$

Topographical Factor [Kzt]:

$$= (1 + K1 \cdot K2 \cdot K3)^2$$

$$= (1 + 0.0 \cdot 0.0 \cdot 0.0)^2$$

$$= 1.0$$

Velocity Pressure evaluated at height z, Imperial Units [qz]:

$$= \max(16, 0.00256 \cdot Kz \cdot Kzt \cdot Kd \cdot Ke \cdot V(\text{mph})^2)$$

$$= \max(16, 0.00256 \cdot 0.849 \cdot 1.0 \cdot 1.0 \cdot 1.0 \cdot 60.399^2)$$

$$= 16.0 \text{ psf } [78.12] \text{ Kgs/m}^2$$

Force on the first element [F]:

$$= qz \cdot G \cdot Cf \cdot \text{WindArea}$$

$$= 16.0 \cdot 0.85 \cdot 0.556 \cdot 58.907$$

$$= 445.4 \text{ lbs. } [202.0] \text{ Kg}$$

Element	Hgt (z) mm.	K1	K2	K3	Kz	Kzt	qz Kgs/m ²
Skirt-2	1300.0	0.000	0.000	0.000	0.849	1.000	78.120
Skirt-1	2798.9	0.000	0.000	0.000	0.849	1.000	78.120
Bottom Dish	3325.0	0.000	0.000	0.000	0.849	1.000	78.120
shell	4850.0	0.000	0.000	0.000	0.859	1.000	78.120
shell	7350.0	0.000	0.000	0.000	0.938	1.000	78.120
Top dish	8613.8	0.000	0.000	0.000	0.970	1.000	78.120

Wind Vibration Calculations

This evaluation is based on work by Kanti Mahajan and Ed Zorilla

Nomenclature

- Cf - Correction factor for natural frequency
- D - Average outer insulated diameter of vessel, mm.
- Df - Damping Factor < 0.75 Unstable, > 0.95 Stable
- Dr - Average outer insulated of top half of vessel, mm.
- f - Natural frequency of vibration (Hertz)
- f1 - Natural frequency of bare vessel based on a unit value of $(D/L^2)(104)$
- L - Total height of structure mm.
- Lc - Total length of conical section(s) of vessel, mm.
- tb - Uncorroded plate thickness at bottom of vessel, mm.
- V30 - Design Wind Speed provided by user, Km/hr
- Vc - Critical wind velocity, Km/hr
- Vw - Maximum wind speed at top of structure, Km/hr
- W - Total corroded weight of structure, Kg
- Ws - Cor. vessel weight excl. weight of parts which do not effect stiff. Kg
- Z - Maximum amplitude of vibration at top of vessel, mm.
- Dl - Logarithmic decrement (taken as 0.03 for Welded Structures)
- Vp - Vib. Chance, <= 0.320E-06 (High); 0.320E-06 < 0.400E-06 (Probable)
- P30 - wind pressure 30 feet above the base

Check other Conditions and Basic Assumptions:

:: Total Cone Length / Total Length < 0.5

:: 0.0/8100.0 = 0.0

:: $[D / L^2] \cdot 10^4 < 8$ (English Units, ft.)

:: $[6.65/26.57^2] \cdot 10^4 = 94.189$ [Geometry Violation]

Compute the vibration possibility. If $Vp > 0.400E-06$ no chance. [Vp]:

$$= W / (L \cdot Dr^2)$$

$$= 21540 / (8100.0 \cdot 2032.642^2)$$

$$= 0.64365E-06$$

Since Vp is > 0.400E-06 no further vibration analysis is required !


Platform Load Calculations

ID	Wind Area cm ²	Elevation mm.	Pressure Kgs/m ²	Force Kgf	Cf
PLAT:[1 OF 1]	42192.00	3300.00	78.12	329.60	1.00
PLAT:[1 OF 1]	62932.50	9050.00	78.12	491.63	1.00

Wind Loads on Masses/Equipment/Piping

ID	Wind Area cm ²	Elevation mm.	Pressure Kgs/m ²	Force Kgf
clip weight	0.00	5850.00	78.12	0.00
davit	0.00	5150.00	78.12	0.00
Exchanger	18240.00	8850.00	78.12	142.49

The Natural Frequency for the Vessel (Ope...) is 7.25829 Hz.

Wind Load Calculation:

From	To	Wind Height mm.	Wind Diameter mm.	Wind Area cm ²	Wind Pressure Kgs/m ²	Element Wind Load Kgf
10	20	1300	2736.34	54726.8	78.12	222.223
20	30	2798.91	2744.15	27441.5	78.12	111.429
30	40	3325	2950.76	1475.38	78.12	365.83
40	50	4850	2959.28	88778.4	78.12	360.493
50	60	7350	2964.96	59299.2	78.12	240.79
60	70	8613.83	2959.28	14657.9	78.12	788.734

The Wind Loads calculated and printed in the Wind Load calculation report have been factored by the input scalar/load reduction factor of: 1.100.

Be sure the wind speed is in accordance with the specified wind design code.

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**Earthquake Load Calculation Results:****Earthquake Load Calculation:****Input Values:**

Seismic Design Code	ASCE/SEI 7-16
Seismic Load Reduction Scale Factor	1.000
Importance Factor	1.250
Table Value Fa	1.600
Table Value Fv	2.400
Max. Mapped Res. Acceleration [Ss]	0.147
Max. Eff. Ground Acceleration [S]	0.065
Force Modification Factor R	1.250
Site Class	D
Component Elevation Ratio z/h	0.100
Amplification Factor Ap	2.500
Force Factor	0.000
Consider Vertical Acceleration	Yes
Minimum Acceleration Multiplier	0.000
User Value of Sds (used if > 0)	0.000
User Value of Sd1 (used if > 0)	0.000
Moment Reduction Factor Tau	1.000

Seismic Analysis Results:

$$\begin{aligned}
 SMS &= Fa * Ss = 1.6 * 0.147 = 0.235 \\
 SM1 &= Fv * S1 = 2.4 * 0.065 = 0.156 \\
 SDS &= 2/3 * Sms = 2/3 * 0.235 = 0.157 \\
 SD1 &= 2/3 * Sm1 = 2/3 * 0.156 = 0.104
 \end{aligned}$$

Check Approximate Fundamental Period from 12.8-7 [Ta]:
 $= Ct \cdot hn^x$ where $Ct = 0.020$, $x = 0.75$ and hn = Structural Height (ft.)
 $= 0.020(28.13320.75)$
 $= 0.244$ seconds

The Coefficient Cu from Table 12.8-1 is : 1.700

Fundamental Period (1/Frequency) [T]:
 $= (1/\text{Natural Frequency}) = (1/7.258)$
 $= 0.138$

Check the Value of T which is the smaller of $Cu \cdot Ta$ and T:
 $= \text{Minimum Value of } (1.7 \cdot 0.244, 0.138)$ per 12.8.2
 $= 0.138$

Loads on Building/Structure Supported Equipment:

$$\begin{aligned}
 &\text{Seismic Force per equation 13.3-1, [Fp]:} \\
 &= 0.4 \cdot Ap \cdot Sds \cdot W(1 + 2 \cdot (z/h)) / (R / Ie) \\
 &= 0.4 \cdot 2.5 \cdot 0.157 \cdot 22891(1 + 2 \cdot 0.1) / (1.25/1.25) \\
 &= 4307.084 \text{ Kgf}
 \end{aligned}$$

Maximum value of Fp per equation 13.3-2:
 $= 1.6 \cdot Sds \cdot I \cdot W$
 $= 1.6 \cdot 0.157 \cdot 1.25 \cdot 22891 = 7178.47 \text{ Kgf}$

Minimum value of Fp per equation 13.3-3:
 $= 0.3 \cdot 0.16 \cdot 1.25 \cdot 22891$
 $= 1345.964 \text{ Kgf}$

Total Base Shear $V = Fp$, [V]:
 $= 4307.084 \text{ Kgf}$

Vertical load per 12.4-4, [Ev]:
 $= 0.2 \cdot Sds \cdot W$
 $= 0.2 \cdot 0.157 \cdot 22891$
 $= 717.8 \text{ Kgf}$



Distribute the Base shear force to each element according to the equations

$F_x = C_{vx} * V$ (eqn. 12.8-11) and the vertical distribution factor

$C_{vx} = W_x * h_x^k / (\text{Sum of } W_i * h_i^k)$ and k is an exponent which is related to the period of Vibration.

In this case, the value of k was 1.0

The Natural Frequency for the Vessel (Ope...) is 7.25829 Hz.

Earthquake Load Calculation:

From	To	Earthquake Height mm.	Earthquake Weight Kgf	Element Ope Load Kgf	Element Emp Load Kgf	Element Vertical Load Kgf
10	20	1000	1549.02	56.6941	56.7881	717.847
20	30	2500	500.914	45.8337	45.9097	669.27
30	40	3025	3502.07	387.732	70.8596	653.562
40	50	4550	9512.77	1584.16	479.079	543.737
50	60	7050	2138.07	551.687	460.773	245.416
60	70	8075	5687.7	1680.97	736.517	178.366



The following table is for the Operating Case.

Wind/Earthquake Shear, Bending:

From	To	Distance to Support mm.	Cumulative Wind Shear Kgf	Earthquake Shear Kgf	Wind Bending Kg-m.	Earthquake Bending Kg-m.
10	20	1000	2089.5	4307.08	11502.9	26417.3
20	30	2500	1867.28	4250.39	7546.02	17859.7
30	40	3025	1755.85	4204.56	5734.42	13632.1
40	50	4550	1390.02	3816.82	5655.77	13431.6
50	60	7050	1029.52	2232.66	2026.39	4357.2
60	70	8075	788.734	1680.97	208.093	443.495

Note:
 The Wind Shears/Moments and the Earthquake Shears/Moments calculated and printed in the Wind/Earthquake Shear and Bending report have been factored by the input Scalar/Load reductions factors of;
 Wind: 1.100; Earthquake: 1.000.

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Wind Deflection Calculations:

The following table is for Wind Deflection; Operating Case.

Wind Load Deflection:

From	To	Cumulative Shear Kg	Centroid Deflection mm.	Elem. End Deflection mm.	Elem. Ang. Rotation
10	20	2089.5	0.011813	0.044329	0.41448E-04
20	30	1867.28	0.067012	0.093284	0.55809E-04
30	40	1755.85	0.094685	0.096097	0.56699E-04
40	50	1390.02	0.19277	0.30685	0.80041E-04
50	60	1029.52	0.38833	0.47161	0.83685E-04
60	70	788.734	0.4737	0.47579	0.83686E-04

Critical Wind Velocity for Tower Vibration:

From	To	1st Crit. Wind Speed Km/hr	2nd Crit. Wind Speed Km/hr
10	20	356.537	2228.36
20	30	357.555	2234.72
30	40	384.476	2402.97
40	50	385.586	2409.91
50	60	386.326	2414.54
60	70	385.586	2409.91

Allowable deflection at the Tower Top (Ope)(6.000"/100ft. Criteria)
 Allowable deflection : 40.500 Actual deflection : 0.476 mm.

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**Longitudinal Stress Constants:**

From	To	Metal Area New cm ²	Metal Area Corroded cm ²	Section Modulus New mm. ³	Section Modulus Corroded mm. ³
10	20	602.24	422.227	28713390	20193570
20	30	607.896	426.186	29256574	20574702
30	40	539.76	360.406	25639728	17173630
40	50	720.811	541.457	34241168	25801468
50	60	841.822	662.468	39990872	31568686
60	70	720.811	541.457	34241168	25801468

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**Longitudinal Allowable Stresses:**

From	To	Tensile N./mm ²	Hydrotest Tensile N./mm ²	Compressive N./mm ²	Hydrotest Compressive N./mm ²
10	20	115.918	196.556	-85.563	-98.7391
20	30	82.7985	140.397	-85.3437	-98.3827
30	40	140.757	238.676	-81.9938	-92.8599
40	50	140.757	238.676	-91.3874	-107.96
50	60	140.757	238.676	-95.7729	-114.563
60	70	140.757	238.676	-91.3874	-107.96

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Longitudinal Stress Report



Longitudinal Operating and Empty Stresses are computed in the corroded condition. Stresses due to loads in the hydrostatic test cases have also been computed in the corroded condition.

Longitudinal Pressure Stresses due to:

From	To	Longitudinal Stress Internal Pressure N./mm ²	Longitudinal Stress External Pressure N./mm ²	Longitudinal Stress Hydrotest Pressure N./mm ²
10	20	...	...	...
20	30	...	...	...
30	40	27.7274	-8.29143	36.0457
40	50	18.4616	-5.55354	24
50	60	15.0922	-4.55799	19.6199
60	70	18.4616	-5.55354	24

Longitudinal Stresses due to Weight Loads for these Conditions:

From	To	Wght. Str. Empty N./mm ²	Wght. Str. Operating N./mm ²	Wght. Str. Hydrotest N./mm ²	Wght. Str. Emp. Mom. N./mm ²	Wght. Str. Opr. Mom. N./mm ²
10	20	-3.66738	-5.31665	-7.44136	1.42902	1.42902
20	30	-3.27688	-4.91083	-7.0158	1.40255	1.40255
30	40	-3.73867	-3.73867	-3.73867	1.68031	1.68031
40	50	-2.03392	-2.03392	-2.03392	0.34507	0.34507
50	60	-1.15849	-1.15849	-1.15849	0.11764	0.11764
60	70	-1.03015	-1.03015	-1.03015	0.0025137	0.0025137

Longitudinal Stresses due to Weight Loads and Bending for these Conditions:

From	To	Wght. Str. Hyd. Mom. N./mm ²	Bend. Str. Oper. Wind N./mm ²	Bend. Str. Oper. Equ. N./mm ²	Bend. Str. Hyd. Wind N./mm ²	Bend. Str. Hyd. Equ. N./mm ²
10	20	1.42902	5.58616	12.8291	2.79308	...
20	30	1.40255	3.5967	8.51257	1.79835	...
30	40	1.68031	3.27452	7.78435	1.63726	...
40	50	0.34507	2.14965	5.10509	1.07482	...
50	60	0.11764	0.62949	1.35354	0.31474	...
60	70	0.0025137	0.079092	0.16856	0.039546	...

Longitudinal Stresses due to these Conditions:

From	To	Vortex Shedding Operating Case N./mm ²	Vortex Shedding Empty Case N./mm ²	Vortex Shedding Test Case N./mm ²	Earthquake Empty Case N./mm ²
10	20	...	...	...	5.79732
20	30	...	...	...	3.95348
30	40	...	...	...	3.72558
40	50	...	...	...	2.44724
50	60	...	...	...	0.6611
60	70	...	...	...	0.073856

Longitudinal Stresses due to Applied Axial Forces:

From	To	Longitudinal Stress Y Forces Wind	Longitudinal Stress Y Forces Seismic
------	----	--------------------------------------	---



		N./mm ²	N./mm ²
10	20	...	0.16673
20	30	...	0.154
30	40	...	0.17784
40	50	...	0.098481
50	60	...	0.03633
60	70	...	0.032306

Longitudinal Stresses due to User Forces and Moments:

From	To	Wind For/Mom Corroded N./mm ²	Earthquake For/Mom Corroded N./mm ²	Wind For/Mom No Corrosion N./mm ²	Earthquake For/Mom No Corrosion N./mm ²
10	20	...	...	...	...
20	30	...	...	...	...
30	40	...	...	...	...
40	50	...	...	...	...
50	60	...	...	...	...
60	70	...	...	...	...

Bending Stresses due to Blast Load:

From	To	Bending Stress N./mm ²
10	20	...
20	30	...
30	40	...
40	50	...
50	60	...
60	70	...

**Stress Combination Load Cases for Vertical Vessels:****Load Case Definition Key**

IP = Longitudinal Stress due to Internal Pressure
 EP = Longitudinal Stress due to External Pressure
 HP = Longitudinal Stress due to Hydrotest Pressure
 NP = No Pressure
 EW = Longitudinal Stress due to Weight (No Liquid)
 OW = Longitudinal Stress due to Weight (Operating)
 HW = Longitudinal Stress due to Weight (Hydrotest)
 WI = Bending Stress due to Wind Moment (Operating)
 EQ = Bending Stress due to Earthquake Moment (Operating)
 EE = Bending Stress due to Earthquake Moment (Empty)
 HI = Bending Stress due to Wind Moment (Hydrotest)
 HE = Bending Stress due to Earthquake Moment (Hydrotest)
 WE = Bending Stress due to Wind Moment (Empty) (no CA)
 WF = Bending Stress due to Wind Moment (Filled) (no CA)
 CW = Longitudinal Stress due to Weight (Empty) (no CA)
 VO = Bending Stress due to Vortex Shedding Loads (Ope)
 VE = Bending Stress due to Vortex Shedding Loads (Emp)
 VF = Bending Stress due to Vortex Shedding Loads (Test No CA.)
 FW = Axial Stress due to Vertical Forces for the Wind Case
 FS = Axial Stress due to Vertical Forces for the Seismic Case
 BW = Bending Stress due to Lat. Forces for the Wind Case, Corroded
 BS = Bending Stress due to Lat. Forces for the Seismic Case, Corroded
 BN = Bending Stress due to Lat. Forces for the Wind Case, UnCorroded
 BU = Bending Stress due to Lat. Forces for the Seismic Case, UnCorroded
 BL = Bending Stress due to Blast Loads, Corroded

General Notes:

Case types HI and HE are in the Corroded condition.

Case types WE, WF, and CW are in the Un-Corroded condition.

A blank stress and stress ratio indicates that the corresponding stress comprising those components that did not contribute to that type of stress.

An asterisk (*) in the final column denotes overstress.

Analysis of Load Case 1 : NP+EW+WI+FW+BW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10	3.35	115.92	-10.68	85.56	0.0289	0.1249
20	1.72	82.80	-8.28	85.34	0.0208	0.0970
30	1.22	140.76	-8.69	81.99	0.0086	0.1060
40	0.46	140.76	-4.53	91.39	0.0033	0.0496
50		140.76	-1.91	95.77		0.0199
60		140.76	-1.11	91.39		0.0122

Analysis of Load Case 2 : NP+EW+EE+FS+BS

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10	3.73	115.92	-10.73	85.56	0.0321	0.1254
20	2.23	82.80	-8.48	85.34	0.0270	0.0994
30	1.85	140.76	-8.97	81.99	0.0131	0.1094
40	0.86	140.76	-4.73	91.39	0.0061	0.0517
50		140.76	-1.90	95.77		0.0198
60		140.76	-1.07	91.39		0.0118

Analysis of Load Case 3 : NP+OW+WI+FW+BW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10	1.70	115.92	-12.33	85.56	0.0147	0.1441



20	0.09	82.80	-9.91	85.34	0.0011	0.1161
30	1.22	140.76	-8.69	81.99	0.0086	0.1060
40	0.46	140.76	-4.53	91.39	0.0033	0.0496
50		140.76	-1.91	95.77		0.0199
60		140.76	-1.11	91.39		0.0122

Analysis of Load Case 4 : NP+OW+EQ+FS+BS

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10	9.11	115.92	-19.41	85.56	0.0786	0.2268
20	5.16	82.80	-14.67	85.34	0.0623	0.1719
30	5.90	140.76	-13.03	81.99	0.0419	0.1589
40	3.51	140.76	-7.39	91.39	0.0250	0.0808
50	0.35	140.76	-2.59	95.77	0.0025	0.0271
60		140.76	-1.17	91.39		0.0128

Analysis of Load Case 5 : NP+HW+HI

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10		196.56	-11.66	98.74		0.1181
20		140.40	-10.22	98.38		0.1038
30		238.68	-7.06	92.86		0.0760
40		238.68	-3.45	107.96		0.0320
50		238.68	-1.59	114.56		0.0139
60		238.68	-1.07	107.96		0.0099

Analysis of Load Case 6 : NP+HW+HE

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10		196.56	-8.87	98.74		0.0898
20		140.40	-8.42	98.38		0.0856
30		238.68	-5.42	92.86		0.0584
40		238.68	-2.38	107.96		0.0220
50		238.68	-1.28	114.56		0.0111
60		238.68	-1.03	107.96		0.0096

Analysis of Load Case 7 : IP+OW+WI+FW+BW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10	1.70	115.92	-12.33	85.56	0.0147	0.1441
20	0.09	82.80	-9.91	85.65	0.0011	0.1157
30	28.94	140.76		81.99	0.2056	
40	18.92	140.76		91.39	0.1344	
50	14.68	140.76		95.77	0.1043	
60	17.51	140.76		91.39	0.1244	

Analysis of Load Case 8 : IP+OW+EQ+FS+BS

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10	9.11	115.92	-19.41	85.56	0.0786	0.2268
20	5.16	82.80	-14.67	85.65	0.0623	0.1713
30	33.63	140.76		81.99	0.2389	
40	21.98	140.76		91.39	0.1561	
50	15.44	140.76		95.77	0.1097	
60	17.63	140.76		91.39	0.1253	

Analysis of Load Case 9 : EP+OW+WI+FW+BW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10	1.70	115.92	-12.33	90.76	0.0147	0.1359
20	0.09	82.80	-9.91	90.87	0.0011	0.1091
30		140.76	-16.98	86.28		0.1969
40		140.76	-10.08	97.93		0.1030



50	140.76	-6.46	103.19	0.0626
60	140.76	-6.67	97.93	0.0681

Analysis of Load Case 10 : EP+OW+EQ+FS+BS

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10	9.11	115.92	-19.41	90.76	0.0786	0.2138
20	5.16	82.80	-14.67	90.87	0.0623	0.1615
30		140.76	-21.32	86.28		0.2471
40		140.76	-12.94	97.93		0.1321
50		140.76	-7.15	103.19		0.0693
60		140.76	-6.72	97.93		0.0686

Analysis of Load Case 11 : HP+HW+HI

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10		196.56	-11.66	98.74		0.1181
20		140.40	-10.22	98.38		0.1038
30	35.62	238.68		92.86	0.1493	
40	23.39	238.68		107.96	0.0980	
50	18.89	238.68		114.56	0.0792	
60	23.01	238.68		107.96	0.0964	

Analysis of Load Case 12 : HP+HW+HE

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10		196.56	-8.87	98.74		0.0898
20		140.40	-8.42	98.38		0.0856
30	33.99	238.68		92.86	0.1424	
40	22.31	238.68		107.96	0.0935	
50	18.58	238.68		114.56	0.0778	
60	22.97	238.68		107.96	0.0962	

Analysis of Load Case 13 : IP+WE+EW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10		115.92	-5.10	85.56		0.0596
20		82.80	-4.68	85.65		0.0546
30	25.67	140.76		81.99	0.1824	
40	16.77	140.76		91.39	0.1192	
50	14.05	140.76		95.77	0.0998	
60	17.43	140.76		91.39	0.1239	

Analysis of Load Case 14 : IP+WF+CW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10		115.92	-3.73	85.56		0.0436
20		82.80	-3.44	85.65		0.0402
30	25.23	140.76		81.99	0.1793	
40	16.93	140.76		91.39	0.1203	
50	14.18	140.76		95.77	0.1007	
60	17.69	140.76		91.39	0.1257	

Analysis of Load Case 15 : IP+VO+OW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10		115.92	-6.75	85.56		0.0788
20		82.80	-6.31	85.65		0.0737
30	25.67	140.76		81.99	0.1824	
40	16.77	140.76		91.39	0.1192	
50	14.05	140.76		95.77	0.0998	
60	17.43	140.76		91.39	0.1239	



Analysis of Load Case 16 : IP+VE+EW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10		115.92	-5.10	85.56		0.0596
20		82.80	-4.68	85.65		0.0546
30	25.67	140.76		81.99	0.1824	
40	16.77	140.76		91.39	0.1192	
50	14.05	140.76		95.77	0.0998	
60	17.43	140.76		91.39	0.1239	

Analysis of Load Case 17 : NP+VO+OW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10		115.92	-6.75	85.56		0.0788
20		82.80	-6.31	85.34		0.0740
30		140.76	-5.42	81.99		0.0661
40		140.76	-2.38	91.39		0.0260
50		140.76	-1.28	95.77		0.0133
60		140.76	-1.03	91.39		0.0113

Analysis of Load Case 18 : FS+BS+IP+OW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10		96.60	-6.58	71.30		0.0923
20		69.00	-6.16	71.37		0.0863
30	25.85	117.30		68.33	0.2204	
40	16.87	117.30		76.16	0.1438	
50	14.09	117.30		79.81	0.1201	
60	17.47	117.30		76.16	0.1489	

Analysis of Load Case 19 : FS+BS+EP+OW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10		96.60	-6.58	75.64		0.0870
20		69.00	-6.16	75.72		0.0813
30		117.30	-13.53	71.90		0.1882
40		117.30	-7.83	81.61		0.0960
50		117.30	-5.80	85.99		0.0674
60		117.30	-6.55	81.61		0.0803

Analysis of Load Case 20 : BL+IP+OW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10		115.92	-6.75	85.56		0.0788
20		82.80	-6.31	85.65		0.0737
30	25.67	140.76		81.99	0.1824	
40	16.77	140.76		91.39	0.1192	
50	14.05	140.76		95.77	0.0998	
60	17.43	140.76		91.39	0.1239	

Absolute Maximum of the all of the Stress Ratio's 0.2471

Governing Element: Bottom Dish

Governing Load Case 10 : EP+OW+EQ+FS+BS

Element Description	Long Pressure Stress	Weight + Axial Load Stress	Eccentric Moment Stress	Wind or Earthquake Stress	Combined Stress	Which Side
Skirt-2	...	-5.1	1.4	12.8	9.1	+ Tens
Skirt-2	...	-5.1	-1.4	-12.8	-19.4	- Comp
Skirt-1	...	-4.8	1.4	8.5	5.2	+ Tens
Skirt-1	...	-4.8	-1.4	-8.5	-14.7	- Comp



Bottom Dish	-8.3	-3.6	1.7	7.8	...	+ Tens
Bottom Dish	-8.3	-3.6	-1.7	-7.8	-21.3	- Comp
shell	-5.6	-1.9	0.3	5.1	...	+ Tens
shell	-5.6	-1.9	-0.3	-5.1	-12.9	- Comp
shell	-4.6	-1.1	0.1	1.4	...	+ Tens
shell	-4.6	-1.1	-0.1	-1.4	-7.2	- Comp
Top dish	-5.6	-1.0	...	0.2	...	+ Tens
Top dish	-5.6	-1.0	...	-0.2	-6.7	- Comp

**Skirt Support Analysis Per EN 13445-3 Section 16.12:**

The analysis of this skirt has been conducted referencing the EN 13445-3 design code. It is important to note that the selected design code does not explicitly cover the analysis of this component. Therefore, the results provided herein should be regarded as informational only.

Item: Skirt-1

Mean skirt diameter at the critical height is computed later.

This is a Conical type Support Skirt , Wider at the Bottom

Half Apex Angle of the skirt a 0.716 deg

Total vessel (empty) weight Fg 13873.886 Kg

Head Weight below section 2-2 DeltaFg 2510.067 Kg

Weight of vessel contents (liquid) Ff 7100.831 Kg

Vessel Gross Weight Fg + Ff 20974.715 Kg

Skirt Mean Diameter Top End DzTop 1913.001 mm.

Skirt Mean Diameter Bottom End DzBot 1938.001 mm.

Skirt Analysis Thickness (Corroded) ez 7.000 mm.

Skirt Height H 1000.000 mm.

Additional Axial Force at section 1-1 F1 0.000 Kg

Moment at section 1-1 M1 13632.147 Kg-m.

Applied Moment - Bottom of Skirt Mb 17859.705 Kg-m.

Weight at Bottom of Skirt W 21341.523 Kg

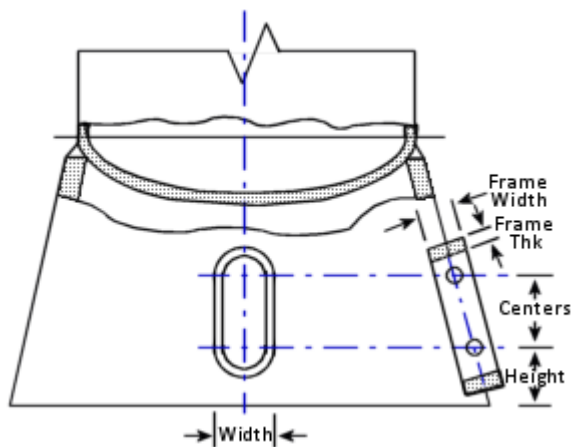
Structural Type B Skirt per EN 13445 Figure 16.12-2:

Type of Head: Elliptical

Head Mean Diameter at Skirt Top Db 1912.000 mm.

Head Corroded Thickness eb 6.000 mm.

Per EN 13445 Section 7.5.4 [R, r]:

 $h_i = 478.0 \text{ mm.}$ $K = D_i / (2 \cdot h_i) = 1906.0 / (2 \cdot 478.0) = 1.9937$ $R = D_i (0.44 \cdot K + 0.02) = 1906.0 \cdot (0.44 \cdot 1.994 + 0.02) = 1710.1367 \text{ mm.}$ $r = D_i ((0.5/K) - 0.08) = 1906.0 \cdot ((0.5/1.994) - 0.08) = 325.52 \text{ mm.}$ 

There are no Access Openings in the Skirt - Plain Skirt Considered

Moment of Inertia Ixx = 1924584.875 cm⁴Moment of Inertia Iyy = 1924584.875 cm⁴Section Modulus W = 20121124.000 mm.³Section Metal Area A = 420.724 cm²**Analysis of the Section at weakest section Modulus [Z min]:**



Radius of Gyration of Skirt at 4-4 [rg]:

$$= \sqrt{\frac{\min(I_{xx}, I_{yy})}{\text{Area}}}$$

$$= \sqrt{\frac{\min(1924585, 1924585)}{420.724}}$$

$$= 676.348 \text{ mm.}$$

Skirt Slenderness Ratio [rg/L]:

$$= r_g / L$$

$$= 676.348 / 1000.0$$

$$= 0.676$$

Correct the Moment [M4]:

top of skirt is the critical end
 Controlling Moment [M4] = 13632.147 Kg-m.

Axial Stress at Residual (Weakest) Surface Area [Sigma]:

Downwind Side:

$$= -M / Z - W / A$$

$$= -13632.147 / 80484496 - 21341.523 / 420.72379$$

$$= -6.636 \text{ N./mm}^2$$

Upwind Side:

$$= +(M) / Z - W / A$$

$$= +13632.147 / 80484496 - 21341.523 / 420.72379$$

$$= -3.314 \text{ N./mm}^2$$

16.12.3.4 Checking at connection areas:

Force at the joint, 16.12-1 [Fzp]:

$$= F_1 - F_g - F_f + 4 \cdot M_1 / D_z$$

$$= 0.0 - 13873.886 - 7100.831 + 4 \cdot 13632.147 / 1913.001$$

$$= 7528.9 \text{ Kg}$$

Force at the joint, 16.12-2 [Fzq]:

$$= F_1 - F_g - F_f - 4 \cdot M_1 / D_z$$

$$= 0.0 - 13873.886 - 7100.831 - 4 \cdot 13632.147 / 1913.001$$

$$= -49478.4 \text{ Kg}$$

16.12.3.4.1 Membrane stresses:

Membrane stress at 1-1, 16.12-3 & 4:

Stress [σ_{1pm}]:

$$= (F_{zp} + \Delta F_g) / (\pi \cdot D_b \cdot e_b) + P \cdot D_b / (4 \cdot e_b)$$

$$= (7528.934 + 2510.067 + 7100.831) / (\pi \cdot 1912.0 \cdot 6.0) + 3.5 \cdot 1912.0 / (4 \cdot 6.0)$$

$$= 32.55 \text{ N./mm}^2$$

Stress [σ_{1qm}]:

$$= (F_{zq} + \Delta F_g + F_f) / (\pi \cdot D_b \cdot e_b) + P \cdot D_b / (4 \cdot e_b)$$

$$= (-49478.363 + 2510.067 + 7100.831) / (\pi \cdot 1912.0 \cdot 6.0) + 3.5 \cdot 1912.0 / (4 \cdot 6.0)$$

$$= 17.04 \text{ N./mm}^2$$

Membrane stress at 2-2, 16.12-9:

Stress [σ_{2pm} & σ_{2qm}]:

$$= (F_f + \Delta F_g) / (\pi \cdot D_b \cdot e_b) + P \cdot D_b / (4 \cdot e_b)$$

$$= (7100.831 + 2510.067) / (\pi \cdot 1912.0 \cdot 6.0) + 3.5 \cdot 1912.0 / (4 \cdot 6.0)$$

$$= 30.50 \text{ N./mm}^2$$

Membrane stress at 3-3, 16.12 12 & 13:

Stress [σ_{3pm}]:

$$= F_{zp} / (\pi \cdot D_z \cdot e_z)$$

$$= 7528.934 / (\pi \cdot 1913.001 \cdot 7.0)$$

$$= 1.76 \text{ N./mm}^2$$

Stress [σ_{3qm}]:

$$= F_{zq} / (\pi \cdot D_z \cdot e_z)$$

$$= -49478.363 / (\pi \cdot 1913.001 \cdot 7.0)$$



$$= -11.53 \text{ N./mm}^2$$

16.12.3.4.2 Bending stresses:

Cosine of angle, 16.12-25 [cos(γ)]:

$$= 1 - 0.5(D_b + e_b - D_z + e_z) / (r + e_b)$$

$$= 1 - 0.5(1912.0 + 6.0 - 1938.0 + 7.0) / (325.52 + 6.0)$$

$$= 0.9819$$

$$\gamma = \arccos(\cos(\gamma)) = \arccos(0.982) = 10.917 \text{ deg}$$

Dimension [a]:

$$= 0.5 \cdot \sqrt{(e_b^2 + e_z^2 + 2 \cdot e_b \cdot e_z \cdot \cos(\gamma))}$$

$$= 0.5 \cdot \sqrt{(6.0^2 + 7.0^2 + 2 \cdot 6.0 \cdot 7.0 \cdot 0.982)}$$

$$= 6.471 \text{ mm.}$$

Bending moment at verification points p and q [Mp & Mq]:

$$M_p = a \cdot F_{zp} = 6.471 \cdot 7528.9 = 48.72 \text{ Kg-m.}$$

$$M_q = a \cdot F_{zq} = 6.471 \cdot -49478.4 = -320.17 \text{ Kg-m.}$$

Correction factor [C]:

$$= 0.63 - 0.057(e_b/e_z)^2$$

$$= 0.63 - 0.057(6.0/7.0)^2$$

$$= 0.588$$

Bending stresses in sections 1-1 to 3-3 at the outer surface (o), 16.12-26 to 29:

Stress [$\sigma_{1pb(o)}$ & $\sigma_{2pb(o)}$]:

$$= C \cdot 6 \cdot M_p / (\pi \cdot D_b \cdot e_b^2)$$

$$= 0.588 \cdot 6 \cdot 48.718 / (\pi \cdot 1912.0 \cdot 6.0^2)$$

$$= 7.80 \text{ N./mm}^2$$

Stress [$\sigma_{1qb(o)}$ & $\sigma_{2qb(o)}$]:

$$= C \cdot 6 \cdot M_q / (\pi \cdot D_b \cdot e_b^2)$$

$$= 0.588 \cdot 6 \cdot -320.166 / (\pi \cdot 1912.0 \cdot 6.0^2)$$

$$= -51.24 \text{ N./mm}^2$$

Stress [$\sigma_{3pb(o)}$]:

$$= C \cdot 6 \cdot M_p / (\pi \cdot D_z \cdot e_z^2)$$

$$= 0.588 \cdot 6 \cdot 48.718 / (\pi \cdot 1913.001 \cdot 7.0^2)$$

$$= 5.72 \text{ N./mm}^2$$

Stress [$\sigma_{3qb(o)}$]:

$$= C \cdot 6 \cdot M_q / (\pi \cdot D_z \cdot e_z^2)$$

$$= 0.588 \cdot 6 \cdot -320.166 / (\pi \cdot 1913.001 \cdot 7.0^2)$$

$$= -37.62 \text{ N./mm}^2$$

Geometry variables, 16.12-32 to 36:

$$e_b/D_b = 0.236/75.276 = 0.0031$$

$$y = 125 \cdot e_b/D_b = 125 \cdot 6.0/1912.0 = 0.3923$$

As this an elliptical head, it will be treated a a Korbbogen type head

Geometric value [a]:

$$= 1.51861 \cdot 2.71828 - 4.2335 \cdot R_{YY} + 3.994$$

$$= 1.51861 \cdot 2.71828 - 4.2335 \cdot 0.392 + 3.994 = 4.2826$$

Bending stress in sections 1-1 and 2-2 Equation 16.12-31 [$\sigma_{1b(p)}$ & $\sigma_{2b(p)}$]:

$$= (P + P_h) \cdot D_b \cdot (\gamma / \gamma_A) \cdot (\alpha - 1) / (4 \cdot e_b)$$

$$= (3.5 + 0.0) \cdot 1912.0 \cdot (10.92/40.0) \cdot (4.283 - 1) / (4 \cdot 6.0)$$

$$= 4.71 \text{ N./mm}^2$$

16.12.3.4.3 Total stresses and strength conditions:

Structural shape B and C:

Stress [σ_{1pitot}]:

$$= \sigma_{1pm} - \sigma_{1pb(o)} + \sigma_{1b(p)}$$

$$= 32.549 - 7.796 + 4.707$$



$$= 29.46 \text{ N./mm}^2$$

Stress [σ_{1potot}]:

$$\begin{aligned} &= \sigma_{1pm} + \sigma_{1pb(o)} - \sigma_{1b(p)} \\ &= 32.549 + 7.796 - 4.707 \\ &= 35.64 \text{ N./mm}^2 \end{aligned}$$

Stress [σ_{1qitot}]:

$$\begin{aligned} &= \sigma_{1qm} - \sigma_{1qb(o)} + \sigma_{1b(p)} \\ &= 17.037 - 51.236 + 4.707 \\ &= 72.98 \text{ N./mm}^2 \end{aligned}$$

Stress [σ_{1qotot}]:

$$\begin{aligned} &= \sigma_{1qm} + \sigma_{1qb(o)} - \sigma_{1b(p)} \\ &= 17.037 + 51.236 - 4.707 \\ &= -38.91 \text{ N./mm}^2 \end{aligned}$$

Stress [σ_{2pitot}]:

$$\begin{aligned} &= \sigma_{2pm} + \sigma_{2pb(o)} + \sigma_{2b(p)} \\ &= 30.5 + 7.796 + 4.707 \\ &= 43.00 \text{ N./mm}^2 \end{aligned}$$

Stress [σ_{2potot}]:

$$\begin{aligned} &= \sigma_{2pm} - \sigma_{2pb(o)} - \sigma_{2b(p)} \\ &= 30.5 - 7.796 - 4.707 \\ &= 18.00 \text{ N./mm}^2 \end{aligned}$$

Stress [σ_{2qitot}]:

$$\begin{aligned} &= \sigma_{2qm} + \sigma_{2qb(o)} + \sigma_{2b(p)} \\ &= 30.5 + 51.236 + 4.707 \\ &= -16.03 \text{ N./mm}^2 \end{aligned}$$

Stress [σ_{2qotot}]:

$$\begin{aligned} &= \sigma_{2pm} - \sigma_{2pb(o)} - \sigma_{2b(p)} \\ &= 30.5 - 51.236 - 4.707 \\ &= 77.03 \text{ N./mm}^2 \end{aligned}$$

Stress [σ_{3pitot}]:

$$\begin{aligned} &= \sigma_{3pm} - \sigma_{3pb} \\ &= 1.755 - 5.725 \\ &= -3.97 \text{ N./mm}^2 \end{aligned}$$

Stress [σ_{3potot}]:

$$\begin{aligned} &= \sigma_{3pm} + \sigma_{3pb} \\ &= 1.755 + 5.725 \\ &= 7.48 \text{ N./mm}^2 \end{aligned}$$

Stress [σ_{3qitot}]:

$$\begin{aligned} &= \sigma_{3qm} - \sigma_{3qb} \\ &= -11.534 - 37.623 \\ &= 26.09 \text{ N./mm}^2 \end{aligned}$$

Stress [σ_{3qotot}]:

$$\begin{aligned} &= \sigma_{3qm} + \sigma_{3qb} \\ &= -11.534 + 37.623 \\ &= -49.16 \text{ N./mm}^2 \end{aligned}$$

Allowable stresses:

For σ_{1pitot} and σ_{1potot} , the stress must not exceed:

$$\begin{aligned} &= fs(3 - (1/1.5)) \cdot (\sigma_{1pm}/f)^2 \\ &= 137.997(3 - (1/1.5)) \cdot (32.549/137.997)^2 \\ &= 408.87 \text{ N./mm}^2 \end{aligned}$$

For σ_{1qitot} and σ_{1qotot} , the stress must not exceed:

$$\begin{aligned} &= fs(3 - (1/1.5)) \cdot (\sigma_{1qm}/f)^2 \\ &= 137.997(3 - (1/1.5)) \cdot (17.037/137.997)^2 \\ &= 412.59 \text{ N./mm}^2 \end{aligned}$$



For σ_{2pitot} and σ_{2potot} , the stress must not exceed:

$$= fs(3 - (1 / 1.5) \cdot (\sigma_{2pm} / f)^2)$$

$$= 137.997(3 - (1/1.5) \cdot (30.5/137.997)^2)$$

$$= 409.50 \text{ N./mm}^2$$

For σ_{2qitot} and σ_{2qotot} , the stress must not exceed:

$$= fs(3 - (1 / 1.5) \cdot (\sigma_{2qm} / f)^2)$$

$$= 137.997(3 - (1/1.5) \cdot (30.5/137.997)^2)$$

$$= 409.50 \text{ N./mm}^2$$

For σ_{3pitot} & σ_{3potot} , the stress must not exceed:

$$= fs(3 - (1 / 1.5) \cdot (\sigma_{3pm} / f)^2)$$

$$= 137.997(3 - (1/1.5) \cdot (1.755/137.997)^2)$$

$$= 413.98 \text{ N./mm}^2$$

For σ_{3qitot} & σ_{3qotot} , the stress must not exceed:

$$= fs(3 - (1 / 1.5) \cdot (\sigma_{3qm} / f)^2)$$

$$= 137.997(3 - (1/1.5) \cdot (-11.534/137.997)^2)$$

$$= 413.35 \text{ N./mm}^2$$

Please note: None of the following stresses in the table below are in the region where there is an access opening.

Summary of Stresses: (N./mm2)

Stress	Actual	Allowed	Result	Equation
Stot1pi	29.46	408.87	Pass	46
Stot1po	35.64	408.87	Pass	47
Stot1qi	72.98	412.59	Pass	48
Stot1qo	-38.91	412.59	Pass	49
Stot2pi	43.00	409.50	Pass	50
Stot2po	18.00	409.50	Pass	51
Stot2qi	-16.03	409.50	Pass	52
Stot2qo	77.03	409.50	Pass	53
Stot3pi	-3.97	413.98	Pass	54
Stot3po	7.48	413.98	Pass	55
Stot3qi	26.09	413.35	Pass	56
Stot3qo	-49.16	413.35	Pass	57

The stresses are satisfactory in the region without openings.

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**Shop/Field Installation Options :**

Platform(s) installed in the Field before being lifted.
 Insulation is installed in the Field before being lifted.

Note : The CG is computed from the first Element From Node

Center of Gravity of the Platforms	6696.938 mm.
Center of Gravity of the Liquid	3979.417 mm.
Center of Gravity of the Insulation	5500.844 mm.
Center of Gravity of the Stiffening Rings	5250.000 mm.
Center of Gravity of the Nozzles	5439.390 mm.
Center of Gravity of the Added Weights (Operating)	7857.526 mm.
Center of Gravity of the Added Weights (Empty)	7857.526 mm.
Center of Gravity of Bare Shell New and Cold	4260.592 mm.
Center of Gravity of Bare Shell Corroded	4229.123 mm.
Vessel CG in the Operating Condition	5222.435 mm.
Vessel CG in the Fabricated (Shop/Empty) Condition	5260.142 mm.
Vessel CG in the Test Condition	5440.692 mm.

Rigging Analysis Results:

- Lifting weight based on the user selected calculated erected weight

Total Effective Length of Vessel for this analysis	8100.00 mm.
Total vessel weight (No Liquid) Twt	15789.71 Kg
Impact weight multiplication factor Imp	1.50
Design lifting weight, DWT = Imp * Twt	23684.57 Kg
Elevation of the Tailing Lug (bottom)	150.00 mm.
Elevation of the Lifting Lug (top)	7650.00 mm.
Design Reaction force at the tailing lug	7547.03 Kg
Design Reaction force at the lifting lug	16137.54 Kg
CG distance during field lifting operation	5260.14 mm.
CG distance from tailing lug	5110.14 mm.
CG distance from the nearer lifting lug/trunnion	2389.86 mm.

Compressive Allowable Increase Factor 1.00

Critical Values: (N./mm², mm.)

	Max Stress	Elevation	Allowables
Bending	5.05	3040.00	90.06 (UG-23)
Shear	1.38	7650.00	96.60 (0.7*S)

The compressive bending allowable stress is the minimum of the tensile allowable and the compressive allowable times the specified compressive allowable increase factor.

Forces and Moments at selected elevations (not all analysis points shown):

Distance mm.	Bending Moment Kg-m.	Bending Stress N./mm ²	Shear Force Kg	Shear Stress N./mm ²
0.00	0.0	0.0	6528.4	1.1
2000.00	10100.1	3.4	68.9	0.0
3000.00	13124.2	5.0	1480.9	0.3
3050.00	13210.0	3.8	1480.9	0.2
6050.00	7978.5	2.0	8399.7	1.0
8050.00	1325.4	0.4	799.1	0.1

Unity Check (Actual Stress / Allowable Stress):

Maximum Unity Check is 0.056 at elevation 3040.0 mm. - Must be <=1

LP Condensate Flash Vessel

PV Elite 28 U1 Licensee: SL User

FileName : LP CONDENSATE FLASH DRUM (V8301C) -7 Percent Page 66 of 171

Center of Gravity Calculation: Step: 23 3:41pm Sep 16,2026



HEXAGON

Note: The rigging analysis is performed using a uniformly distributed load.

--- Plot data successfully generated ...----

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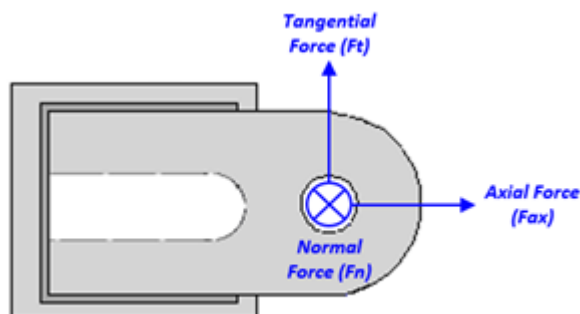
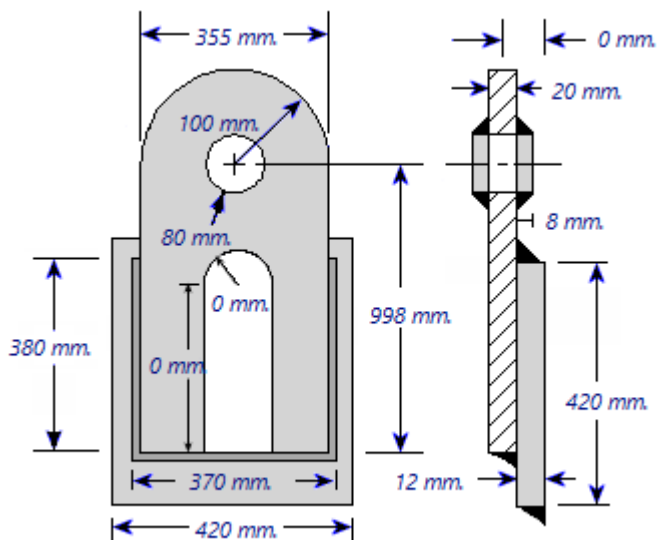
Lifting Lug Calculations:

Input Values:

Lifting Lug Material		SA-516 70
Lifting Lug Yield Stress	Yield	262.00 N./mm ²
Width of Lifting Lug	w	355.0000 mm.
Thickness of Lifting Lug	t	20.0000 mm.
Diameter of Hole in Lifting Lug	dh	80.0000 mm.
Radius of Semi-Circular Arc of Lifting Lug	r	100.0000 mm.
Height of Lug from Bottom to Center of Hole	h	998.0000 mm.
Offset from Vessel OD to Center of Hole	off	0.0000 mm.
Lug Fillet Weld Size	tw	16.0000 mm.
Length of weld along side of Lifting Lug	wl	380.0000 mm.
Length of Weld along Bottom of Lifting Lug	wb	370.0000 mm.
Thickness of Collar (if any)	tc	0.0000 mm.
Diameter of Collar (if any)	dc	0.0000 mm.
Impact Factor	Impfac	1.50
Number of Lugs in Group		2
Reinforcing Pad Circumferential Width	wpb	420.00 mm.
Reinforcing Pad Longitudinal Length	wpl	420.00 mm.
Reinforcing Pad Thickness	thkp	12.00 mm.
Reinforcing Pad Fillet Weld Size	tfp	8.00 mm.

Lifting Lug Orientation to Vessel: Flat

Lift Orientation : From Horizontal to Vertical





PV Elite computes the forces and stresses from 0 to 90 degrees at 1 degree increments. The table below provides the results in 5 degree increments.

The 0 degree angle position is in the horizontal direction and the 90 degree angle position is in the vertical direction.

PV Elite does not compute weak axis bending forces on the lugs. It is assumed that a spreader bar is used.

Lifting Lug and Weld Stresses at various Angles: N./mm²

Angle	----- Weld Stresses			----- Lug Stresses		Maximum
	Primary	Secondary	Allowable	Combined	Allowable	Ratio
0	6.2	46.7	104.8	116.4	172.9	0.673
5	6.2	46.8	104.8	117.5	172.9	0.680
10	6.3	46.5	104.8	117.8	172.9	0.681
15	6.3	45.8	104.8	117.1	172.9	0.677
20	6.3	44.8	104.8	115.5	172.9	0.668
25	6.4	43.5	104.8	113.1	172.9	0.654
30	6.4	41.8	104.8	109.8	172.9	0.635
35	6.4	39.7	104.8	105.6	172.9	0.611
40	6.5	37.4	104.8	100.7	172.9	0.582
45	6.5	34.8	104.8	95.0	172.9	0.549
50	6.6	31.9	104.8	88.5	172.9	0.512
55	6.6	28.7	104.8	81.4	172.9	0.470
60	6.7	25.3	104.8	73.6	172.9	0.426
65	6.8	21.7	104.8	65.2	172.9	0.377
70	6.9	17.9	104.8	56.4	172.9	0.326
75	7.1	13.9	104.8	47.0	172.9	0.272
80	7.4	9.7	104.8	37.3	172.9	0.216
85	7.9	5.2	104.8	27.1	172.9	0.157
90	9.1	1.2	104.8	16.4	172.9	0.095

Computed Results:

Total vessel weight (no liquid)	15789.71 Kg
Design reaction force at the tailing lug	7547.03 Kg
Design reaction force at the lifting lug	16137.54 Kg

Worst case lift angle for primary weld stresses	90 deg
Tangential force at this angle	0.0 Kg
Axial force at this angle	11842.3 Kg

Worst case lift angle for secondary weld stresses	3 deg
Tangential force at this angle	8082.7 Kg
Axial force at this angle	423.6 Kg

Worst case lift angle for lug stresses	9 deg
Tangential force at this angle	8043.0 Kg
Axial force at this angle	1273.9 Kg

Converting the weld leg dimension (tw) to the weld throat dimension.

Weld Group Inertia Calculations:

Weld group inertia about the longitudinal axis	I _{ll}	21125.770 cm ⁴
Weld group centroid distance in the long. direction	Y _{ll}	254.064 mm.
Dist. of weld group centroid from lug bottom	Y _{llb}	125.936 mm.
Weld group inertia about the circumferential axis	I _{lc}	32436.039 cm ⁴
Weld group centroid distance in the circ. direction	Y _{lc}	185.000 mm.

Note: The Impact Factor is applied to the Forces acting on the Lug.

Primary Shear Stress in the Welds due to Shear Loads [S_{sl}]:

$$= \sqrt{(F_{ax}^2 + F_t^2 + F_n^2)} / ((2 \cdot w_l + w_b) \cdot t_w)$$

$$= \sqrt{(11842^2 + 0^2 + 0^2)} / ((2 \cdot 380.0 + 370.0) \cdot 11.312)$$

$$= 9.09 \text{ N./mm}^2$$



Shear Stress in the Welds due to Bending Loads [Sblf]:

$$= (F_n(h-Y_{IIb}))Y_{II/III} + (F_{ax} \cdot \text{off} \cdot Y_{II/III}) + (F_t \cdot \text{off} \cdot Y_{Ic/IIc})$$

$$= (0(998.0-125.936))254.064/21125.77 +$$

$$(11842 \cdot 0.0 \cdot 254.064/21125.77) +$$

$$(0 \cdot 0.0 \cdot 185.0/32436.039)$$

$$= 0.00 \text{ N./mm}^2$$

Total Shear Stress for Combined Loads [St]:

$$= S_{sII} + S_{blf}$$

$$= 9.085 + 0.0$$

$$= 9.09 \text{ N./mm}^2$$

Allowable Shear Stress for Combined Loads [Sta]:

$$= 0.4 \cdot \text{Yield} \cdot \text{Occfac (AISC Shear Allowable)}$$

$$= 0.4 \cdot 262 \cdot 1.0$$

$$= 104.80 \text{ N./mm}^2$$

Secondary Shear Stress in the Welds due to Shear Loads:

Unit Weld Polar Section Modulus [Uwsm]:

$$= (2 \cdot w_l + w)^3 / 12 - w_l^2(w_l + w)^2 / (2 \cdot w_l + w)$$

$$= 45553568.00 \text{ mm}^3$$

Loads on Welds due to Torsional Moment:

Shear Stress in Normal Direction [Fth]:

$$= F_t \cdot (h - (w_l - \text{Cent})) \cdot (w_b / 2) / U_{wsm}$$

$$= 28.51 \text{ Kg/mm.}$$

Shear Stress in Circumferential Direction [Ftv]:

$$= F_t \cdot (h - (w_l - \text{Cent})) \cdot \text{Cent} / U_{wsm}$$

$$= 38.60 \text{ Kg/mm.}$$

Primary Shear Stress [Fsv]:

$$= F_t / (2 \cdot w_l + w)$$

$$= 7.25 \text{ Kg/mm.}$$

Resultant Load on Weld Group [Fr]:

$$= \sqrt{(F_{th}^2 + (F_{tv} + F_{sv})^2)}$$

$$= \sqrt{(28.5^2 + (38.6 + 7.2)^2)}$$

$$= 53.99 \text{ Kg/mm.}$$

Resultant Secondary Weld Stress [Fws]:

$$= F_r / T_{weld}$$

$$= 53.99/11.312$$

$$= 46.81 \text{ N./mm}^2$$

Allowable Resultant Secondary Weld Stress [Psa]:

$$= 0.4 \cdot \text{Yield} \cdot \text{Occfac}$$

$$= 0.4 \cdot 262.0 \cdot 1.0$$

$$= 104.80 \text{ N./mm}^2$$

Shear Stress in Lug above Hole [Shs]:

$$= \sqrt{(P_l^2 + F_n^2)} / S_{ha}$$

$$= \sqrt{(11842^2 + 0^2)} / 24.0$$

$$= 48.39 \text{ N./mm}^2$$

Allowable Shear Stress in Lug above Hole [Sas]:

$$= 0.4 \cdot \text{Yield} \cdot \text{Occfac}$$

$$= 0.4 \cdot 262 \cdot 1.0$$

$$= 104.80 \text{ N./mm}^2$$

Pin Hole Bearing Width [Pbw]:

$$= \text{Lug Thickness} + 2 \cdot \text{Collar Thickness}$$

$$= 20.0 + 2 \cdot 0.0$$

$$= 20.000 \text{ mm.}$$

Pin Hole Bearing Stress [Pbs]:



$$= \sqrt{(P_l^2 + F_n^2)} / (P_{bw} \cdot d_h)$$

$$= \sqrt{(11842^2 + 0^2)} / (20.0 \cdot 80.0)$$

$$= 72.58 \text{ N./mm}^2$$

Allowable Bearing Stress [Pba]:

$$= \min(0.75 \cdot \text{Yield} \cdot \text{Occfac}, 0.9 \cdot \text{Yield}) \text{ AISC Bearing All.}$$

$$= \min(0.75 \cdot 262 \cdot 1.0, 235.8)$$

$$= 196.50 \text{ N./mm}^2$$

Bending Stress at the Base of the Lug [Fbs]:

$$= F_n \cdot (h - w_l) / (w \cdot t^2 / 6) + F_t \cdot (h - w_l) / (w^2 \cdot t / 6)$$

$$= 0 \cdot (998.0 - 380.0) / (355.0 \cdot 20.0^2 / 6) +$$

$$8043 \cdot (998.0 - 380.0) / (355.0^2 \cdot 20.0 / 6)$$

$$= 116.04 \text{ N./mm}^2$$

Tensile Stress at the Base of the Lug [Fa]:

$$= F_{ax} / (w \cdot t)$$

$$= 1274 / (355.0 \cdot 20.0)$$

$$= 1.76 \text{ N./mm}^2$$

Total Combined Stress at the Base of the Lug:

$$= F_{bs} + F_a$$

$$= 116.0 + 1.8$$

$$= 117.80 \text{ N./mm}^2$$

Lug Allowable Stress for Bending and Tension:

$$= \min(0.66 \cdot \text{Yield} \cdot \text{Occfac}, 0.75 \cdot \text{Yield})$$

$$= \min(0.66 \cdot 262 \cdot 1.0, 196.5)$$

$$= 172.92 \text{ N./mm}^2$$

Required Shackle Pin Diameter [Spd]:

$$= \sqrt{((2(P_l^2 + F_n^2))^{1/2} / (\pi \cdot S_{ta}))}$$

$$= \sqrt{(2(11842^2 + 0^2))^{1/2} / (\pi \cdot 105)}$$

$$= 26.5611 \text{ mm.}$$

WRC 107/537 Stress Analysis for the Lifting Lug to Shell Junction in the new and Cold Condition (no corrosion applied).

The full lug load is applied in both the circ. and long. directions simultaneously.

Attachment dimensions (C11P, C22P, etc.) may be modified accounting for attachment welds where applicable.

Input Echo, WRC107/537 Item 1, Description: Lift Lug

Diameter Basis for Vessel	Vbasis	ID
Cylindrical or Spherical Vessel	Cylsph	Cylindrical
Internal Corrosion Allowance	Cas	0.0000 mm.
Vessel Diameter	Dv	1900.000 mm.
Vessel Thickness	Tv	14.000 mm.

Note: Lifting lug calculations are performed at ambient temperature conditions.

Attachment Type	Type	Rectangular
Parameter C11	C11	387.00 mm.
Parameter C22	C22	396.00 mm.
Thickness of Reinforcing Pad	Tpad	12.000 mm.
Pad Parameter C11P	C11p	436.000 mm.
Pad Parameter C22P	C22p	428.000 mm.
Design Internal Pressure	Dp	0.000 bars
Include Pressure Thrust		No

External Forces and Moments in WRC 107/537 Convention:

Radial Load (SUS)	P	0.0	Kgf
Longitudinal Shear (SUS)	Vl	11842.3	Kgf
Circumferential Shear (SUS)	Vc	11842.3	Kgf
Circumferential Moment (SUS)	Mc	0.0	Kg-m.

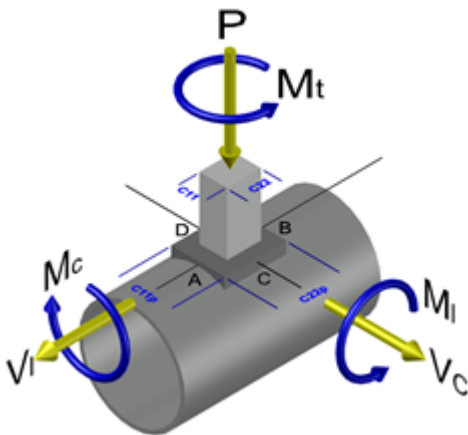


Longitudinal Moment (SUS) Ml 0.0 Kg-m.
Torsional Moment (SUS) Mt 9568.8 Kg-m.

Use Interactive Control No
WRC107 Version Version March 1979

Include Pressure Stress Indices per Div. 2 No
Compute Pressure Stress per WRC-368 No
Local Loads applied at end of Nozzle/Attachment No

WRC Bulletin 537 provides equations for the dimensionless curves found in bulletin 107. As noted in the foreword to bulletin 537, "537 is equivalent to WRC 107". Where 107 is printed in the results below, "537" can be interchanged with "107".



WRC 107/537 has dimensional limitations that can be found in the bulletin's Foreword and which can also be inferred by review of the non-dimensional curve limits.

WRC 107 Stress Calculation for SUSained loads:

Radial Load	P	0.0	Kgf
Circumferential Shear	VC	11842.3	Kgf
Longitudinal Shear	VL	11842.3	Kgf
Circumferential Moment	MC	0.0	Kg-m.
Longitudinal Moment	ML	0.0	Kg-m.
Torsional Moment	MT	9568.8	Kg-m.

Dimensionless Parameters used : Gamma = 37.04

Dimensionless Loads for Cylindrical Shells at Attachment Junction:

Curves read for 1979	Beta	Figure	Value	Location
$N(\text{PHI}) / (P/R_m)$	0.206	4C	5.137	(A,B)
$N(\text{PHI}) / (P/R_m)$	0.206	3C	3.283	(C,D)
$M(\text{PHI}) / (P)$	0.203	2C1	0.036	(A,B)
$M(\text{PHI}) / (P)$	0.203	1C !	0.067	(C,D)
$N(\text{PHI}) / (MC/(R_m^{*2} * \text{Beta}))$	0.202	3A	1.505	(A,B,C,D)
$M(\text{PHI}) / (MC/(R_m * \text{Beta}))$	0.225	1A	0.074	(A,B,C,D)
$N(\text{PHI}) / (ML/(R_m^{*2} * \text{Beta}))$	0.204	3B	3.730	(A,B,C,D)
$M(\text{PHI}) / (ML/(R_m * \text{Beta}))$	0.209	1B	0.024	(A,B,C,D)
$N(x) / (P/R_m)$	0.204	3C	3.313	(A,B)
$N(x) / (P/R_m)$	0.204	4C	5.157	(C,D)
$M(x) / (P)$	0.205	1C1	0.067	(A,B)
$M(x) / (P)$	0.205	2C !	0.036	(C,D)
$N(x) / (MC/(R_m^{*2} * \text{Beta}))$	0.202	4A	2.852	(A,B,C,D)
$M(x) / (MC/(R_m * \text{Beta}))$	0.245	2A	0.032	(A,B,C,D)
$N(x) / (ML/(R_m^{*2} * \text{Beta}))$	0.204	4B	1.511	(A,B,C,D)
$M(x) / (ML/(R_m * \text{Beta}))$	0.216	2B	0.036	(A,B,C,D)



Note - The ! mark next to the figure name denotes curve value exceeded.

Stress Concentration Factors: $K_n = 1.00$, $K_b = 1.00$

Stresses in the Vessel at the Attachment Junction (N./mm²)

Type of Stress	Load	Stress Intensity Values at							
		Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Memb. P		-0.0	-0.0	-0.0	-0.0	-0.0	-0.0	-0.0	-0.0
Circ. Bend. P		-0.0	0.0	-0.0	0.0	-0.0	0.0	-0.0	0.0
Circ. Memb. MC		0.0	0.0	0.0	0.0	-0.0	-0.0	0.0	0.0
Circ. Bend. MC		0.0	0.0	0.0	0.0	-0.0	0.0	0.0	-0.0
Circ. Memb. ML		-0.0	-0.0	0.0	0.0	0.0	0.0	0.0	0.0
Circ. Bend. ML		-0.0	0.0	0.0	-0.0	0.0	0.0	0.0	0.0
Tot. Circ. Str.		0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Long. Memb. P		-0.0	-0.0	-0.0	-0.0	-0.0	-0.0	-0.0	-0.0
Long. Bend. P		-0.0	0.0	-0.0	0.0	-0.0	0.0	-0.0	0.0
Long. Memb. MC		0.0	0.0	0.0	0.0	-0.0	-0.0	0.0	0.0
Long. Bend. MC		0.0	0.0	0.0	0.0	-0.0	0.0	0.0	-0.0
Long. Memb. ML		-0.0	-0.0	0.0	0.0	0.0	0.0	0.0	0.0
Long. Bend. ML		-0.0	0.0	0.0	-0.0	0.0	0.0	0.0	0.0
Tot. Long. Str.		0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Shear VC		5.8	5.8	-5.8	-5.8	0.0	0.0	0.0	0.0
Shear VL		0.0	0.0	0.0	0.0	-5.6	-5.6	5.6	5.6
Shear MT		94.2	94.2	94.2	94.2	94.2	94.2	94.2	94.2
Tot. Shear		100.0	100.0	88.4	88.4	88.6	88.6	99.8	99.8
Str. Int.		199.9	199.9	176.9	176.9	177.1	177.1	199.7	199.7

Dimensionless Parameters used : $\Gamma = 68.36$

Dimensionless Loads for Cylindrical Shells at Pad Edge:

Curves read for 1979	Beta	Figure	Value	Location
$N(\Phi) / (P/R_m)$	0.228	4C	7.728	(A,B)
$N(\Phi) / (P/R_m)$	0.228	3C	3.764	(C,D)
$M(\Phi) / (P)$	0.228	2C1	0.018	(A,B)
$M(\Phi) / (P)$	0.228	1C !	0.067	(C,D)
$N(\Phi) / (MC/(R_m^{*2} * \beta))$	0.228	3A	2.363	(A,B,C,D)
$M(\Phi) / (MC/(R_m * \beta))$	0.228	1A	0.061	(A,B,C,D)
$N(\Phi) / (ML/(R_m^{*2} * \beta))$	0.228	3B	5.212	(A,B,C,D)
$M(\Phi) / (ML/(R_m * \beta))$	0.228	1B	0.014	(A,B,C,D)
$N(x) / (P/R_m)$	0.228	3C	3.764	(A,B)
$N(x) / (P/R_m)$	0.228	4C	7.728	(C,D)
$M(x) / (P)$	0.228	1C1	0.043	(A,B)
$M(x) / (P)$	0.228	2C !	0.035	(C,D)
$N(x) / (MC/(R_m^{*2} * \beta))$	0.228	4A	6.129	(A,B,C,D)
$M(x) / (MC/(R_m * \beta))$	0.228	2A	0.026	(A,B,C,D)
$N(x) / (ML/(R_m^{*2} * \beta))$	0.228	4B	2.570	(A,B,C,D)
$M(x) / (ML/(R_m * \beta))$	0.228	2B	0.018	(A,B,C,D)

Note - The ! mark next to the figure name denotes curve value exceeded.

Stress Concentration Factors: $K_n = 1.00$, $K_b = 1.00$

Stresses in the Vessel at the Edge of Reinforcing Pad (N./mm²)



Type of Stress	Load	Stress Intensity Values at							
		Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Memb. P		-0.0	-0.0	-0.0	-0.0	-0.0	-0.0	-0.0	-0.0
Circ. Bend. P		-0.0	0.0	-0.0	0.0	-0.0	0.0	-0.0	0.0
Circ. Memb. MC		0.0	0.0	0.0	0.0	-0.0	-0.0	0.0	0.0
Circ. Bend. MC		0.0	0.0	0.0	0.0	-0.0	0.0	0.0	-0.0
Circ. Memb. ML		-0.0	-0.0	0.0	0.0	0.0	0.0	0.0	0.0
Circ. Bend. ML		-0.0	0.0	0.0	-0.0	0.0	0.0	0.0	0.0
Tot. Circ. Str.		0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Long. Memb. P		-0.0	-0.0	-0.0	-0.0	-0.0	-0.0	-0.0	-0.0
Long. Bend. P		-0.0	0.0	-0.0	0.0	-0.0	0.0	-0.0	0.0
Long. Memb. MC		0.0	0.0	0.0	0.0	-0.0	-0.0	0.0	0.0
Long. Bend. MC		0.0	0.0	0.0	0.0	-0.0	0.0	0.0	-0.0
Long. Memb. ML		-0.0	-0.0	0.0	0.0	0.0	0.0	0.0	0.0
Long. Bend. ML		-0.0	0.0	0.0	-0.0	0.0	0.0	0.0	0.0
Tot. Long. Str.		0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Shear VC		9.5	9.5	-9.5	-9.5	0.0	0.0	0.0	0.0
Shear VL		0.0	0.0	0.0	0.0	-9.5	-9.5	9.5	9.5
Shear MT		141.0	141.0	141.0	141.0	141.0	141.0	141.0	141.0
Tot. Shear		150.6	150.6	131.5	131.5	131.5	131.5	150.6	150.6
Str. Int.		301.1	301.1	263.0	263.0	263.0	263.0	301.1	301.1

WRC 107/537 Stress Summations:**Vessel Stress Summation at Attachment Junction (N./mm²)**

Type of Stress	Load	Stress Intensity Values at							
		Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)		0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Circ. Pl (SUS)		0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Circ. Q (SUS)		0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Long. Pm (SUS)		0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Long. Pl (SUS)		0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Long. Q (SUS)		0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Shear Pm (SUS)		0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Shear Pl (SUS)		5.8	5.8	-5.8	-5.8	-5.6	-5.6	5.6	5.6
Shear Q (SUS)		94.2	94.2	94.2	94.2	94.2	94.2	94.2	94.2
Pm (SUS)		0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Pm+Pl (SUS)		11.5	11.5	11.5	11.5	11.3	11.3	11.3	11.3
Pm+Pl+Q (Total)		199.9	199.9	176.9	176.9	177.1	177.1	199.7	199.7

Vessel Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	0.00	138.00	Passed
Pm+Pl (SUS)	11.54	207.00	Passed
Pm+Pl+Q (TOTAL)	199.94	413.99	Passed

The Pm+Pl+Q allowable above was based on the rules of ASME VIII-2 Part 4, 4.1.6.3 (a)&(b) or per ASME VIII-2, 5.5.6.1(1)&(2) Part 5 as applicable.

The Pm+Pl+Q allowable was based on a temperature range cycling from ambient to design temperature.

WRC 107/537 Stress Summations:

Vessel Stress Summation at Reinforcing Pad Edge (N/mm²)

Type of Stress	Load	Stress Intensity Values at							
		Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)		0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Circ. Pl (SUS)		0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Circ. Q (SUS)		0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Long. Pm (SUS)		0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Long. Pl (SUS)		0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Long. Q (SUS)		0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Shear Pm (SUS)		0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Shear Pl (SUS)		9.5	9.5	-9.5	-9.5	-9.5	-9.5	9.5	9.5
Shear Q (SUS)		141.0	141.0	141.0	141.0	141.0	141.0	141.0	141.0
Pm (SUS)		0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Pm+Pl (SUS)		19.0	19.0	19.0	19.0	19.0	19.0	19.0	19.0
Pm+Pl+Q (Total)		301.1	301.1	263.0	263.0	263.0	263.0	301.1	301.1

Vessel Stress Summation Comparison (N/mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	0.00	138.00	Passed
Pm+Pl (SUS)	19.03	207.00	Passed
Pm+Pl+Q (TOTAL)	301.10	413.99	Passed

The Pm+Pl+Q allowable above was based on the rules of ASME VIII-2 Part 4, 4.1.6.3 (a)&(b) or per ASME VIII-2, 5.5.6.1(1)&(2) Part 5 as applicable.

The Pm+Pl+Q allowable was based on a temperature range cycling from ambient to design temperature.

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**Skirt Data :**

Skirt Outside Diameter at Base	SOD, D6	1927.0000	mm.
Skirt Thickness	STHK	10.0000	mm.
Skirt Internal Corrosion Allowance	SCA	3.0000	mm.
Skirt External Corrosion Allowance		0.0000	mm.
Skirt Material		SA-516 70	

Basing Input: Type of Geometry: Continuous Top Ring W/Gussets

Thickness of Basing	TBA, ea4	25.0000	mm.
Design Temperature of the Basing		38.00	°C
Basing Material		SA-516 70	
Basing Ope. All. Stress	BASOPE, f4	138.00	N./mm ²
Basing Yield Stress		262.00	N./mm ²
Inside Diameter of Basing	DI, D4	1783.0000	mm.
Outside Diameter of Basing	DOU, D5	2183.0000	mm.
Nominal Diameter of Bolts	BND, d(b0)	36.0000	mm.
Bolt Corrosion Allowance	BCA	1.5000	mm.
Bolt Material		SA-307 A	
Bolt Material Class/Thickness Range		6 <= t <= 100	
Bolt Operating Allowable Stress	SA, fB	138.00	N./mm ²
Number of Bolts	RN, nb	12	
Diameter of Bolt Circle	DC, DBC	2055.0000	mm.
Thickness of Gusset Plates	TGA, e7	16.0000	mm.
Width of Gussets at Top Plate	TWDT	118.0000	mm.
Width of Gussets at Base Plate	BWDT	118.0000	mm.
Gusset Plate Elastic Modulus	E	201733296.0	KPa.
Gusset Plate Yield Stress	SY	262.0	N./mm ²
Height of Chair	h1	250.0000	mm.
Distance between Gussets	RG, b6	76.0000	mm.
Dist. from Bolt Center to Gusset (Rg/2)	CG	38.0000	mm.
Number of Gussets per bolt	NG	2	
Thickness of Top Plate or Ring	TTA, e5	32.0000	mm.
Radial Width of the Top Plate	TOPWTH, b5	128.0000	mm.
Anchor Bolt Hole Dia. in Top Plate	dia,H	50.0000	mm.
External Corrosion Allowance	CA	1.0000	mm.
Dead Weight of Vessel	DW	15789.7	Kgf
Operating Weight of Vessel	ROW	22890.5	Kgf
Earthquake Moment on Basing	EQMOM	26492.3	Kg-m.
Wind Moment on Basing	WIMOM	11577.9	Kg-m.
Percent Bolt Preload	ppl	100.0	
Use AISC A5.2 Increase in Fc and Bolt Stress		No	
Use Allowable Weld Stress per AISC J2.5		No	
Factor for Increase of Allowables	Fact	1.0000	

Results for Basing Analysis : Analyze Option

Basing Thickness Calculation method used : Simplified (Steel on Steel)

Calculation of Bolt Loads for Multiple Loading Conditions:

Bolt Load, Earthquake Operating Case [FB,e]:

$$= ((4 \cdot M/DC) - \text{abs}(W - \text{vertical force}))/nb \quad \text{per Jawad \& Farr, Eq. 12.3}$$

$$= ((4 \cdot 26492/2055.0) - 22173)/12$$

$$= 2449.4 \text{ Kgf}$$

Bolt Load, Windload Empty Case [FB,w]:

$$= ((4 \cdot M/DC) - W)/nb \quad \text{per Jawad \& Farr, Eq. 12.3}$$

$$= ((4 \cdot 11578/2055.0) - 15790)/12$$

$$= 562.2 \text{ Kgf}$$



Bolt Load, Windload Vortex Shedding Case [FB,v]:

$$= ((4 \cdot M/DC) - W) / n_b \quad \text{per Jawad \& Farr, Eq. 12.3}$$

$$= ((4 \cdot 0/2055.0) - 22891) / 12$$

$$= -1907.5 \text{ Kgf}$$

Calculation of Maximum Load per Bolt [W/Bolt], Earthquake + Operating Condition:

where: W = ROW - Yforce (from gy Acceleration or user force), M = EQMOM

$$= ((4 \cdot M/DC) - W) / n_b \quad \text{per Jawad \& Farr, Eq. 12.3}$$

$$= ((4 \cdot 26492/2055.0) - 22173) / 12$$

$$= 2449.4 \text{ Kgf}$$

Required Area for Each Bolt, Based on Max Load	1.7406	cm ²
Area Available in a Single Bolt (Corr)	6.0063	cm ²
Area Available in all the Bolts (Corr)	72.0756	cm ²
Bolt Stress Based on Simplified Analysis	40.0	N./mm ²
Allowable Bolt Stress 138.0 [Fact]	138.00	N./mm ²

Concrete Contact Area of Base Ring	CCA	12459.56	cm ²
Concrete Contact Section Modulus of Base Ring		0.5668E+09	mm. ³

Concrete Load (Simplified method), Earthquake in Operating Condition [Sc]:

$$= ((ppl/100 \cdot (Abt \cdot Sa) + W) / Cca) + M/CZ \quad \text{per Jawad \& Farr Eq. 12.1}$$

$$= (1.0(72.0756 \cdot 138 + 23608) / 12459.56) + 26492 / 0.56680E+09$$

$$= 1.44 \text{ N./mm}^2$$

Concrete Allowable Stress	22.50	N./mm ²
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Determine Maximum Bending Width of Basing Section [Rw1]:

$$= (D_{ou} - SkirtOD) / 2$$

$$= (2183.0 - 1927.0) / 2$$

$$= 128.0 \text{ mm.}$$

Determine Maximum Bending Width of Basing Section [Rw2]:

$$= (SkirtID - D_i + 2 \cdot S_{ca}) / 2$$

$$= (1907.0 - 1783.0 + 2 \cdot 3.0) / 2$$

$$= 65.0 \text{ mm.}$$

Calculation of required Basing Thickness, (Simplified) [Tb]:

$$= \max(Rw1, Rw2) \cdot \sqrt{(3 \cdot Sc / S)} + CA \quad \text{per Jawad \& Farr Eq. 12.12}$$

$$= \max(128.0, 65.0) \cdot \sqrt{(3 \cdot 1.442 / 174.663)} + 1.0$$

$$= 21.1477 \text{ mm.}$$

Basing Stress at given Thickness [Sb]:

$$= 3 \cdot Sc (\max[Rw1, Rw2] / (Tb - Ca))^2$$

$$= 3 \cdot 1.442 (\max[128.0, 65.0] / (25.0 - 1.0))^2$$

$$= 123.092, \text{ must be less than } 174.663 \text{ N./mm}^2$$

Required Thickness of Top Plate in Tension:

(Calculated as a fixed beam per Megyesy)

$$F_t = (S_a \cdot Ab_{ss}), \quad \text{Bolt Allowable Stress} \cdot \text{Area}$$

$$R_m = (F_t \cdot 2 \cdot C_g) / 8, \quad \text{Bending Moment}$$

$$S_b \quad \text{Allowable Bending Stress}$$

$$W_t = (Topw_{th} - B_{nd}), \quad \text{Width of Section}$$

Required Thickness of Top Plate [T]:

$$= \sqrt{(6 \cdot R_m / (S_b \cdot W_t))} + CA$$

$$= \sqrt{(6 \cdot 80 / (175 \cdot 92.0))} + 1.0$$

$$= 18.147 \text{ mm.}$$

Required Thickness of Continuous Top Ring per Moss:

$$a = (D_c - SkirtOD) / 2 \quad \text{Skirt Distance to Bolt Circle}$$

$$P = S_a \cdot Ab_{ss} \quad \text{Bolt Allowable Stress} \cdot \text{Area}$$

$$l = Avg_{wdt} \quad \text{Average Gusset Width}$$

$$g_1 = \text{Gamma } 1 \quad \text{Constant Term } f(b/l)$$

$$g_2 = \text{Gamma } 2 \quad \text{Constant Term } f(b/l)$$

$$g = \text{Flat distance} / 2 \quad \text{Nut } 1/2 \text{ Dimension (from Tema)}$$

$$F_b = \quad \text{Allowable Bending Stress}$$



$$M_o = P/(4\pi)[1.3(\ln((2\sin(\pi a/l)/(\pi g))) + 1) - ((0.7 - g^2)P/(4\pi))] \text{ Moment Term}$$

Required Thickness of Top Ring [T]:

$$= \sqrt{(6 \cdot \text{Abs}(M_o) / F_b)} + CA \text{ Required Thickness}$$

$$= \sqrt{(6 \cdot 26/175)} + 1.0$$

$$= 19.720 \text{ mm.}$$

Required Thickness of Gusset in Compression, per AISC E2-1:

1. Allowed Compression at Given Thickness:

Factor K1/r Per E2-1	41.7850	
Factor Cc Per E2-1	123.2875	
Allowable Buckling Str. per E2-1	138.05	N./mm ²
Actual Buckling Str. at Given Thickness	21.95	N./mm ²

$$\text{Required Gusset/Support Plate thickness, + CA} = 6.3610 \text{ mm.}$$

2. Allowed Compression at Calculated Thickness:

Factor K1/r Per E2-1	124.7090	
Factor Cc Per E2-1	123.2875	
Allowable Buckling Str. per E2-2	66.80	N./mm ²
Act. Buckling Str. at Calculated Thickness	65.51	N./mm ²

Summary of Basing Thickness Calculations:

Required Basing Thickness (simplified)	21.1477	mm.
Actual Basing Thickness	25.0000	mm.

$$\text{Required Top Ring/Plate Thickness as a Fixed Beam} = 18.1469 \text{ mm.}$$

$$\text{Required Thickness of Continuous Top Ring (Moss)} = 19.7196 \text{ mm.}$$

$$\text{Actual Top Plate/Ring Thickness} = 32.0000 \text{ mm.}$$

$$\text{Required Gusset/Support Plate thickness, + CA} = 6.3610 \text{ mm.}$$

$$\text{Actual Gusset/Support Plate Thickness} = 16.0000 \text{ mm.}$$

Weld Size Calculations per Steel Plate Engineering Data - Vol. 2

Compute the Weld load at the Skirt/Base Junction [W]:

$$= \text{SkirtStress} \cdot (\text{SkirtThickness} - CA)$$

$$= 19.408 \cdot (10.0 - 3.0)$$

$$= 13.85 \text{ Kg/mm.}$$

Results for Computed Minimum Basing Weld Size [BWeld]:

$$= W / [(0.4 \cdot \text{Yield}) \cdot 2 \cdot 0.707]$$

$$= 14 / [(0.4 \cdot 220) \cdot 2 \cdot 0.707]$$

$$= 1.092 \text{ mm.}$$

Results for Computed Minimum Gusset and Top Plate to Skirt Weld Size

Vertical Plate Load [Wv]:

$$= \text{Bolt Load} / (Cmwh + 2 \cdot (Hg + Tta))$$

$$= 8452.0 / (108.0 + 2 \cdot (193.0 + 32.0))$$

$$= 15.147 \text{ Kg/mm.}$$

Horizontal Plate Load [Wh]:

$$= \text{Bolt Load} \cdot e / (Cmwh \cdot (Hg + Tta) + 0.6667 \cdot (Hg + Tta)^2)$$

$$= 8452.0 \cdot 64.0 / (108.0 \cdot (225.0) + 0.6667 \cdot (225.0)^2)$$

$$= 9.318 \text{ Kg/mm.}$$

Resultant Weld Load [Wr]:

$$= \sqrt{Wv^2 + Wh^2}$$

$$= \sqrt{15.15^2 + 9.32^2}$$

$$= 17.784 \text{ Kg/mm.}$$

Results for Computed Min Gusset and Top Plate to Skirt Weld Size [GsWeld]:

$$= Wr / [(0.4 \cdot \text{Yield}) \cdot 2 \cdot 0.707]$$

$$= 17.78 / [(0.4 \cdot 220) \cdot 2 \cdot 0.707]$$

$$= 1.402 \text{ mm.}$$

Results for Computed Minimum Gusset to Top Plate Weld Size



Weld Load [Wv]:

$$= \text{Bolt Load} / (2 \cdot \text{TopWth})$$

$$= 8452.0 / (2 \cdot 118.0)$$

$$= 35.813 \text{ Kg/mm.}$$

Weld Load [Wh]:

$$= \text{Bolt Load} \cdot e / (2 \cdot \text{Hgt} \cdot \text{TopWth})$$

$$= 8452.0 \cdot 64.0 / (2 \cdot 225.0 \cdot 118.0)$$

$$= 10.187 \text{ Kg/mm.}$$

Resultant Weld Load [Wr]:

$$= \sqrt{Wv^2 + Wh^2}$$

$$= \sqrt{35.81^2 + 10.19^2}$$

$$= 37.234 \text{ Kg/mm.}$$

Results for Computed Min Gusset to Top Plate Weld Size [GtpWeld]:

$$= Wr / [(0.4 \cdot \text{Yield}) \cdot 2 \cdot 0.707]$$

$$= 37.23 / [(0.4 \cdot 220) \cdot 2 \cdot 0.707]$$

$$= 2.935 \text{ mm.}$$

Note: The calculated weld sizes need not exceed the component thickness framing into the weld. At the same time, the weld must meet a minimum size specification which is 3/16 in. (4.76 mm) or 1/4 in. (6.35 mm), depending on the component thickness.

Summary of Required Weld Sizes:

Required Basing to Skirt Double Fillet Weld Size	4.7625	mm.
Required Gusset to Skirt Double Fillet Weld Size	6.3500	mm.
Required Top Plate to Skirt Weld Size	6.3500	mm.
Required Gusset to Top Plate Double Fillet Weld Size	2.9345	mm.

Review notes about nozzle loadings and supports in the Warnings report.

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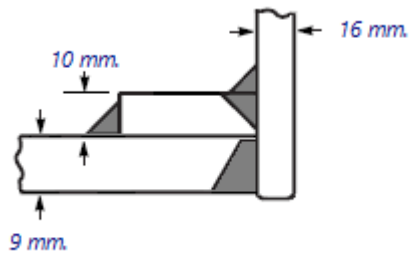
**Input, Nozzle Desc: N2-6"****From: 30**

Pressure for Reinforcement Calculations	P	7.409	bars
Temperature for Internal Pressure	Temp	230	°C
Design External Pressure	Pext	1.03	bars
Temperature for External Pressure	Tempex	198	°C
Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	138.00	N./mm2
Parent Allowable Stress At Ambient	Sva	138.00	N./mm2
Inside Diameter of Elliptical Head	D	1900.00	mm.
Aspect Ratio of Elliptical Head	Ar	2.00	
Head Finished (Minimum) Thickness	t	9.0000	mm.
Head Internal Corrosion Allowance	cas	3.0000	mm.
Head External Corrosion Allowance	co	0.0000	mm.
Distance from Head Centerline	L1	300.0000	mm.
User Entered Minimum Design Metal Temperature		-1.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-106 B	
Material UNS Number		K03006	
Material Specification/Type		Smls. pipe	
Allowable Stress at Temperature	Sn	118.00	N./mm2
Allowable Stress At Ambient	Sna	118.00	N./mm2
Diameter Basis (for tr calc only)		Outside	
Layout Angle		270.00	deg
Diameter		6.0000	in.
Size and Thickness Basis		Minimum	
Nominal Thickness		160	
Flange Material		SA-105	
Flange Type		Weld Neck Flange	
Corrosion Allowance	can	1.6000	mm.
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	200.0000	mm.
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm.
Groove weld depth between Nozzle and Vessel	Wgmv	9.0000	mm.
Inside Projection	h	0.0000	mm.
Weld leg size, Inside Element to Shell	Wi	0.0000	mm.
Pad Material		SA-516 70	
Pad Allowable Stress at Temperature	Sp	138.00	N./mm2
Pad Allowable Stress At Ambient	Spa	138.00	N./mm2
Diameter of Pad along vessel surface	Dp	318.2750	mm.
Thickness of Pad	te	10.0000	mm.
Weld leg size between Pad and Shell	Wp	8.0000	mm.
Groove weld depth between Pad and Nozzle	Wgpn	10.0000	mm.
Reinforcing Pad Width		75.0000	mm.
Flange Class		300	
Flange Grade		GR 1.1	
Piping Exit Angle [North Clockwise reference]		270.0	deg
Bend Radius Multiplier		1.50	
Horizontal Run Length		1000.0	mm.
Centerline Distance to Tangent		800.0	mm.

The Pressure Design option was MAWP + static head (to the nozzle).



Note : Checking Nozzle in the Meridional direction.

Reinforcement CALCULATION, Description: N2-6"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Outside Diameter Used in Calculation	6.625 in.
Actual Thickness Used in Calculation	0.628 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Elliptical Head, tr [Int. Press]
 $= P \cdot D \cdot K1 / (2 \cdot Sv \cdot E - 0.2 \cdot P)$ per Appendix 1-4(c)
 $= 7.41 \cdot 1906.0 \cdot 0.8973 / (2 \cdot 138.0 \cdot 1.0 - 0.2 \cdot 7.41)$
 $= 4.5936 \text{ mm.}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot Ro / (Sn \cdot E + 0.4 \cdot P)$ per Appendix 1-1 (a)(1)
 $= 7.41 \cdot 84.1375 / (118 \cdot 1.0 + 0.4 \cdot 7.41)$
 $= 0.5270 \text{ mm.}$

Required Nozzle thickness under External Pressure per UG-28 : 0.5874 mm.

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	283.0299 mm.
Parallel to Vessel Wall, opening length	d	141.5149 mm.
Normal to Vessel Wall (Thickness Limit), pad side Tlwp		15.0000 mm.

The Pad diameter is greater than the Diameter Limit. The excess will not be considered.

Weld Strength Reduction Factor [fr1]:
 $= \min(1, Sn/Sv)$
 $= \min(1, 118.0/138.0)$
 $= 0.855$

Weld Strength Reduction Factor [fr2]:
 $= \min(1, Sn/Sv)$
 $= \min(1, 118.0/138.0)$
 $= 0.855$

Weld Strength Reduction Factor [fr4]:
 $= \min(1, Sp/Sv)$
 $= \min(1, 138.0/138.0)$
 $= 1.000$

Weld Strength Reduction Factor [fr3]:
 $= \min(fr2, fr4)$
 $= \min(0.855, 1.0)$
 $= 0.855$

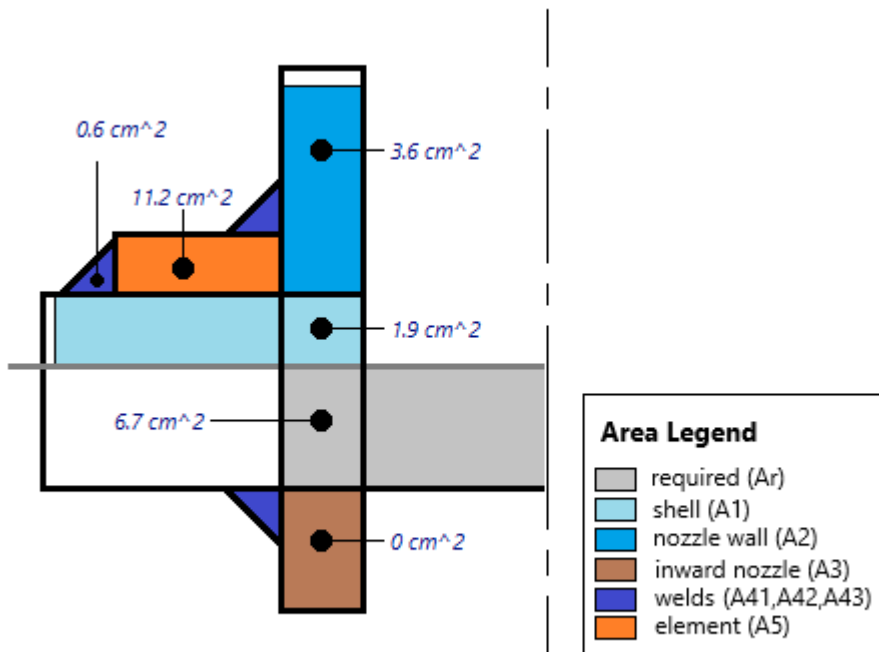


Figure UG 37.1 Nozzle Areas

Results of Nozzle Reinforcement Area Calculations: (cm²)

AREA AVAILABLE, A1 to A5		MAWP	External	Mapnc
Area Required	Ar	6.692	3.674	NA
Area in Shell	A1	1.932	1.314	NA
Area in Nozzle Wall	A2	3.575	3.559	NA
Area in Inward Nozzle	A3	0.000	0.000	NA
Area in Welds	A41+A42+A43	0.641	0.641	NA
Area in Element	A5	11.240	11.240	NA
TOTAL AREA AVAILABLE	Atot	17.388	16.754	NA

The MAWP Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 80.47 degs.

The area available without a pad is Insufficient.
 The area available with the given pad is Sufficient.

SELECTION OF POSSIBLE REINFORCING PADS: Diameter Thickness
 Based on given Pad Thickness: 173.7136 10.0000 mm.
 Based on given Pad Diameter: 318.2750 0.4839 mm.
 Based on Shell or Nozzle Thickness: 176.6752 9.0000 mm.

Area Required [A]:

$$= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1)) \text{ UG-37(c)}$$

$$= (141.5149 \cdot 4.5936 \cdot 1.0 + 2 \cdot 14.3575 \cdot 4.5936 \cdot 1.0 \cdot (1 - 0.86))$$

$$= 6.692 \text{ cm}^2$$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:

$$= d (E1 \cdot t - F \cdot tr) - 2 \cdot tn (E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$$

$$= 141.515 (1.0 \cdot 6.0 - 1.0 \cdot 4.594) - 2 \cdot 14.358$$

$$(1.0 \cdot 6.0 - 1.0 \cdot 4.5936) \cdot (1 - 0.855)$$

$$= 1.932 \text{ cm}^2$$

Area Available in Nozzle Wall Projecting Outward [A2]:

$$= (2 \cdot Tlwp) \cdot (tn - trn) \cdot fr2 / \sin(\alpha3)$$



$$= (2 \cdot 15.0) \cdot (14.36 - 0.53) \cdot 0.8551 / \sin(83.0)$$

$$= 3.575 \text{ cm}^2$$

See ASME VIII-1 2011(a) Appendix L, L-7.7.7(b) for more information.

Area Available in Welds [A41 + A42 + A43]:

$$= (W_o^2 - A_r \text{ Lost}) \cdot Fr_3 + ((W_i - c_{an} / 0.707)^2 - A_r \text{ Lost}) \cdot fr_2 + W_p^2 \cdot fr_4$$

$$= (0.75) \cdot 0.86 + (0.0) \cdot 0.86 + 0.0^2 \cdot 1.0$$

$$= 0.641 \text{ cm}^2$$

Area Available in Element [A5]:

$$= (\min(D_p, D_L) - (\text{Nozzle OD})) \cdot (\min(t_p, t_{lwp}, t_e)) \cdot fr_4$$

$$= (283.0299 - 170.6323) \cdot 10.0 \cdot 1.0$$

$$= 11.240 \text{ cm}^2$$

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, min(Curve:B, Curve:A)

Govrn. thk, tg = 15.958, tr = 0.527, c = 1.6 mm., E* = 1.0
 Thickness Ratio = tr * E* / (tg - c) = 0.037, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve A	6 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C

Nozzle Neck to Pad Weld for the Nozzle, Curve: B

Govrn. thk, tg = 10.0, c = 3.0 mm., E* = 1.0
 Thickness Ratio = tr * E* / (tg - c) = 0.766, Temp. Reduction = 13 °C
 Attachment governing, Attachment stress = Shell stress per Code

Min Metal Temp. w/o impact per UCS-66, Curve B	-29 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-42 °C

Nozzle Neck to Pad Weld for Reinforcement pad, Curve: B

Govrn. thk, tg = 10.0, c = 3.0 mm., E* = 1.0
 Thickness Ratio = tr * E* / (tg - c) = 0.766, Temp. Reduction = 13 °C
 Attachment governing, Attachment stress = Shell stress per Code

Min Metal Temp. w/o impact per UCS-66, Curve B	-29 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-42 °C

Shell to Pad Weld Junction at Pad OD, Curve: B

Govrn. thk, tg = 9.0, tr = 4.594, c = 3.0 mm., E* = 1.0
 Thickness Ratio = tr * E* / (tg - c) = 0.766, Temp. Reduction = 13 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-29 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-42 °C

Nozzle-Shell/Head Weld (UCS-66(a)(b)), Curve: B

Govrn. thk, tg = 9.0, tr = 4.594, c = 3.0 mm., E* = 1.0
 Thickness Ratio = tr * E* / (tg - c) = 0.766, Temp. Reduction = 13 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-29 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-42 °C

Gov. MDMT of the Nozzle	:	-42 °C
Gov. MDMT of the Reinforcement Pad	:	-42 °C
Gov. MDMT of the nozzle to shell joint welded assembly	:	-42 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c)	-18 °C
Flange MDMT with Temp reduction per UCS-66(b)(1)(-b)	-104 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :



Design Pressure/Ambient Rating = 7.41/51.00 = 0.145

Weld Size Calculations, Description: N2-6"

Intermediate Calc. for nozzle/shell Welds T_{min} 10.0000 mm.
 Intermediate Calc. for pad/shell Welds T_{min}Pad 9.0000 mm.

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * W _o mm.
Pad Weld	4.5000 = 0.5 * T _{min} Pad	5.6560 = 0.7 * W _p mm.

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:
 $= \max(0, (A - A1 + 2 \cdot t_n \cdot f_{r1} \cdot (E1 \cdot t - t_r)) \cdot S_v)$
 $= \max(0, (6.6918 - 1.9318 + 2 \cdot 14.3575 \cdot 0.8551 \cdot (1.0 \cdot 6.0 - 4.5936)) \cdot 138)$
 $= 7184.03 \text{ Kg}$

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:
 $= (A2 + A5 + A4 - (W_i - \text{Can} / .707)^2 \cdot f_{r2}) \cdot S_v$
 $= (3.5749 + 11.2398 + 0.6413 - 0.0 \cdot 0.86) \cdot 138$
 $= 21748.91 \text{ Kg}$

Weld Load [W2]:
 $= (A2 + A3 + A4 + (2 \cdot t_n \cdot t \cdot f_{r1})) \cdot S_v$
 $= (3.5749 + 0.0 + 0.8551 + (1.4732)) \cdot 138$
 $= 8306.65 \text{ Kg}$

Weld Load [W3]:
 $= (A2 + A3 + A4 + A5 + (2 \cdot t_n \cdot t \cdot f_{r1})) \cdot S$
 $= (3.5749 + 0.0 + 0.6413 + 11.2398 + (1.4732)) \cdot 138$
 $= 23821.95 \text{ Kg}$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [S_{onw}]:
 $= (\pi/2) \cdot D_{lo} \cdot W_o \cdot 0.49 \cdot S_{nw}$
 $= (3.1416/2.0) \cdot 170.6323 \cdot 10.0 \cdot 0.49 \cdot 118$
 $= 15802. \text{ Kg}$

Shear, Pad Element Weld [S_{pew}]:
 $= (\pi/2) \cdot D_P \cdot W_P \cdot 0.49 \cdot S_{EW}$
 $= (3.1416/2.0) \cdot 318.275 \cdot 8.0 \cdot 0.49 \cdot 138$
 $= 27577. \text{ Kg}$

Shear, Nozzle Wall [S_{nw}]:
 $= (\pi \cdot (D_{lr} + D_{lo}) / 4) \cdot (Thk - \text{Can}) \cdot 0.7 \cdot S_n$
 $= (3.1416 \cdot 78.0368) \cdot (15.9576 - 1.6) \cdot 0.7 \cdot 118$
 $= 29646. \text{ Kg}$

Tension, Pad Groove Weld [T_{pgw}]:
 $= (\pi/2) \cdot D_{lo} \cdot W_{gpn} \cdot 0.74 \cdot S_{eg}$
 $= (3.1416/2) \cdot 170.6323 \cdot 10.0 \cdot 0.74 \cdot 138$
 $= 27910. \text{ Kg}$

Tension, Shell Groove Weld [T_{ngw}]:
 $= (\pi/2) \cdot D_{lo} \cdot (T - \text{cas}) \cdot 0.74 \cdot S_{ng}$
 $= (3.1416/2.0) \cdot 170.6323 \cdot (9.0 - 3.0) \cdot 0.74 \cdot 138$
 $= 16746. \text{ Kg}$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:



$$= (\text{SPEW} + \text{SNW}) = (27577 + 29646) = 57224 \text{ Kg}$$

Failure Path [Path 2-2]:

$$= (\text{Sonw} + \text{Tpgw} + \text{Tngw} + \text{Sinw}) \\ = (15802 + 27910 + 16746 + 0) = 60458 \text{ Kg}$$

Failure Path [Path 3-3]:

$$= (\text{Spew} + \text{Tngw} + \text{Sinw}) \\ = (27577 + 16746 + 0) = 44323 \text{ Kg}$$

Summary of Failure Path Calculations:

Path 1-1 = 57223	Kgf, must exceed W = 7184 Kg	or W1 = 21748 Kg
Path 2-2 = 60457	Kgf, must exceed W = 7184 Kg	or W2 = 8306 Kg
Path 3-3 = 44323	Kgf, must exceed W = 7184 Kg	or W3 = 23821 Kg

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 7 bars

Note: The MAWP of this junction was limited by the parent Shell/Head.

Nozzle is O.K. for the External Pressure: 1 bars

Note : Checking Nozzle in the Latitudinal direction.

Reinforcement CALCULATION, Description: N2-6"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Outside Diameter Used in Calculation	6.625 in.
Actual Thickness Used in Calculation	0.628 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Elliptical Head, tr [Int. Press]
 $= P \cdot D \cdot K1 / (2 \cdot Sv \cdot E - 0.2 \cdot P)$ per Appendix 1-4(c)
 $= 7.41 \cdot 1906.0 \cdot 0.8973 / (2 \cdot 138.0 \cdot 1.0 - 0.2 \cdot 7.41)$
 $= 4.5936 \text{ mm.}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot Ro / (Sn \cdot E + 0.4 \cdot P)$ per Appendix 1-1 (a)(1)
 $= 7.41 \cdot 84.1375 / (118 \cdot 1.0 + 0.4 \cdot 7.41)$
 $= 0.5270 \text{ mm.}$

Required Nozzle thickness under External Pressure per UG-28 : 0.5874 mm.

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	279.1198 mm.
Parallel to Vessel Wall, opening length	d	139.5599 mm.
Normal to Vessel Wall (Thickness Limit), pad side Tlwp		15.0000 mm.

The Pad diameter is greater than the Diameter Limit. The excess will not be considered.

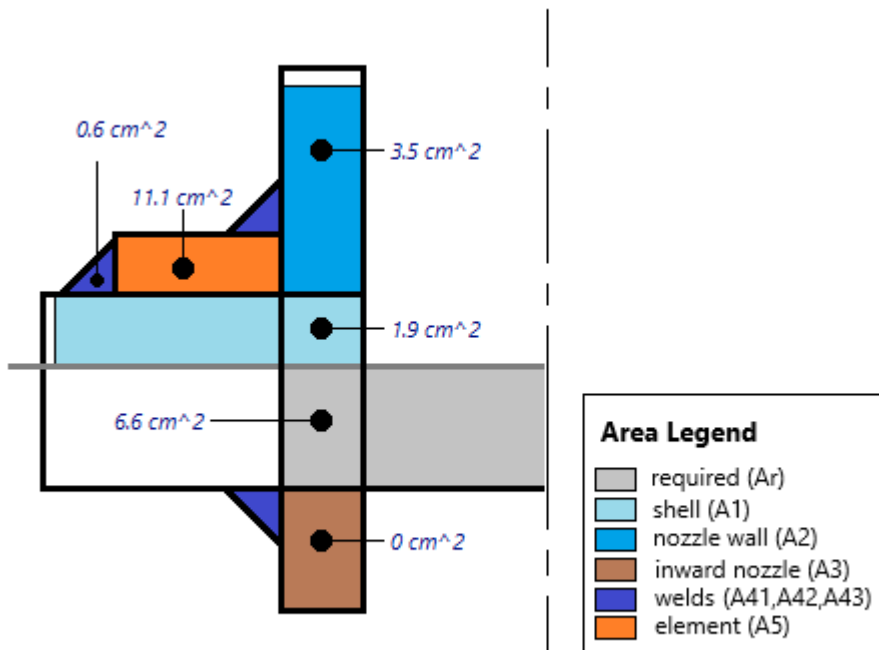


Figure UG 37.1 Nozzle Areas

Results of Nozzle Reinforcement Area Calculations: (cm²)

AREA AVAILABLE, A1 to A5		MAWP	External	Mapnc
Area Required	Ar	6.602	3.624	NA
Area in Shell	A1	1.904	1.295	NA
Area in Nozzle Wall	A2	3.548	3.532	NA
Area in Inward Nozzle	A3	0.000	0.000	NA
Area in Welds	A41+A42+A43	0.641	0.641	NA
Area in Element	A5	11.084	11.084	NA
TOTAL AREA AVAILABLE	Atot	17.178	16.553	NA

The MAWP Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Insufficient.
 The area available with the given pad is Sufficient.

SELECTION OF POSSIBLE REINFORCING PADS:	Diameter	Thickness
Based on given Pad Thickness:	173.3607	10.0000 mm.
Based on given Pad Diameter:	318.2750	0.4588 mm.
Based on Shell or Nozzle Thickness:	173.9258	9.0000 mm.

Area Required [A]:
 $= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1)) \text{ UG-37(c)}$
 $= (139.5599 \cdot 4.5936 \cdot 1.0 + 2 \cdot 14.3575 \cdot 4.5936 \cdot 1.0 \cdot (1 - 0.86))$
 $= 6.602 \text{ cm}^2$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:
 $= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$
 $= 139.56(1.0 \cdot 6.0 - 1.0 \cdot 4.594) - 2 \cdot 14.358$
 $(1.0 \cdot 6.0 - 1.0 \cdot 4.5936) \cdot (1 - 0.855)$
 $= 1.904 \text{ cm}^2$

Area Available in Nozzle Wall Projecting Outward [A2]:
 $= (2 \cdot tlwp) \cdot (tn - trn) \cdot fr2$



$$= (2 \cdot 15.0) \cdot (14.36 - 0.53) \cdot 0.8551$$

$$= 3.548 \text{ cm}^2$$

Area Available in Welds [A41 + A42 + A43]:

$$= (W_o^2 - \text{Ar Lost}) \cdot \text{Fr}_3 + ((W_i - \text{can}/0.707)^2 - \text{Ar Lost}) \cdot \text{fr}_2 + W_p^2 \cdot \text{fr}_4$$

$$= (0.75) \cdot 0.86 + (0.0) \cdot 0.86 + 0.0^2 \cdot 1.0$$

$$= 0.641 \text{ cm}^2$$

Area Available in Element [A5]:

$$= (\min(D_p, D_L) - (\text{Nozzle OD})) \cdot (\min(t_p, t_{lp}, t_e)) \cdot \text{fr}_4$$

$$= (279.1198 - 168.275) \cdot 10.0 \cdot 1.0$$

$$= 11.084 \text{ cm}^2$$

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	$t_a = 2.1874 \text{ mm.}$
Wall Thickness per UG16(b),	$tr_{16b} = 5.5000 \text{ mm.}$
Wall Thickness, shell/head, internal pressure	$trb_1 = 8.0979 \text{ mm.}$
Wall Thickness	$tb_1 = \max(trb_1, tr_{16b}) = 8.0979 \text{ mm.}$
Wall Thickness, shell/head, external pressure	$trb_2 = 3.7113 \text{ mm.}$
Wall Thickness	$tb_2 = \max(trb_2, tr_{16b}) = 5.5000 \text{ mm.}$
Wall Thickness per table UG-45	$tb_3 = 7.8200 \text{ mm.}$

Determine Nozzle Thickness candidate [tb]:

$$= \min[tb_3, \max(tb_1, tb_2)]$$

$$= \min[7.82, \max(8.0979, 5.5)]$$

$$= 7.8200 \text{ mm.}$$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

$$= \max(t_a, t_b)$$

$$= \max(2.1874, 7.82)$$

$$= 7.8200 \text{ mm.}$$

Available Nozzle Neck Thickness = 15.9576 mm. -> OK

**Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code,
 (see 319.4.4 and 302.3.5):**

Sustained	:	17.8,	Allowable	:	118.0 N./mm ²	Passed
Expansion	:	0.0,	Allowable	:	277.2 N./mm ²	Passed
Occasional	:	1.6,	Allowable	:	156.9 N./mm ²	Passed
Shear	:	13.4,	Allowable	:	82.6 N./mm ²	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Weld Size Calculations, Description: N2-6"

Intermediate Calc. for nozzle/shell Welds	T_{min}	10.0000 mm.
Intermediate Calc. for pad/shell Welds	T_{minPad}	9.0000 mm.

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	$6.0000 = \text{Min per Code}$	$7.0700 = 0.7 \cdot W_o \text{ mm.}$
Pad Weld	$4.5000 = 0.5 \cdot T_{minPad}$	$5.6560 = 0.7 \cdot W_p \text{ mm.}$

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:

$$= \max(0, (A - A_1 + 2 \cdot t_n \cdot \text{fr}_1 \cdot (E_1 \cdot t - tr)) \cdot S_v)$$

$$= \max(0, (6.602 - 1.9043 + 2 \cdot 14.3575 \cdot 0.8551 \cdot (1.0 \cdot 6.0 - 4.5936)) \cdot 138)$$

$$= 7096.35 \text{ Kg}$$

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:

$$= (A_2 + A_5 + A_4 - (W_i - \text{Can}/0.707)^2 \cdot \text{fr}_2) \cdot S_v$$

$$= (3.5478 + 11.0845 + 0.6413 - 0.0 \cdot 0.86) \cdot 138$$

$$= 21492.39 \text{ Kg}$$



Weld Load [W2]:

$$= (A2 + A3 + A4 + (2 \cdot t_n \cdot t \cdot fr1)) \cdot S_v$$

$$= (3.5478 + 0.0 + 0.8551 + (1.4732)) \cdot 138$$

$$= 8268.63 \text{ Kg}$$

Weld Load [W3]:

$$= (A2 + A3 + A4 + A5 + (2 \cdot t_n \cdot t \cdot fr1)) \cdot S$$

$$= (3.5478 + 0.0 + 0.6413 + 11.0845 + (1.4732)) \cdot 138$$

$$= 23565.43 \text{ Kg}$$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:

$$= (\pi/2) \cdot D_{lo} \cdot W_o \cdot 0.49 \cdot S_{nw}$$

$$= (3.1416/2.0) \cdot 168.275 \cdot 10.0 \cdot 0.49 \cdot 118$$

$$= 15584. \text{ Kg}$$

Shear, Pad Element Weld [Spew]:

$$= (\pi/2) \cdot D_P \cdot W_P \cdot 0.49 \cdot S_{EW}$$

$$= (3.1416/2.0) \cdot 318.275 \cdot 8.0 \cdot 0.49 \cdot 138$$

$$= 27577. \text{ Kg}$$

Shear, Nozzle Wall [Snw]:

$$= (\pi \cdot (D_{lr} + D_{lo})/4) \cdot (Thk - Can) \cdot 0.7 \cdot S_n$$

$$= (3.1416 \cdot 76.9587) \cdot (15.9576 - 1.6) \cdot 0.7 \cdot 118$$

$$= 29237. \text{ Kg}$$

Tension, Pad Groove Weld [Tpgw]:

$$= (\pi/2) \cdot D_{lo} \cdot W_{gpn} \cdot 0.74 \cdot S_{eg}$$

$$= (3.1416/2) \cdot 168.275 \cdot 10.0 \cdot 0.74 \cdot 138$$

$$= 27524. \text{ Kg}$$

Tension, Shell Groove Weld [Tngw]:

$$= (\pi/2) \cdot D_{lo} \cdot (T - cas) \cdot 0.74 \cdot S_{ng}$$

$$= (3.1416/2.0) \cdot 168.275 \cdot (9.0 - 3.0) \cdot 0.74 \cdot 138$$

$$= 16514. \text{ Kg}$$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:

$$= (S_{PEW} + S_{NW}) = (27577 + 29237) = 56814 \text{ Kg}$$

Failure Path [Path 2-2]:

$$= (S_{onw} + T_{pgw} + T_{ngw} + S_{inw})$$

$$= (15584 + 27524 + 16514 + 0) = 59623 \text{ Kg}$$

Failure Path [Path 3-3]:

$$= (S_{pew} + T_{ngw} + S_{inw})$$

$$= (27577 + 16514 + 0) = 44092 \text{ Kg}$$

Summary of Failure Path Calculations:

Path 1-1 = 56814	Kgf, must exceed W = 7096 Kg	or W1 = 21492 Kg
Path 2-2 = 59622	Kgf, must exceed W = 7096 Kg	or W2 = 8268 Kg
Path 3-3 = 44091	Kgf, must exceed W = 7096 Kg	or W3 = 23565 Kg

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 7 bars

Note: The MAWP of this junction was limited by the parent Shell/Head.

Nozzle is O.K. for the External Pressure: 1 bars

The Drop for this Nozzle is : 17.1088 mm.

The Cut Length for this Nozzle is, Drop + Ho + H + T : 226.2486 mm.

Warning:



The equations for nozzle local stress analysis pertain to radial nozzles only and this nozzle is not specified as radial. Please check the specific analysis code for all assumptions and limitations.

Input Echo, WRC107/537 Item 1, Description: N2-6" :

Diameter Basis for Vessel	Vbasis	ID	
Cylindrical or Spherical Vessel	Cylsph	Spherical	
Internal Corrosion Allowance	Cas	3.0000	mm.
Vessel Diameter	Dv	3420.000	mm.
Vessel Thickness	Tv	9.000	mm.
Design Temperature	T1	230.0	C
Vessel Material		SA-516 70	
Vessel UNS Number		K02700	
Vessel Cold S.I. Allowable	Smc	138.00	N./mm ²
Vessel Hot S.I. Allowable	Smh	138.00	N./mm ²

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Attachment Type	Type	Round	
WRC107 Attachment Classification	Holsol	Hollow	
Diameter Basis for Nozzle	Nbasis	OD	
Corrosion Allowance for Nozzle	Can	1.6000	mm.
Nozzle Diameter	Dn	168.275	mm.
Nozzle Thickness	Tn	15.958	mm.
Nozzle Material		SA-106 B	
Nozzle UNS Number		K03006	
Nozzle Cold S.I. Allowable	SNmc	118.00	N./mm ²
Nozzle Hot S.I. Allowable	SNmh	118.00	N./mm ²
Thickness of Reinforcing Pad	Tpad	10.000	mm.
Diameter of Reinforcing Pad	Dpad	318.275	mm.
Design Internal Pressure	Dp	7.409	bars
Include Pressure Thrust		Yes	

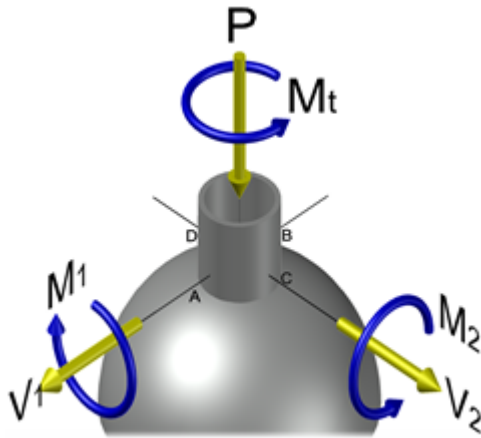
External Forces and Moments in WRC 107/537 Convention:

Radial Load (SUS)	P	-1125.0	Kgf
Longitudinal Shear (SUS)	(V1) V1	750.0	Kgf
Circumferential Shear (SUS)	(Vc) V2	750.0	Kgf
Circumferential Moment (SUS)	(Mc) M1	315.0	Kg-m.
Longitudinal Moment (SUS)	(Ml) M2	315.0	Kg-m.
Torsional Moment (SUS)	Mt	450.0	Kg-m.

Use Interactive Control	No
WRC107 Version	Version March 1979

Include Pressure Stress Indices per Div. 2	No
Compute Pressure Stress per WRC-368	No
Local Loads applied at end of Nozzle/Attachment	No

WRC Bulletin 537 provides equations for the dimensionless curves found in bulletin 107. As noted in the foreword to bulletin 537, "537 is equivalent to WRC 107". Where 107 is printed in the results below, "537" can be interchanged with "107".



WRC 107/537 has dimensional limitations that can be found in the bulletin's Foreword and which can also be inferred by review of the non-dimensional curve limits.

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{(R_{\text{mean}}(t - ca))}$$

$$= 168.275 + 2 \cdot 1.65 \cdot \sqrt{(1716.0(9.0 - 3.0))}$$

$$= 503.123 \text{ mm.}$$

WRC 107 Stress Calculation for SUSTained loads:

Radial Load		P	-2228.3	Kgf
Circumferential Shear	(VC)	V2	750.0	Kgf
Longitudinal Shear	(VL)	V1	750.0	Kgf
Circumferential Moment	(MC)	M1	315.0	Kg-m.
Longitudinal Moment	(ML)	M2	315.0	Kg-m.
Torsional Moment		MT	450.0	Kg-m.

Dimensionless Parameters: U = 0.51 TAU = 5.36 RHO = 1.11

Dimensionless Loads for Spherical Shells at Attachment Junction:

Curves read for 1979	Figure	Value	Location
$N(x) * T / P$	SP 2	0.07142	(A,B,C,D)
$M(x) / P$	SP 2	0.08190	(A,B,C,D)
$N(x) * T * \text{SQRT}(R_m * T) / MC$	SM 2	0.13715	(A,B,C,D)
$M(x) * \text{SQRT}(R_m * T) / MC$	SM 2	0.20679	(A,B,C,D)
$N(x) * T * \text{SQRT}(R_m * T) / ML$	SM 2	0.13715	(A,B,C,D)
$M(x) * \text{SQRT}(R_m * T) / ML$	SM 2	0.20679	(A,B,C,D)
$N(y) * T / P$	SP 2	0.14410	(A,B,C,D)
$M(y) / P$	SP 2	0.03411	(A,B,C,D)
$N(y) * T * \text{SQRT}(R_m * T) / MC$	SM 2	0.10165	(A,B,C,D)
$M(y) * \text{SQRT}(R_m * T) / MC$	SM 2	0.10662	(A,B,C,D)
$N(y) * T * \text{SQRT}(R_m * T) / ML$	SM 2	0.10165	(A,B,C,D)
$M(y) * \text{SQRT}(R_m * T) / ML$	SM 2	0.10662	(A,B,C,D)

Stress Concentration Factors: Kn = 1.00, Kb = 1.00

Stresses in the Vessel at the Attachment Junction (N./mm2)

Type of Stress	Load	Stress Intensity Values at							
		Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Rad.	Memb. P	6.1	6.1	6.1	6.1	6.1	6.1	6.1	6.1
Rad.	Bend. P	41.9	-41.9	41.9	-41.9	41.9	-41.9	41.9	-41.9
Rad.	Memb. MC	0.0	0.0	0.0	0.0	-10.0	-10.0	10.0	10.0
Rad.	Bend. MC	0.0	0.0	0.0	0.0	-90.2	90.2	90.2	-90.2
Rad.	Memb. ML	-10.0	-10.0	10.0	10.0	0.0	0.0	0.0	0.0
Rad.	Bend. ML	-90.2	90.2	90.2	-90.2	0.0	0.0	0.0	0.0

Type of Stress Load		Stress Intensity Values at							
		Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Rad.	Memb. P	21.7	21.7	21.7	21.7	21.7	21.7	21.7	21.7
Rad.	Bend. P	63.4	-63.4	63.4	-63.4	63.4	-63.4	63.4	-63.4
Rad.	Memb. MC	0.0	0.0	0.0	0.0	-29.7	-29.7	29.7	29.7
Rad.	Bend. MC	0.0	0.0	0.0	0.0	-129.0	129.0	129.0	-129.0
Rad.	Memb. ML	-29.7	-29.7	29.7	29.7	0.0	0.0	0.0	0.0
Rad.	Bend. ML	-129.0	129.0	129.0	-129.0	0.0	0.0	0.0	0.0
Tot. Rad.	Str.	-73.7	57.6	243.8	-141.0	-73.7	57.6	243.8	-141.0
Tang.	Memb. P	6.5	6.5	6.5	6.5	6.5	6.5	6.5	6.5
Tang.	Bend. P	19.0	-19.0	19.0	-19.0	19.0	-19.0	19.0	-19.0
Tang.	Memb. MC	0.0	0.0	0.0	0.0	-8.9	-8.9	8.9	8.9
Tang.	Bend. MC	0.0	0.0	0.0	0.0	-38.3	38.3	38.3	-38.3
Tang.	Memb. ML	-8.9	-8.9	8.9	8.9	0.0	0.0	0.0	0.0
Tang.	Bend. ML	-38.3	38.3	38.3	-38.3	0.0	0.0	0.0	0.0
Tot. Tang.	Str.	-21.7	16.9	72.6	-41.9	-21.7	16.9	72.6	-41.9
	Shear VC	2.5	2.5	-2.5	-2.5	0.0	0.0	0.0	0.0
	Shear VL	0.0	0.0	0.0	0.0	-2.5	-2.5	2.5	2.5
	Shear MT	4.6	4.6	4.6	4.6	4.6	4.6	4.6	4.6



Tot. Shear	7.1	7.1	2.2	2.2	2.2	2.2	7.1	7.1
Str. Int.	74.6	58.8	243.9	141.0	73.7	57.8	244.1	141.5

WRC 107/537 Stress Summations:**Vessel Stress Summation at Attachment Junction (N./mm²)**

Type of Stress	Load	Stress Intensity Values at							
		Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Rad. Pm (SUS)		39.7	39.7	39.7	39.7	39.7	39.7	39.7	39.7
Rad. Pl (SUS)		-3.9	-3.9	16.1	16.1	-3.9	-3.9	16.1	16.1
Rad. Q (SUS)		-48.3	48.3	132.2	-132.2	-48.3	48.3	132.2	-132.2
Long. Pm (SUS)		39.7	39.7	39.7	39.7	39.7	39.7	39.7	39.7
Long. Pl (SUS)		4.9	4.9	19.7	19.7	4.9	4.9	19.7	19.7
Long. Q (SUS)		-29.0	29.0	64.0	-64.0	-29.0	29.0	64.0	-64.0
Shear Pm (SUS)		0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Shear Pl (SUS)		1.7	1.7	-1.7	-1.7	-1.7	-1.7	1.7	1.7
Shear Q (SUS)		6.2	6.2	6.2	6.2	6.2	6.2	6.2	6.2
Pm (SUS)		39.7	39.7	39.7	39.7	39.7	39.7	39.7	39.7
Pm+Pl (SUS)		44.9	44.9	60.1	60.1	44.9	44.9	60.1	60.1
Pm+Pl+Q (Total)		32.2	88.3	188.2	76.7	29.4	85.7	188.9	77.3

Vessel Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	39.66	138.00	Passed
Pm+Pl (SUS)	60.06	207.00	Passed
Pm+Pl+Q (TOTAL)	188.87	413.99	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 107/537 Stress Summations:**Vessel Stress Summation at Reinforcing Pad Edge (N./mm²)**

Type of Stress	Load	Stress Intensity Values at							
		Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Rad. Pm (SUS)		105.8	105.8	105.8	105.8	105.8	105.8	105.8	105.8
Rad. Pl (SUS)		-8.0	-8.0	51.4	51.4	-8.0	-8.0	51.4	51.4
Rad. Q (SUS)		-65.6	65.6	192.4	-192.4	-65.6	65.6	192.4	-192.4
Long. Pm (SUS)		105.8	105.8	105.8	105.8	105.8	105.8	105.8	105.8
Long. Pl (SUS)		-2.4	-2.4	15.4	15.4	-2.4	-2.4	15.4	15.4
Long. Q (SUS)		-19.3	19.3	57.3	-57.3	-19.3	19.3	57.3	-57.3
Shear Pm (SUS)		0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Shear Pl (SUS)		2.5	2.5	-2.5	-2.5	-2.5	-2.5	2.5	2.5
Shear Q (SUS)		4.6	4.6	4.6	4.6	4.6	4.6	4.6	4.6
Pm (SUS)		105.8	105.8	105.8	105.8	105.8	105.8	105.8	105.8
Pm+Pl (SUS)		104.3	104.3	157.4	157.4	104.3	104.3	157.4	157.4
Pm+Pl+Q (Total)		85.0	164.6	349.6	99.2	84.2	163.5	349.9	100.1

**Vessel Stress Summation Comparison (N./mm²):**

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	105.76	138.00	Passed
Pm+Pl (SUS)	157.37	207.00	Passed
Pm+Pl+Q (TOTAL)	349.90	413.99	Passed

*Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.*

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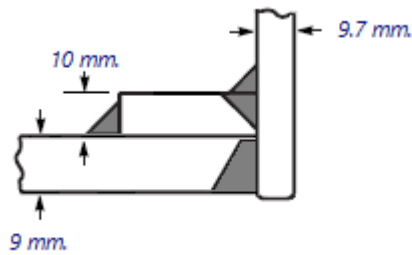
**Input, Nozzle Desc: D1-3"****From: 30**

Pressure for Reinforcement Calculations	P	7.410	bars
Temperature for Internal Pressure	Temp	230	°C
Design External Pressure	Pext	1.03	bars
Temperature for External Pressure	Tempex	198	°C
Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	138.00	N./mm2
Parent Allowable Stress At Ambient	Sva	138.00	N./mm2
Inside Diameter of Elliptical Head	D	1900.00	mm.
Aspect Ratio of Elliptical Head	Ar	2.00	
Head Finished (Minimum) Thickness	t	9.0000	mm.
Head Internal Corrosion Allowance	cas	3.0000	mm.
Head External Corrosion Allowance	co	0.0000	mm.
Distance from Head Centerline	L1	200.0000	mm.
User Entered Minimum Design Metal Temperature		-1.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-106 B	
Material UNS Number		K03006	
Material Specification/Type		Smls. pipe	
Allowable Stress at Temperature	Sn	118.00	N./mm2
Allowable Stress At Ambient	Sna	118.00	N./mm2
Diameter Basis (for tr calc only)		Outside	
Layout Angle		90.00	deg
Diameter		3.0000	in.
Size and Thickness Basis		Minimum	
Nominal Thickness		160	
Flange Material		SA-105	
Flange Type		Weld Neck Flange	
Corrosion Allowance	can	1.6000	mm.
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	200.0000	mm.
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm.
Groove weld depth between Nozzle and Vessel	Wgnv	9.0000	mm.
Inside Projection	h	0.0000	mm.
Weld leg size, Inside Element to Shell	Wi	0.0000	mm.
Pad Material		SA-516 70	
Pad Allowable Stress at Temperature	Sp	138.00	N./mm2
Pad Allowable Stress At Ambient	Spa	138.00	N./mm2
Diameter of Pad along vessel surface	Dp	190.0000	mm.
Thickness of Pad	te	10.0000	mm.
Weld leg size between Pad and Shell	Wp	8.0000	mm.
Groove weld depth between Pad and Nozzle	Wgpn	10.0000	mm.
Reinforcing Pad Width		50.5500	mm.
Flange Class		300	
Flange Grade		GR 1.1	
Piping Exit Angle [North Clockwise reference]		90.0	deg
Bend Radius Multiplier		1.50	
Horizontal Run Length		1100.0	mm.
Centerline Distance to Tangent		800.0	mm.

The Pressure Design option was MAWP + static head (to the nozzle).



Note : Checking Nozzle in the Meridional direction.

Reinforcement CALCULATION, Description: D1-3"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Outside Diameter Used in Calculation	3.500 in.
Actual Thickness Used in Calculation	0.383 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Elliptical Head, tr [Int. Press]
 $= P \cdot D \cdot K1 / (2 \cdot Sv \cdot E - 0.2 \cdot P)$ per Appendix 1-4(c)
 $= 7.41 \cdot 1906.0 \cdot 0.8973 / (2 \cdot 138.0 \cdot 1.0 - 0.2 \cdot 7.41)$
 $= 4.5944 \text{ mm.}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot Ro / (Sn \cdot E + 0.4 \cdot P)$ per Appendix 1-1 (a)(1)
 $= 7.41 \cdot 44.45 / (118 \cdot 1.0 + 0.4 \cdot 7.41)$
 $= 0.2784 \text{ mm.}$

Required Nozzle thickness under External Pressure per UG-28 : 0.4040 mm.

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	146.1078 mm.
Parallel to Vessel Wall, opening length	d	73.0539 mm.
Normal to Vessel Wall (Thickness Limit), pad side Tlwp		15.0000 mm.

The Pad diameter is greater than the Diameter Limit. The excess will not be considered.

*Taking a UG-36(c)(3)(a) exemption for nozzle: D1-3".
 This calculation is valid for nozzles that meet all the requirements of paragraph UG-36. Please check the Code carefully, especially for nozzles that are not isolated or do not meet Code spacing requirements. To force the computation of areas for small nozzles go to Tools->Configuration and check the box to force the UG-37 small nozzle area calculation.*

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, min(Curve:B, Curve:A)

Govrn. thk, tg = 9.735, tr = 0.278, c = 1.6 mm., E* = 1.0
 Thickness Ratio = $tr \cdot E* / (tg - c) = 0.034$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve A	-8 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Nozzle Neck to Pad Weld for the Nozzle, Curve: B

Govrn. thk, tg = 9.735, tr = 0.278, c = 1.6 mm., E* = 1.0
 Thickness Ratio = $tr \cdot E* / (tg - c) = 0.034$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-29 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C

Nozzle Neck to Pad Weld for Reinforcement pad, Curve: B



Govrn. thk, $t_g = 9.735$, $t_r = 0.278$, $c = 1.6$ mm., $E^* = 1.0$
 Thickness Ratio = $t_r * E^* / (t_g - c) = 0.034$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -29 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -104 °C

Shell to Pad Weld Junction at Pad OD, Curve: B

Govrn. thk, $t_g = 9.0$, $t_r = 4.594$, $c = 3.0$ mm., $E^* = 1.0$
 Thickness Ratio = $t_r * E^* / (t_g - c) = 0.766$, Temp. Reduction = 13 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -29 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -42 °C

Nozzle-Shell/Head Weld (UCS-66(a)(b)), Curve: B

Govrn. thk, $t_g = 9.0$, $t_r = 4.594$, $c = 3.0$ mm., $E^* = 1.0$
 Thickness Ratio = $t_r * E^* / (t_g - c) = 0.766$, Temp. Reduction = 13 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -29 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -42 °C

Gov. MDMT of the Nozzle : -42 °C
 Gov. MDMT of the Reinforcement Pad : -42 °C
 Gov. MDMT of the nozzle to shell joint welded assembly : -42 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c) -18 °C
 Flange MDMT with Temp reduction per UCS-66(b)(1)(-b) -104 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
 Design Pressure/Ambient Rating = $7.41/51.00 = 0.145$

Weld Size Calculations, Description: D1-3"

Intermediate Calc. for nozzle/shell Welds T_{min} 8.1346 mm.
 Intermediate Calc. for pad/shell Welds T_{minPad} 9.0000 mm.

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	$5.6942 = 0.7 * t_{min}$	$7.0700 = 0.7 * W_o$ mm.
Pad Weld	$4.5000 = 0.5 * T_{minPad}$	$5.6560 = 0.7 * W_p$ mm.

Skipping the nozzle attachment weld strength calculations. Per UW-15(b)(2) the nozzles exempted by UG-36(c)(3)(a) (small nozzles) do not require a weld strength check.

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 7 bars

Note: The MAWP of this junction was limited by the parent Shell/Head.

Note : Checking Nozzle in the Latitudinal direction.

Reinforcement CALCULATION, Description: D1-3"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Outside Diameter Used in Calculation 3.500 in.
 Actual Thickness Used in Calculation 0.383 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Elliptical Head, t_r [Int. Press]
 $= P \cdot D \cdot K1 / (2 \cdot S \cdot E - 0.2 \cdot P)$ per Appendix 1-4(c)
 $= 7.41 \cdot 1906.0 \cdot 0.8973 / (2 \cdot 138.0 \cdot 1.0 - 0.2 \cdot 7.41)$
 $= 4.5944$ mm.



Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot Ro / (Sn \cdot E + 0.4 \cdot P)$ per Appendix 1-1 (a)(1)
 $= 7.41 \cdot 44.45 / (118 \cdot 1.0 + 0.4 \cdot 7.41)$
 $= 0.2784$ mm.

Required Nozzle thickness under External Pressure per UG-28 : 0.4040 mm.

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	145.2618	mm.
Parallel to Vessel Wall, opening length	d	72.6309	mm.
Normal to Vessel Wall (Thickness Limit), pad side Tlwp		15.0000	mm.

The Pad diameter is greater than the Diameter Limit. The excess will not be considered.

Taking a UG-36(c)(3)(a) exemption for nozzle: D1-3".

This calculation is valid for nozzles that meet all the requirements of paragraph UG-36. Please check the Code carefully, especially for nozzles that are not isolated or do not meet Code spacing requirements. To force the computation of areas for small nozzles go to Tools->Configuration and check the box to force the UG-37 small nozzle area calculation.

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta	= 2.0040	mm.
Wall Thickness per UG16(b),	tr16b	= 5.5000	mm.
Wall Thickness, shell/head, internal pressure	trb1	= 8.0988	mm.
Wall Thickness	tb1 = max(trb1, tr16b)	= 8.0988	mm.
Wall Thickness, shell/head, external pressure	trb2	= 3.7113	mm.
Wall Thickness	tb2 = max(trb2, tr16b)	= 5.5000	mm.
Wall Thickness per table UG-45	tb3	= 6.4000	mm.

Determine Nozzle Thickness candidate [tb]:

$= \min[tb3, \max(tb1, tb2)]$
 $= \min[6.4, \max(8.0988, 5.5)]$
 $= 6.4000$ mm.

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

$= \max(ta, tb)$
 $= \max(2.004, 6.4)$
 $= 6.4000$ mm.

Available Nozzle Neck Thickness = 9.7345 mm. -> OK

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained	:	31.7,	Allowable	:	118.0 N./mm ²	Passed
Expansion	:	0.0,	Allowable	:	263.3 N./mm ²	Passed
Occasional	:	1.5,	Allowable	:	156.9 N./mm ²	Passed
Shear	:	25.1,	Allowable	:	82.6 N./mm ²	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Weld Size Calculations, Description: D1-3"

Intermediate Calc. for nozzle/shell Welds	Tmin	8.1346	mm.
Intermediate Calc. for pad/shell Welds	TminPad	9.0000	mm.

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	$5.6942 = 0.7 \cdot t_{min}$	$7.0700 = 0.7 \cdot W_o$ mm.
Pad Weld	$4.5000 = 0.5 \cdot T_{minPad}$	$5.6560 = 0.7 \cdot W_p$ mm.

Skipping the nozzle attachment weld strength calculations. Per UW-15(b)(2) the nozzles exempted by UG-36(c)(3)(a) (small nozzles) do not require a weld strength check.

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 7 bars



Note: The MAWP of this junction was limited by the parent Shell/Head.

The Drop for this Nozzle is : 5.8016 mm.

The Cut Length for this Nozzle is, Drop + Ho + H + T : 214.8630 mm.

Warning:

The equations for nozzle local stress analysis pertain to radial nozzles only and this nozzle is not specified as radial. Please check the specific analysis code for all assumptions and limitations.

Input Echo, WRC107/537 Item 1, Description: D1-3" :

Diameter Basis for Vessel	Vbasis	ID	
Cylindrical or Spherical Vessel	Cylsph	Spherical	
Internal Corrosion Allowance	Cas	3.0000	mm.
Vessel Diameter	Dv	3420.000	mm.
Vessel Thickness	Tv	9.000	mm.
Design Temperature	T1	230.0	C
Vessel Material		SA-516 70	
Vessel UNS Number		K02700	
Vessel Cold S.I. Allowable	Smc	138.00	N./mm2
Vessel Hot S.I. Allowable	Smh	138.00	N./mm2

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Attachment Type	Type	Round	
WRC107 Attachment Classification	Holsol	Hollow	
Diameter Basis for Nozzle	Nbasis	OD	
Corrosion Allowance for Nozzle	Can	1.6000	mm.
Nozzle Diameter	Dn	88.900	mm.
Nozzle Thickness	Tn	9.735	mm.
Nozzle Material		SA-106 B	
Nozzle UNS Number		K03006	
Nozzle Cold S.I. Allowable	SNmc	118.00	N./mm2
Nozzle Hot S.I. Allowable	SNmh	118.00	N./mm2
Thickness of Reinforcing Pad	Tpad	10.000	mm.
Diameter of Reinforcing Pad	Dpad	190.000	mm.
Design Internal Pressure	Dp	7.410	bars
Include Pressure Thrust		No	

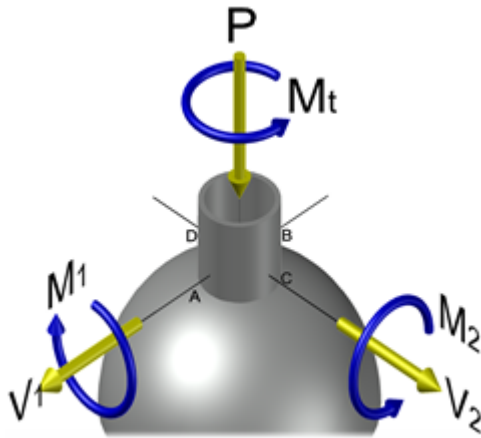
External Forces and Moments in WRC 107/537 Convention:

Radial Load (SUS)	P	-611.9	Kgf
Longitudinal Shear (SUS)	(V1) V1	407.9	Kgf
Circumferential Shear (SUS)	(Vc) V2	407.9	Kgf
Circumferential Moment (SUS)	(Mc) M1	91.3	Kg-m.
Longitudinal Moment (SUS)	(Ml) M2	91.3	Kg-m.
Torsional Moment (SUS)	Mt	130.5	Kg-m.

Use Interactive Control	No
WRC107 Version	Version March 1979

Include Pressure Stress Indices per Div. 2	No
Compute Pressure Stress per WRC-368	No
Local Loads applied at end of Nozzle/Attachment	No

WRC Bulletin 537 provides equations for the dimensionless curves found in bulletin 107. As noted in the foreword to bulletin 537, "537 is equivalent to WRC 107". Where 107 is printed in the results below, "537" can be interchanged with "107".



WRC 107/537 has dimensional limitations that can be found in the bulletin's Foreword and which can also be inferred by review of the non-dimensional curve limits.

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{(R_{\text{mean}}(t - c_a))}$$

$$= 88.9 + 2 \cdot 1.65 \cdot \sqrt{(1716.0 (9.0 - 3.0))}$$

$$= 423.748 \text{ mm.}$$

WRC 107 Stress Calculation for SUSTained loads:

Radial Load		P	-611.9	Kgf
Circumferential Shear	(VC)	V2	407.9	Kgf
Longitudinal Shear	(VL)	V1	407.9	Kgf
Circumferential Moment	(MC)	M1	91.3	Kg-m.
Longitudinal Moment	(ML)	M2	91.3	Kg-m.
Torsional Moment		MT	130.5	Kg-m.

Dimensionless Param: U = 0.27 TAU = 5.00 (4.96) RHO = 1.97

Dimensionless Loads for Spherical Shells at Attachment Junction:

Curves read for 1979	Figure	Value	Location
$N(x) * T / P$	SP 2	0.11717	(A,B,C,D)
$M(x) / P$	SP 2	0.08917	(A,B,C,D)
$N(x) * T * \text{SQRT}(R_m * T) / MC$	SM 2	0.32642	(A,B,C,D)
$M(x) * \text{SQRT}(R_m * T) / MC$	SM 2	0.28284	(A,B,C,D)
$N(x) * T * \text{SQRT}(R_m * T) / ML$	SM 2	0.32642	(A,B,C,D)
$M(x) * \text{SQRT}(R_m * T) / ML$	SM 2	0.28284	(A,B,C,D)
$N(y) * T / P$	SP 2	0.23096	(A,B,C,D)
$M(y) / P$	SP 2	0.08647	(A,B,C,D)
$N(y) * T * \text{SQRT}(R_m * T) / MC$	SM 2	0.08531	(A,B,C,D)
$M(y) * \text{SQRT}(R_m * T) / MC$	SM 2	0.41192	(A,B,C,D)
$N(y) * T * \text{SQRT}(R_m * T) / ML$	SM 2	0.08531	(A,B,C,D)
$M(y) * \text{SQRT}(R_m * T) / ML$	SM 2	0.41192	(A,B,C,D)

Stress Concentration Factors: Kn = 1.00, Kb = 1.00

Stresses in the Vessel at the Attachment Junction (N./mm2)

Type of Stress	Load	Stress Intensity Values at							
		Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Rad.	Memb. P	2.7	2.7	2.7	2.7	2.7	2.7	2.7	2.7
Rad.	Bend. P	12.5	-12.5	12.5	-12.5	12.5	-12.5	12.5	-12.5
Rad.	Memb. MC	0.0	0.0	0.0	0.0	-6.9	-6.9	6.9	6.9
Rad.	Bend. MC	0.0	0.0	0.0	0.0	-35.8	35.8	35.8	-35.8
Rad.	Memb. ML	-6.9	-6.9	6.9	6.9	0.0	0.0	0.0	0.0
Rad.	Bend. ML	-35.8	35.8	35.8	-35.8	0.0	0.0	0.0	0.0

Type of Stress Load		Stress Intensity Values at							
		Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Rad.	Memb. P	11.9	11.9	11.9	11.9	11.9	11.9	11.9	11.9
Rad.	Bend. P	37.7	-37.7	37.7	-37.7	37.7	-37.7	37.7	-37.7
Rad.	Memb. MC	0.0	0.0	0.0	0.0	-20.5	-20.5	20.5	20.5
Rad.	Bend. MC	0.0	0.0	0.0	0.0	-121.0	121.0	121.0	-121.0
Rad.	Memb. ML	-20.5	-20.5	20.5	20.5	0.0	0.0	0.0	0.0
Rad.	Bend. ML	-121.0	121.0	121.0	-121.0	0.0	0.0	0.0	0.0
Tot. Rad. Str.		-91.9	74.7	191.1	-126.4	-91.9	74.7	191.1	-126.4
Tang. Memb. P		3.6	3.6	3.6	3.6	3.6	3.6	3.6	3.6
Tang. Bend. P		11.7	-11.7	11.7	-11.7	11.7	-11.7	11.7	-11.7
Tang. Memb. MC		0.0	0.0	0.0	0.0	-6.1	-6.1	6.1	6.1
Tang. Bend. MC		0.0	0.0	0.0	0.0	-36.5	36.5	36.5	-36.5
Tang. Memb. ML		-6.1	-6.1	6.1	6.1	0.0	0.0	0.0	0.0
Tang. Bend. ML		-36.5	36.5	36.5	-36.5	0.0	0.0	0.0	0.0
Tot. Tang. Str.		-27.2	22.3	57.9	-38.5	-27.2	22.3	57.9	-38.5
Shear VC		2.2	2.2	-2.2	-2.2	0.0	0.0	0.0	0.0
Shear VL		0.0	0.0	0.0	0.0	-2.2	-2.2	2.2	2.2
Shear MT		3.8	3.8	3.8	3.8	3.8	3.8	3.8	3.8



Tot. Shear	6.0	6.0	1.5	1.5	1.5	1.5	6.0	6.0
Str. Int.	92.5	75.4	191.1	126.5	92.0	74.8	191.4	126.8

WRC 107/537 Stress Summations:**Vessel Stress Summation at Attachment Junction (N./mm²)**

Type of Stress	Load	Stress Intensity Values at							
		Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Rad. Pm (SUS)		39.7	39.7	39.7	39.7	39.7	39.7	39.7	39.7
Rad. Pl (SUS)		-4.1	-4.1	9.6	9.6	-4.1	-4.1	9.6	9.6
Rad. Q (SUS)		-23.2	23.2	48.3	-48.3	-23.2	23.2	48.3	-48.3
Long. Pm (SUS)		39.7	39.7	39.7	39.7	39.7	39.7	39.7	39.7
Long. Pl (SUS)		3.6	3.6	7.2	7.2	3.6	3.6	7.2	7.2
Long. Q (SUS)		-40.0	40.0	64.3	-64.3	-40.0	40.0	64.3	-64.3
Shear Pm (SUS)		0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Shear Pl (SUS)		1.8	1.8	-1.8	-1.8	-1.8	-1.8	1.8	1.8
Shear Q (SUS)		6.4	6.4	6.4	6.4	6.4	6.4	6.4	6.4
Pm (SUS)		39.7	39.7	39.7	39.7	39.7	39.7	39.7	39.7
Pm+Pl (SUS)		43.7	43.7	50.3	50.3	43.7	43.7	50.3	50.3
Pm+Pl+Q (Total)		18.7	85.7	112.6	20.6	14.3	84.1	115.0	24.7

Vessel Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	39.67	138.00	Passed
Pm+Pl (SUS)	50.25	207.00	Passed
Pm+Pl+Q (TOTAL)	115.05	413.99	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 107/537 Stress Summations:**Vessel Stress Summation at Reinforcing Pad Edge (N./mm²)**

Type of Stress	Load	Stress Intensity Values at							
		Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Rad. Pm (SUS)		105.8	105.8	105.8	105.8	105.8	105.8	105.8	105.8
Rad. Pl (SUS)		-8.6	-8.6	32.3	32.3	-8.6	-8.6	32.3	32.3
Rad. Q (SUS)		-83.3	83.3	158.8	-158.8	-83.3	83.3	158.8	-158.8
Long. Pm (SUS)		105.8	105.8	105.8	105.8	105.8	105.8	105.8	105.8
Long. Pl (SUS)		-2.5	-2.5	9.7	9.7	-2.5	-2.5	9.7	9.7
Long. Q (SUS)		-24.8	24.8	48.2	-48.2	-24.8	24.8	48.2	-48.2
Shear Pm (SUS)		0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Shear Pl (SUS)		2.2	2.2	-2.2	-2.2	-2.2	-2.2	2.2	2.2
Shear Q (SUS)		3.8	3.8	3.8	3.8	3.8	3.8	3.8	3.8
Pm (SUS)		105.8	105.8	105.8	105.8	105.8	105.8	105.8	105.8
Pm+Pl (SUS)		104.0	104.0	138.3	138.3	104.0	104.0	138.3	138.3
Pm+Pl+Q (Total)		79.1	181.2	296.9	87.9	78.6	180.5	297.2	88.7

**Vessel Stress Summation Comparison (N./mm²):**

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	105.78	138.00	Passed
Pm+Pl (SUS)	138.34	207.00	Passed
Pm+Pl+Q (TOTAL)	297.16	413.99	Passed

*Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.*

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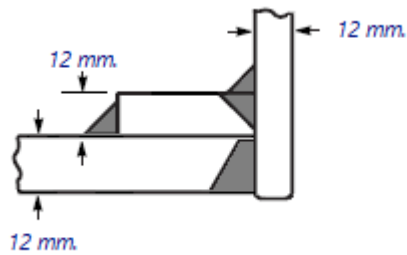
**Input, Nozzle Desc: M1-24"****From: 40**

Pressure for Reinforcement Calculations	P	7.223	bars
Temperature for Internal Pressure	Temp	230	°C
Design External Pressure	Pext	1.03	bars
Temperature for External Pressure	Tempex	198	°C
Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	138.00	N./mm2
Parent Allowable Stress At Ambient	Sva	138.00	N./mm2
Inside Diameter of Cylindrical Shell	D	1900.00	mm.
Design Length of Section	L	2408.3333	mm.
Shell Finished (Minimum) Thickness	t	12.0000	mm.
Shell Internal Corrosion Allowance	cas	3.0000	mm.
Shell External Corrosion Allowance	co	0.0000	mm.
Distance from Bottom/Left Tangent		1550.00	mm.
User Entered Minimum Design Metal Temperature		-1.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-516 70	
Material UNS Number		K02700	
Material Specification/Type		Plate	
Allowable Stress at Temperature	Sn	138.00	N./mm2
Allowable Stress At Ambient	Sna	138.00	N./mm2
Diameter Basis (for tr calc only)		Outside	
Layout Angle		90.00	deg
Diameter		24.0000	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	12.0000	mm.
Flange Material		SA-105	
Flange Type		Weld Neck Flange	
Corrosion Allowance	can	3.0000	mm.
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	300.0000	mm.
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm.
Groove weld depth between Nozzle and Vessel	Wgnv	9.0000	mm.
Inside Projection	h	0.0000	mm.
Weld leg size, Inside Element to Shell	Wi	0.0000	mm.
Pad Material		SA-516 70	
Pad Allowable Stress at Temperature	Sp	138.00	N./mm2
Pad Allowable Stress At Ambient	Spa	138.00	N./mm2
Diameter of Pad along vessel surface	Dp	909.6000	mm.
Thickness of Pad	te	12.0000	mm.
Weld leg size between Pad and Shell	Wp	8.0000	mm.
Groove weld depth between Pad and Nozzle	Wgpn	12.0000	mm.
Reinforcing Pad Width		150.0000	mm.
This is a Manway, Coupling or Access Opening.			
Flange Class		150	
Flange Grade		GR 1.1	

The Pressure Design option was MAWP + static head (to the nozzle).

**Reinforcement CALCULATION, Description: M1-24"**

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Outside Diameter Used in Calculation	24.000 in.
Actual Thickness Used in Calculation	0.472 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 7.22 \cdot 953.0 / (138 \cdot 1.0 - 0.6 \cdot 7.22)$
 $= 5.0042 \text{ mm.}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R_o / (S_n \cdot E + 0.4 \cdot P)$ per Appendix 1-1 (a)(1)
 $= 7.22 \cdot 304.8 / (138 \cdot 1.0 + 0.4 \cdot 7.22)$
 $= 1.5921 \text{ mm.}$

Required Nozzle thickness under External Pressure per UG-28 : 1.4766 mm.

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	1183.2001 mm.
Parallel to Vessel Wall, opening length	d	591.6000 mm.
Normal to Vessel Wall (Thickness Limit), pad side Tlwp		22.5000 mm.

Weld Strength Reduction Factor [fr1]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 138.0 / 138.0)$
 $= 1.000$

Weld Strength Reduction Factor [fr2]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 138.0 / 138.0)$
 $= 1.000$

Weld Strength Reduction Factor [fr4]:
 $= \min(1, S_p / S_v)$
 $= \min(1, 138.0 / 138.0)$
 $= 1.000$

Weld Strength Reduction Factor [fr3]:
 $= \min(fr2, fr4)$
 $= \min(1.0, 1.0)$
 $= 1.000$

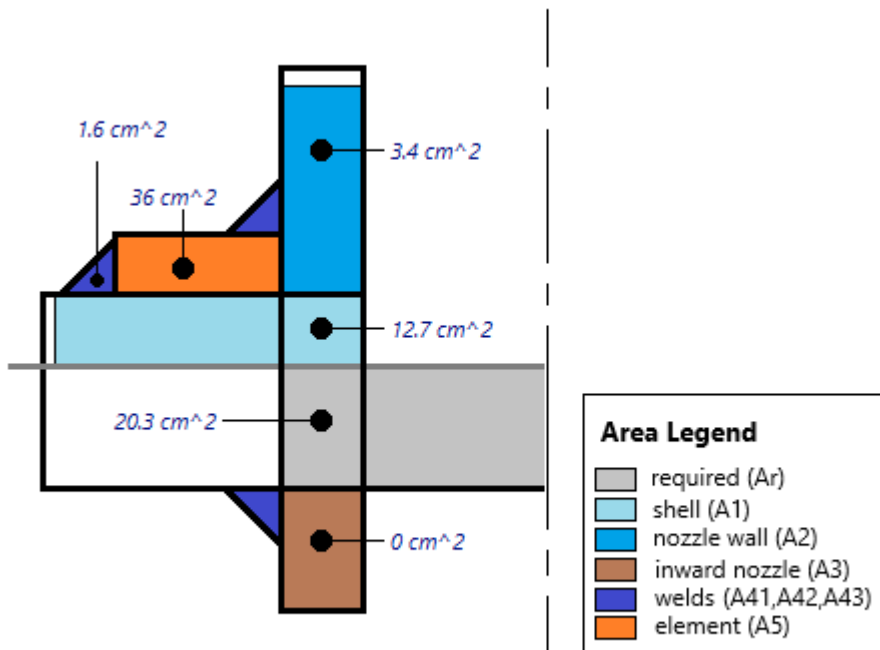


Figure UG 37.1 Nozzle Areas

Results of Nozzle Reinforcement Area Calculations: (cm²)

AREA AVAILABLE, A1 to A5		MAWP	External	Mapnc
Area Required	Ar	29.605	20.286	NA
Area in Shell	A1	23.639	12.672	NA
Area in Nozzle Wall	A2	3.334	3.386	NA
Area in Inward Nozzle	A3	0.000	0.000	NA
Area in Welds A41+A42+A43		1.640	1.640	NA
Area in Element	A5	36.000	36.000	NA
TOTAL AREA AVAILABLE	Atot	64.613	53.698	NA

The External Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Insufficient.
 The area available with the given pad is Sufficient.

SELECTION OF POSSIBLE REINFORCING PADS:	Diameter	Thickness
Based on given Pad Thickness:	631.1688	12.0000 mm.
Based on given Pad Diameter:	909.6000	0.8628 mm.
Based on Shell or Nozzle Thickness:	631.1688	12.0000 mm.

Area Required [A]:
 $= 0.5(d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F(1 - fr1))$ per UG-37(d)
 $= 0.5(591.6 \cdot 6.858 \cdot 1 + 2 \cdot 9.0 \cdot 6.858 \cdot 1(1 - 1.0))$
 $= 20.286 \text{ cm}^2$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:
 $= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$
 $= 591.6(1.0 \cdot 9.0 - 1.0 \cdot 6.858) - 2 \cdot 9.0$
 $(1.0 \cdot 9.0 - 1.0 \cdot 6.858) \cdot (1 - 1.0)$
 $= 12.672 \text{ cm}^2$

Area Available in Nozzle Wall Projecting Outward [A2]:
 $= (2 \cdot Tlwp) \cdot (tn - trn) \cdot fr2$



$$= (2 \cdot 22.5) \cdot (9.0 - 1.48) \cdot 1.0$$

$$= 3.386 \text{ cm}^2$$

Area Available in Welds [A41 + A42 + A43]:

$$= W_o^2 \cdot fr_3 + (W_i - can/0.707)^2 \cdot fr_2 + W_p^2 \cdot fr_4$$

$$= 10.0^2 \cdot 1.0 + (0.0)^2 \cdot 1.0 + 8.0^2 \cdot 1.0$$

$$= 1.640 \text{ cm}^2$$

Area Available in Element [A5]:

$$= (\min(D_p, DL) - (\text{Nozzle OD})) \cdot (\min(t_p, T_{lp}, t_e)) \cdot fr_4$$

$$= (909.6 - 609.6) \cdot 12.0 \cdot 1.0$$

$$= 36.000 \text{ cm}^2$$

UG-45 Minimum Nozzle Neck Thickness Requirement:

Wall Thickness for Internal/External pressures $t_a = 4.5921 \text{ mm}$.

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

$$= t_a$$

$$= 4.5921 \text{ mm}$$

Available Nozzle Neck Thickness = 12.0000 mm. -> OK

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:**Nozzle Neck to Flange Weld, min(Curve:B, Curve:A)**

Govrn. thk, $t_g = 12.0$, $t_r = 1.592$, $c = 3.0 \text{ mm.}$, $E^* = 1.0$
 Thickness Ratio = $t_r \cdot E^* / (t_g - c) = 0.177$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve A	-3 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Nozzle Neck to Pad Weld for the Nozzle, Curve: B

Govrn. thk, $t_g = 12.0$, $t_r = 1.592$, $c = 3.0 \text{ mm.}$, $E^* = 1.0$
 Thickness Ratio = $t_r \cdot E^* / (t_g - c) = 0.177$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-23 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Nozzle Neck to Pad Weld for Reinforcement pad, Curve: B

Govrn. thk, $t_g = 12.0$, $t_r = 1.592$, $c = 3.0 \text{ mm.}$, $E^* = 1.0$
 Thickness Ratio = $t_r \cdot E^* / (t_g - c) = 0.177$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-23 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Shell to Pad Weld Junction at Pad OD, Curve: B

Govrn. thk, $t_g = 12.0$, $t_r = 5.004$, $c = 3.0 \text{ mm.}$, $E^* = 1.0$
 Thickness Ratio = $t_r \cdot E^* / (t_g - c) = 0.556$, Temp. Reduction = 26 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-23 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-48 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Nozzle-Shell/Head Weld (UCS-66(a)1(b)), Curve: B

Govrn. thk, $t_g = 12.0$, $t_r = 5.004$, $c = 3.0 \text{ mm.}$, $E^* = 1.0$
 Thickness Ratio = $t_r \cdot E^* / (t_g - c) = 0.556$, Temp. Reduction = 26 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-23 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-48 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C



Gov. MDMT of the Nozzle : -48 °C
 Gov. MDMT of the Reinforcement Pad : -48 °C
 Gov. MDMT of the nozzle to shell joint welded assembly : -48 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c) -18 °C
 Flange MDMT with Temp reduction per UCS-66(b)(1)(-b) -48 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
 Design Pressure/Ambient Rating = $7.22/19.60 = 0.369$

Weld Size Calculations, Description: M1-24"

Intermediate Calc. for nozzle/shell Welds T_{min} 9.0000 mm.
 Intermediate Calc. for pad/shell Welds T_{minPad} 9.0000 mm.

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	$6.0000 = \text{Min per Code}$	$7.0700 = 0.7 * W_o \text{ mm.}$
Pad Weld	$4.5000 = 0.5 * T_{minPad}$	$5.6560 = 0.7 * W_p \text{ mm.}$

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:
 $= \max(0, (A - A1 + 2 \cdot t_n \cdot f_{r1} \cdot (E1 \cdot t - t_r)) \cdot S_v)$
 $= \max(0, (20.2859 - 12.6722 + 2 \cdot 9.0 \cdot 1.0 \cdot (1.0 \cdot 9.0 - 6.858)) \cdot 138)$
 $= 11256.32 \text{ Kgf}$

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:
 $= (A2 + A5 + A4 - (W_i - \text{Can} / .707)^2 \cdot f_{r2}) \cdot S_v$
 $= (3.3855 + 36.0 + 1.64 - 0.0 \cdot 1.0) \cdot 138$
 $= 57729.36 \text{ Kgf}$

Weld Load [W2]:
 $= (A2 + A3 + A4 + (2 \cdot t_n \cdot t \cdot f_{r1})) \cdot S_v$
 $= (3.3855 + 0.0 + 1. + (1.62)) \cdot 138$
 $= 8450.72 \text{ Kgf}$

Weld Load [W3]:
 $= (A2 + A3 + A4 + A5 + (2 \cdot t_n \cdot t \cdot f_{r1})) \cdot S$
 $= (3.3855 + 0.0 + 1.64 + 36.0 + (1.62)) \cdot 138$
 $= 60008.95 \text{ Kgf}$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:
 $= (\pi/2) \cdot D_{lo} \cdot W_o \cdot 0.49 \cdot S_{nw}$
 $= (3.1416/2.0) \cdot 609.6 \cdot 10.0 \cdot 0.49 \cdot 138$
 $= 66024. \text{ Kgf}$

Shear, Pad Element Weld [Spew]:
 $= (\pi/2) \cdot D_P \cdot W_P \cdot 0.49 \cdot S_{EW}$
 $= (3.1416/2.0) \cdot 909.6 \cdot 8.0 \cdot 0.49 \cdot 138$
 $= 78813. \text{ Kgf}$

Shear, Nozzle Wall [Snw]:
 $= (\pi \cdot (D_{lr} + D_{lo}) / 4) \cdot (Thk - \text{Can}) \cdot 0.7 \cdot S_n$
 $= (3.1416 \cdot 300.3) \cdot (12.0 - 3.0) \cdot 0.7 \cdot 138$
 $= 83635. \text{ Kgf}$

Tension, Pad Groove Weld [Tpgw]:
 $= (\pi/2) \cdot D_{lo} \cdot W_{gpn} \cdot 0.74 \cdot S_{eg}$
 $= (3.1416/2) \cdot 609.6 \cdot 12.0 \cdot 0.74 \cdot 138$
 $= 119652. \text{ Kgf}$



Tension, Shell Groove Weld [Tngw]:

$$= (\pi/2) \cdot D_o \cdot (T - c_{as}) \cdot 0.74 \cdot S_{ng}$$

$$= (3.1416/2.0) \cdot 609.6 \cdot (12.0 - 3.0) \cdot 0.74 \cdot 138$$

$$= 89739. \text{ Kg}$$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:

$$= (\text{SPEW} + \text{SNW}) = (78813 + 83635) = 162448 \text{ Kg}$$

Failure Path [Path 2-2]:

$$= (\text{Sonw} + \text{Tpgw} + \text{Tngw} + \text{Sinw})$$

$$= (66024 + 119652 + 89739 + 0) = 275415 \text{ Kg}$$

Failure Path [Path 3-3]:

$$= (\text{Spew} + \text{Tngw} + \text{Sinw})$$

$$= (78813 + 89739 + 0) = 168552 \text{ Kg}$$

Summary of Failure Path Calculations:

Path 1-1 = 162448 Kg, must exceed W = 11256 Kg	or W1 = 57729 Kg
Path 2-2 = 275415 Kg, must exceed W = 11256 Kg	or W2 = 8450 Kg
Path 3-3 = 168552 Kg, must exceed W = 11256 Kg	or W3 = 60008 Kg

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 11 bars

Note: The MAWP of this junction was limited by the parent Shell/Head.

Nozzle is O.K. for the External Pressure: 1 bars

The Drop for this Nozzle is : 50.2240 mm.

The Cut Length for this Nozzle is, Drop + Ho + H + T : 362.2240 mm.

Percent Elongation Calculations:

% Elongation per Table UG-79-1 ($50 \cdot t_{nom} / R_f \cdot (1 - R_f / R_o)$) 2.008 %

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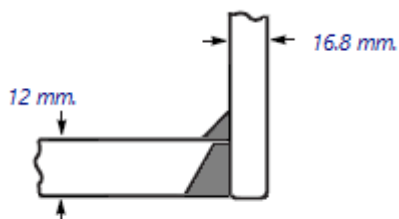
**Input, Nozzle Desc: K1-2"****From: 40**

Pressure for Reinforcement Calculations	P	7.218	bars
Temperature for Internal Pressure	Temp	230	°C
Design External Pressure	Pext	1.03	bars
Temperature for External Pressure	Tempex	198	°C
Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	138.00	N./mm2
Parent Allowable Stress At Ambient	Sva	138.00	N./mm2
Inside Diameter of Cylindrical Shell	D	1900.00	mm.
Design Length of Section	L	2408.3333	mm.
Shell Finished (Minimum) Thickness	t	12.0000	mm.
Shell Internal Corrosion Allowance	cas	3.0000	mm.
Shell External Corrosion Allowance	co	0.0000	mm.
Distance from Bottom/Left Tangent		1600.00	mm.
User Entered Minimum Design Metal Temperature		-1.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-105	
Material UNS Number		K03504	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	136.80	N./mm2
Allowable Stress At Ambient	Sna	138.00	N./mm2
Diameter Basis (for tr calc only)		Inside	
Layout Angle		270.00	deg
Diameter		2.0000	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	16.7640	mm.
Flange Material		SA-105	
Flange Type		Long Weld Neck	
Corrosion Allowance	can	3.0000	mm.
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	206.3750	mm.
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm.
Groove weld depth between Nozzle and Vessel	Wgnv	9.0000	mm.
Inside Projection	h	0.0000	mm.
Weld leg size, Inside Element to Shell	Wi	0.0000	mm.
Flange Class		300	
Flange Grade		GR 1.1	

The Pressure Design option was MAWP + static head (to the nozzle).

**Reinforcement CALCULATION, Description: K1-2"**



ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation 2.000 in.
 Actual Thickness Used in Calculation 0.660 in.

Messages regarding the Nozzle Input Data:

For this nozzle, the actual wall thickness was specified. In these cases, the weight of the attached nozzle flange cannot be determined. It is required to manually compute the weight of the assembly and enter it into the nozzle dialog's overriding weight field.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 7.22 \cdot 953.0 / (138 \cdot 1.0 - 0.6 \cdot 7.22)$
 $= 5.0009$ mm.

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R / (S_n \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 7.22 \cdot 28.4 / (137 \cdot 1.0 - 0.6 \cdot 7.22)$
 $= 0.1503$ mm.

Required Nozzle thickness under External Pressure per UG-28 : 0.3975 mm.

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	113.6000	mm.
Parallel to Vessel Wall, opening length	d	56.8000	mm.
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	22.5000	mm.

*Taking a UG-36(c)(3)(a) exemption for nozzle: K1-2".
 This calculation is valid for nozzles that meet all the requirements of paragraph UG-36. Please check the Code carefully, especially for nozzles that are not isolated or do not meet Code spacing requirements. To force the computation of areas for small nozzles go to Tools->Configuration and check the box to force the UG-37 small nozzle area calculation.*

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta	= 3.3975	mm.
Wall Thickness per UG16(b),	tr16b	= 5.5000	mm.
Wall Thickness, shell/head, internal pressure	trb1	= 8.0009	mm.
Wall Thickness	tb1 = max(trb1, tr16b)	= 8.0009	mm.
Wall Thickness, shell/head, external pressure	trb2	= 3.7146	mm.
Wall Thickness	tb2 = max(trb2, tr16b)	= 5.5000	mm.
Wall Thickness per table UG-45	tb3	= 7.8000	mm.

Determine Nozzle Thickness candidate [tb]:
 $= \min[tb3, \max(tb1, tb2)]$
 $= \min[7.8, \max(8.0009, 5.5)]$
 $= 7.8000$ mm.

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
 $= \max(ta, tb)$
 $= \max(3.3975, 7.8)$
 $= 7.8000$ mm.

Available Nozzle Neck Thickness = 16.7640 mm. -> OK

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle-Shell/Head Weld (UCS-66(a)1(b)), min(Curve:A, Curve:B)

Govrn. thk, tg = 12.0, tr = 5.001, c = 3.0 mm., E* = 1.0
 Thickness Ratio = $tr \cdot E^* / (tg - c) = 0.556$, Temp. Reduction = 26 °C

Min Metal Temp. w/o impact per UCS-66, Curve A	-3 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-29 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C



Gov. MDMT of the nozzle to shell joint welded assembly : -29 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c) -18 °C
Flange MDMT with Temp reduction per UCS-66(b)(1)(-b) -104 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
Design Pressure/Ambient Rating = $7.22/51.00 = 0.142$

Weld Size Calculations, Description: K1-2"

Intermediate Calc. for nozzle/shell Welds Tmin 9.0000 mm.

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	$6.0000 = \text{Min per Code}$	$7.0700 = 0.7 * W_o \text{ mm.}$

Skipping the nozzle attachment weld strength calculations. Per UW-15(b)(2) the nozzles exempted by UG-36(c)(3)(a) (small nozzles) do not require a weld strength check.

The Drop for this Nozzle is : 0.9362 mm.
The Cut Length for this Nozzle is, Drop + Ho + H + T : 219.3112 mm.

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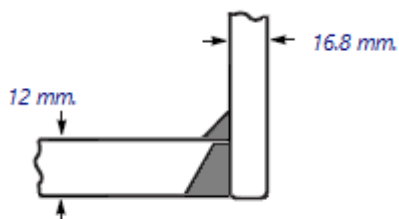
**Input, Nozzle Desc: K2-2"****From: 40**

Pressure for Reinforcement Calculations	P	7.340	bars
Temperature for Internal Pressure	Temp	230	°C
Design External Pressure	Pext	1.03	bars
Temperature for External Pressure	Tempex	198	°C
Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	138.00	N./mm2
Parent Allowable Stress At Ambient	Sva	138.00	N./mm2
Inside Diameter of Cylindrical Shell	D	1900.00	mm.
Design Length of Section	L	2408.3333	mm.
Shell Finished (Minimum) Thickness	t	12.0000	mm.
Shell Internal Corrosion Allowance	cas	3.0000	mm.
Shell External Corrosion Allowance	co	0.0000	mm.
Distance from Bottom/Left Tangent		300.00	mm.
User Entered Minimum Design Metal Temperature		-1.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-105	
Material UNS Number		K03504	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	136.80	N./mm2
Allowable Stress At Ambient	Sna	138.00	N./mm2
Diameter Basis (for tr calc only)		Inside	
Layout Angle		270.00	deg
Diameter		2.0000	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	16.7640	mm.
Flange Material		SA-105	
Flange Type		Long Weld Neck	
Corrosion Allowance	can	3.0000	mm.
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	206.3750	mm.
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm.
Groove weld depth between Nozzle and Vessel	Wgnv	9.0000	mm.
Inside Projection	h	0.0000	mm.
Weld leg size, Inside Element to Shell	Wi	0.0000	mm.
Flange Class		300	
Flange Grade		GR 1.1	

The Pressure Design option was MAWP + static head (to the nozzle).

**Reinforcement CALCULATION, Description: K2-2"**



ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation 2.000 in.
 Actual Thickness Used in Calculation 0.660 in.

Messages regarding the Nozzle Input Data:

For this nozzle, the actual wall thickness was specified. In these cases, the weight of the attached nozzle flange cannot be determined. It is required to manually compute the weight of the assembly and enter it into the nozzle dialog's overriding weight field.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 7.34 \cdot 953.0 / (138 \cdot 1.0 - 0.6 \cdot 7.34)$
 $= 5.0858$ mm.

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R / (S_n \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 7.34 \cdot 28.4 / (137 \cdot 1.0 - 0.6 \cdot 7.34)$
 $= 0.1529$ mm.

Required Nozzle thickness under External Pressure per UG-28 : 0.3975 mm.

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	113.6000	mm.
Parallel to Vessel Wall, opening length	d	56.8000	mm.
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	22.5000	mm.

*Taking a UG-36(c)(3)(a) exemption for nozzle: K2-2".
 This calculation is valid for nozzles that meet all the requirements of paragraph UG-36. Please check the Code carefully, especially for nozzles that are not isolated or do not meet Code spacing requirements. To force the computation of areas for small nozzles go to Tools->Configuration and check the box to force the UG-37 small nozzle area calculation.*

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta	= 3.3975	mm.
Wall Thickness per UG16(b),	tr16b	= 5.5000	mm.
Wall Thickness, shell/head, internal pressure	trb1	= 8.0858	mm.
Wall Thickness	tb1 = max(trb1, tr16b)	= 8.0858	mm.
Wall Thickness, shell/head, external pressure	trb2	= 3.7146	mm.
Wall Thickness	tb2 = max(trb2, tr16b)	= 5.5000	mm.
Wall Thickness per table UG-45	tb3	= 7.8000	mm.

Determine Nozzle Thickness candidate [tb]:
 $= \min[tb3, \max(tb1, tb2)]$
 $= \min[7.8, \max(8.0858, 5.5)]$
 $= 7.8000$ mm.

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
 $= \max(ta, tb)$
 $= \max(3.3975, 7.8)$
 $= 7.8000$ mm.

Available Nozzle Neck Thickness = 16.7640 mm. -> OK

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle-Shell/Head Weld (UCS-66(a)1(b)), min(Curve:A, Curve:B)

Govrn. thk, tg = 12.0, tr = 5.086, c = 3.0 mm., E* = 1.0
 Thickness Ratio = $tr \cdot E^* / (tg - c) = 0.565$, Temp. Reduction = 25 °C

Min Metal Temp. w/o impact per UCS-66, Curve A	-3 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-28 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C



Gov. MDMT of the nozzle to shell joint welded assembly : -29 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c) -18 °C
Flange MDMT with Temp reduction per UCS-66(b)(1)(-b) -104 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
Design Pressure/Ambient Rating = $7.34/51.00 = 0.144$

Weld Size Calculations, Description: K2-2"

Intermediate Calc. for nozzle/shell Welds Tmin 9.0000 mm.

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	$6.0000 = \text{Min per Code}$	$7.0700 = 0.7 * W_o \text{ mm.}$

Skipping the nozzle attachment weld strength calculations. Per UW-15(b)(2) the nozzles exempted by UG-36(c)(3)(a) (small nozzles) do not require a weld strength check.

The Drop for this Nozzle is : 0.9362 mm.
The Cut Length for this Nozzle is, Drop + Ho + H + T : 219.3112 mm.

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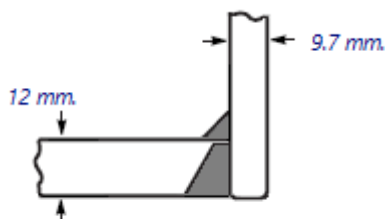
**Input, Nozzle Desc: N5-2"****From: 40**

Pressure for Reinforcement Calculations	P	7.340	bars
Temperature for Internal Pressure	Temp	230	°C
Design External Pressure	Pext	1.03	bars
Temperature for External Pressure	Tempex	198	°C
Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	138.00	N./mm2
Parent Allowable Stress At Ambient	Sva	138.00	N./mm2
Inside Diameter of Cylindrical Shell	D	1900.00	mm.
Design Length of Section	L	2408.3333	mm.
Shell Finished (Minimum) Thickness	t	12.0000	mm.
Shell Internal Corrosion Allowance	cas	3.0000	mm.
Shell External Corrosion Allowance	co	0.0000	mm.
Distance from Bottom/Left Tangent		300.00	mm.
User Entered Minimum Design Metal Temperature		-1.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-106 B	
Material UNS Number		K03006	
Material Specification/Type		Smls. pipe	
Allowable Stress at Temperature	Sn	118.00	N./mm2
Allowable Stress At Ambient	Sna	118.00	N./mm2
Diameter Basis (for tr calc only)		Outside	
Layout Angle		90.00	deg
Diameter		2.0000	in.
Size and Thickness Basis		Minimum	
Nominal Thickness		XXS	
Flange Material		SA-105	
Flange Type		Weld Neck Flange	
Corrosion Allowance	can	3.0000	mm.
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	200.0000	mm.
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm.
Groove weld depth between Nozzle and Vessel	Wgnv	9.0000	mm.
Inside Projection	h	0.0000	mm.
Weld leg size, Inside Element to Shell	Wi	0.0000	mm.
Flange Class		600	
Flange Grade		GR 1.1	

The Pressure Design option was MAWP + static head (to the nozzle).

**Reinforcement CALCULATION, Description: N5-2"**



ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Outside Diameter Used in Calculation 2.375 in.
 Actual Thickness Used in Calculation 0.382 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 7.34 \cdot 953.0 / (138 \cdot 1.0 - 0.6 \cdot 7.34)$
 $= 5.0858$ mm.

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R_o / (S_n \cdot E + 0.4 \cdot P)$ per Appendix 1-1 (a)(1)
 $= 7.34 \cdot 30.1625 / (118 \cdot 1.0 + 0.4 \cdot 7.34)$
 $= 0.1872$ mm.

Required Nozzle thickness under External Pressure per UG-28 : 0.3223 mm.

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	93.8896	mm.
Parallel to Vessel Wall, opening length	d	46.9448	mm.
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	16.7253	mm.

Taking a UG-36(c)(3)(a) exemption for nozzle: N5-2".

This calculation is valid for nozzles that meet all the requirements of paragraph UG-36. Please check the Code carefully, especially for nozzles that are not isolated or do not meet Code spacing requirements. To force the computation of areas for small nozzles go to Tools->Configuration and check the box to force the UG-37 small nozzle area calculation.

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta	= 3.3223	mm.
Wall Thickness per UG16(b),	tr16b	= 5.5000	mm.
Wall Thickness, shell/head, internal pressure	trb1	= 8.0858	mm.
Wall Thickness	tb1 = max(trb1, tr16b)	= 8.0858	mm.
Wall Thickness, shell/head, external pressure	trb2	= 3.7146	mm.
Wall Thickness	tb2 = max(trb2, tr16b)	= 5.5000	mm.
Wall Thickness per table UG-45	tb3	= 6.4200	mm.

Determine Nozzle Thickness candidate [tb]:
 $= \min[tb3, \max(tb1, tb2)]$
 $= \min[6.42, \max(8.0858, 5.5)]$
 $= 6.4200$ mm.

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
 $= \max(ta, tb)$
 $= \max(3.3223, 6.42)$
 $= 6.4200$ mm.

Available Nozzle Neck Thickness = 9.6901 mm. -> OK

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained	:	34.1,	Allowable	:	118.0 N./mm2	Passed
Expansion	:	0.0,	Allowable	:	260.9 N./mm2	Passed
Occasional	:	1.1,	Allowable	:	156.9 N./mm2	Passed
Shear	:	28.7,	Allowable	:	82.6 N./mm2	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, min(Curve:B, Curve:A)

Govern. thk, tg = 9.69, tr = 0.187, c = 3.0 mm., E* = 1.0
 Thickness Ratio = $tr \cdot E^* / (tg - c) = 0.028$, Temp. Reduction = 78 °C



Min Metal Temp. w/o impact per UCS-66, Curve A -8 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -104 °C
 Min Metal Temp. w/o impact per UG-20(f) -29 °C

Nozzle-Shell/Head Weld (UCS-66(a)(b)), Curve: B

Govrn. thk, $t_g = 9.69$, $t_r = 0.187$, $c = 3.0$ mm., $E^* = 1.0$
 Thickness Ratio = $t_r * E^* / (t_g - c) = 0.028$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -29 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -104 °C

Gov. MDMT of the nozzle to shell joint welded assembly : -104 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c) -18 °C
 Flange MDMT with Temp reduction per UCS-66(b)(1)(-b) -104 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
 Design Pressure/Ambient Rating = $7.34/102.00 = 0.072$

Weld Size Calculations, Description: N5-2"

Intermediate Calc. for nozzle/shell Welds T_{min} 6.6901 mm.

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	$4.6831 = 0.7 * t_{min}$	$7.0700 = 0.7 * W_o$ mm.

[Skipping the nozzle attachment weld strength calculations. Per UW-15\(b\)\(2\) the nozzles exempted by UG-36\(c\)\(3\)\(a\) \(small nozzles\) do not require a weld strength check.](#)

The Drop for this Nozzle is : 0.4789 mm.
 The Cut Length for this Nozzle is, Drop + Ho + H + T : 212.4790 mm.

Input Echo, WRC297 Item 1, Description: N5-2" :

Diameter Basis for Cylindrical Shell	ID	
Shell Internal Corrosion Allowance	3.0000	mm.
Shell External Corrosion Allowance	0.0000	mm.
Shell Diameter	1900.000	mm.
Shell Thickness	12.0000	mm.
Shell Stress Concentration Factor	1.000	
Vessel Material	SA-516 70	
Vessel Cold S.I. Allowable	Smc	138.00 N./mm ²
Vessel Hot S.I. Allowable	Smh	138.00 N./mm ²

[See ASME VIII Div. 1, UG-23\(e\), for determination of the SPS allowable.](#)

Diameter Basis for Nozzle	OD	
Nozzle Corrosion Allowance	3.0000	mm.
Nozzle Diameter	60.325	mm.
Nozzle Thickness	9.6901	mm.
Nozzle Stress Concentration Factor	1.000	
Nozzle Material	SA-106 B	
Nozzle Cold S.I. Allowable	SNmc	118.00 N./mm ²
Nozzle Hot S.I. Allowable	SNmh	118.00 N./mm ²

[See ASME VIII Div. 1, UG-23\(e\), for determination of the SPS allowable.](#)

Note:

*External Forces and Moments in WRC 107/537 Convention:
 These loads are assumed to be SUSstained loads.*

Design Internal Pressure	Dp	7.34 bars
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Radial Load	P	-382.42	Kgf
Circumferential Shear	Vc	254.95	Kgf
Longitudinal Shear	Vl	254.95	Kgf
Circumferential Moment	Mc	35.68	Kg-m.
Longitudinal Moment	Ml	35.68	Kg-m.
Torsional Moment	Mt	50.97	Kg-m.

Include Axial Pressure Thrust	No
Include Pressure Stress Indices per Div. 2	No
Local Loads applied at end of Nozzle/Attachment	No

Warning - The ratio, D_n/T_n (6.225) is < 10
check the Limitations of WRC bulletin 297.

Stress Computations at the Edge of the Nozzle:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{R_{\text{mean}}(t - c_a)}$$

$$= 60.325 + 2 \cdot 1.65 \cdot \sqrt{957.5(12.0 - 3.0)}$$

$$= 366.666 \text{ mm.}$$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter	(D) =	1915.000 mm.
Nozzle Outside Diameter	(d) =	60.325 mm.
Vessel Thickness used	(T) =	9.000 mm.
Nozzle Thickness used	(t) =	6.690 mm.
T / t	=	1.345
d / t	=	10.000

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 0.460$$

Nr/P	=	0.191
Mr/P	=	0.260
M0/P	=	0.108
N0/P	=	0.173
MrD/Mc	=	0.269
NrDL/Mc	=	0.163
M0d/Mc	=	0.138
N0DL/Mc	=	-0.106
MrD/Ml	=	0.247
NrDL/Ml	=	0.158
M0D/Ml	=	0.124
N0DL/Ml	=	-0.028

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Membrane (P)	8	8	8	8
Outplane Bending (P)	30	-30	30	-30
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (Ml)	2	2	-2	-2
Outplane Bending (Ml)	-53	53	53	-53
Normal Pressure Stress	77	78	77	78
Outplane Stress Summary	64	111	166	0

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses parallel to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Inplane Membrane (P)	8	8	8	8
Inplane Bending (P)	72	-72	72	-72
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0



Inplane Membrane (ML)	-11	-11	11	11
Inplane Bending (ML)	-106	106	106	-106
Inplane Pressure Stress	38	38	38	38

Inplane Stress Summary	2	70	237	-119
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Vessel Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Shear stress normal to longitudinal plane)	Top Outside	Top Inside	Bottom Outside	Bottom Inside

Outplane Shear (Vc)	2	2	-2	-2
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	9	9	9	9

Shear Stress Summary	12	12	6	6
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Vessel Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stress Intensities)	Top Outside	Top Inside	Bottom Outside	Bottom Inside

Two * Max Shear Stress	66	115	237	120
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Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stresses Normal to circumferential plane)	Left Outside	Left Inside	Right Outside	Right Inside

Outplane Membrane (P)	8	8	8	8
Outplane Bending (P)	30	-30	30	-30
Outplane Membrane (Mc)	7	7	-7	-7
Outplane Bending (Mc)	-59	59	59	-59
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	38	38	38	38

Outplane Stress Summary	25	83	128	-50
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Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stresses parallel to circumferential plane)	Left Outside	Left Inside	Right Outside	Right Inside

Inplane Membrane (P)	8	8	8	8
Inplane Bending (P)	72	-72	72	-72
Inplane Membrane (Mc)	-11	-11	11	11
Inplane Bending (Mc)	-115	115	115	-115
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	77	78	77	78

Inplane Stress Summary	31	118	285	-88
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Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Shear stress normal to circumferential plane)	Left Outside	Left Inside	Right Outside	Right Inside

Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	-2	-2	2	2
Torsional Shear (Mt)	9	9	9	9

Shear Stress Summary	6	6	12	12
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Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stress Intensities)	Left Outside	Left Inside	Right Outside	Right Inside



Two * Max Shear Stress	35	119	286	92
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Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses in the hoop direction)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
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Hoop Membrane (P)	8	8	8	8
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	2	2	-2	-2
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	2	3	2	3

Hoop Stress Summary	12	13	8	9
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Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses Normal to pipe cross-section)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
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Axial Membrane (P)	3	3	3	3
Axial Bending (P)	82	-82	82	-82
Axial Membrane (Mc)	0	0	0	0
Axial Bending (Mc)	0	0	0	0
Axial Membrane (ML)	-25	-25	25	25
Axial Bending (ML)	-131	131	131	-131
Axial Pressure Stress	1	1	1	1

Axial Stress Summary	-69	27	243	-183
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Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Shear stress)	Au Outside	Al Inside	Bu Outside	Bl Inside
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Shear due to (Vc)	3	3	-3	-3
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	13	13	13	13

Shear Stress Summary	17	17	9	9
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Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Stress Intensities)	Au Outside	Al Inside	Bu Outside	Bl Inside
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Two * Max Shear Stress	89	39	244	193
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Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses in the hoop direction)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
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Hoop Membrane (P)	8	8	8	8
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	7	7	-7	-7
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	2	3	2	3

Hoop Stress Summary	18	19	3	4
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Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses Normal to pipe cross-section)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
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Axial Membrane (P)	3	3	3	3
Axial Bending (P)	82	-82	82	-82
Axial Membrane (Mc)	-25	-25	25	25
Axial Bending (Mc)	-145	145	145	-145
Axial Membrane (ML)	0	0	0	0
Axial Bending (ML)	0	0	0	0
Axial Pressure Stress	1	1	1	1
Axial Stress Summary	-84	42	258	-197

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Shear stress)	Cu	Cl	Du	Dl
	Outside	Inside	Outside	Inside
Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	-3	-3	3	3
Shear due to Torsion	13	13	13	13
Shear Stress Summary	9	9	17	17

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stress Intensities)	Cu	Cl	Du	Dl
	Outside	Inside	Outside	Inside
Two * Max Shear Stress	104	45	259	204

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:**Vessel Stress Summation at Vessel-Nozzle Junction (N./mm²):**

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	77	78	77	78	77	78	77	78
Circ. Pl (SUS)	10	10	6	6	-3	-3	19	19
Circ. Q (SUS)	-23	23	83	-83	-43	43	187	-187
Long. Pm (SUS)	38	38	38	38	38	38	38	38
Long. Pl (SUS)	-3	-3	19	19	15	15	1	1
Long. Q (SUS)	-34	34	178	-178	-29	29	89	-89
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	2	2	-2	-2	-2	-2	2	2
Shear Q (SUS)	9	9	9	9	9	9	9	9
Pm (SUS)	77.0	78.0	77.0	78.0	77.0	78.0	77.0	78.0
Pm+Pl (SUS)	87.1	88.1	83.2	84.1	74.2	75.2	96.1	97.1
Pm+Pl+Q (Total)	66.7	113.7	235.7	122.8	35.3	119.3	283.8	92.8

Vessel Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	78.00	138.00	Passed
Pm+Pl (SUS)	97.07	207.00	Passed
Pm+Pl+Q (TOTAL)	283.78	413.99	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:**Nozzle Stress Summation at Vessel-Nozzle Junction (N./mm²):**

Type of	Stress Values at
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Stress Int. Location	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	2	3	2	3	2	3	2	3
Circ. Pl (SUS)	10	10	6	6	15	15	1	1
Circ. Q (SUS)	0	0	0	0	0	0	0	0
Long. Pm (SUS)	1	1	1	1	1	1	1	1
Long. Pl (SUS)	-22	-22	28	28	-22	-22	28	28
Long. Q (SUS)	-49	49	213	-213	-63	63	227	-227
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	3	3	-3	-3	-3	-3	3	3
Shear Q (SUS)	13	13	13	13	13	13	13	13
Pm (SUS)	2.0	3.0	2.0	3.0	2.0	3.0	2.0	3.0
Pm+Pl (SUS)	33.5	34.5	29.4	29.4	38.5	39.5	29.3	29.4
Pm+Pl+Q (Total)	88.0	38.2	242.4	194.0	103.0	45.6	257.0	204.5

Nozzle Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	3.00	118.00	Passed
Pm+Pl (SUS)	39.46	177.00	Passed
Pm+Pl+Q (TOTAL)	257.01	353.99	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

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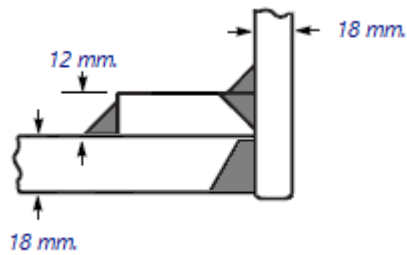
**Input, Nozzle Desc: N1-16"****From: 50**

Pressure for Reinforcement Calculations	P	7.153	bars
Temperature for Internal Pressure	Temp	230	°C
Design External Pressure	Pext	1.03	bars
Temperature for External Pressure	Tempex	198	°C
Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	138.00	N./mm2
Parent Allowable Stress At Ambient	Sva	138.00	N./mm2
Inside Diameter of Cylindrical Shell	D	1900.00	mm.
Design Length of Section	L	3008.3333	mm.
Shell Finished (Minimum) Thickness	t	18.0000	mm.
Shell Internal Corrosion Allowance	cas	3.0000	mm.
Shell External Corrosion Allowance	co	0.0000	mm.
Distance from Bottom/Left Tangent		3650.00	mm.
User Entered Minimum Design Metal Temperature		-1.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-516 70	
Material UNS Number		K02700	
Material Specification/Type		Plate	
Allowable Stress at Temperature	Sn	138.00	N./mm2
Allowable Stress At Ambient	Sna	138.00	N./mm2
Diameter Basis (for tr calc only)		Outside	
Layout Angle		270.00	deg
Diameter		20.0000	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	18.0000	mm.
Flange Material		SA-105	
Flange Type		Weld Neck Flange	
Corrosion Allowance	can	3.0000	mm.
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	250.0000	mm.
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm.
Groove weld depth between Nozzle and Vessel	Wgnv	10.0000	mm.
Inside Projection	h	0.0000	mm.
Weld leg size, Inside Element to Shell	Wi	0.0000	mm.
Pad Material		SA-516 70	
Pad Allowable Stress at Temperature	Sp	138.00	N./mm2
Pad Allowable Stress At Ambient	Spa	138.00	N./mm2
Diameter of Pad along vessel surface	Dp	950.0001	mm.
Thickness of Pad	te	12.0000	mm.
Weld leg size between Pad and Shell	Wp	8.0000	mm.
Groove weld depth between Pad and Nozzle	Wgpn	12.0000	mm.
Reinforcing Pad Width		221.0000	mm.
Flange Class		300	
Flange Grade		GR 1.1	

The Pressure Design option was MAWP + static head (to the nozzle).

**Reinforcement CALCULATION, Description: N1-16"**

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Outside Diameter Used in Calculation	20.000 in.
Actual Thickness Used in Calculation	0.709 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 7.15 \cdot 953.0 / (138 \cdot 1.0 - 0.6 \cdot 7.15)$
 $= 4.9552 \text{ mm.}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R_o / (S_n \cdot E + 0.4 \cdot P)$ per Appendix 1-1 (a)(1)
 $= 7.15 \cdot 254.0 / (138 \cdot 1.0 + 0.4 \cdot 7.15)$
 $= 1.3139 \text{ mm.}$

Required Nozzle thickness under External Pressure per UG-28 : 1.2305 mm.

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	956.0001 mm.
Parallel to Vessel Wall, opening length	d	478.0000 mm.
Normal to Vessel Wall (Thickness Limit), pad side Tlwp		37.5000 mm.

Weld Strength Reduction Factor [fr1]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 138.0 / 138.0)$
 $= 1.000$

Weld Strength Reduction Factor [fr2]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 138.0 / 138.0)$
 $= 1.000$

Weld Strength Reduction Factor [fr4]:
 $= \min(1, S_p / S_v)$
 $= \min(1, 138.0 / 138.0)$
 $= 1.000$

Weld Strength Reduction Factor [fr3]:
 $= \min(fr2, fr4)$
 $= \min(1.0, 1.0)$
 $= 1.000$

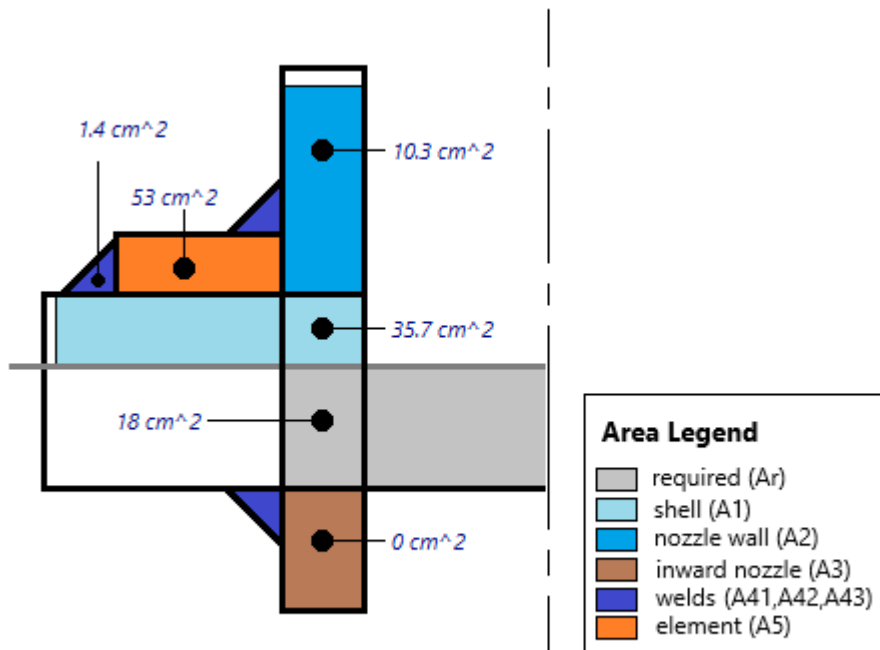


Figure UG 37.1 Nozzle Areas

Results of Nozzle Reinforcement Area Calculations: (cm²)

AREA AVAILABLE, A1 to A5		MAWP	External	Mapnc
Area Required	Ar	23.686	18.024	NA
Area in Shell	A1	48.014	35.653	NA
Area in Nozzle Wall	A2	10.265	10.327	NA
Area in Inward Nozzle	A3	0.000	0.000	NA
Area in Welds	A41+A42+A43	1.390	1.390	NA
Area in Element	A5	53.040	53.040	NA
TOTAL AREA AVAILABLE	Atot	112.709	100.410	NA

The External Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations

90.00 degs.

The area available without a pad is Sufficient.

The area available with the given pad is Sufficient.

Area Required [A]:

$$\begin{aligned}
 &= 0.5(d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F(1 - fr1)) \text{ per UG-37(d)} \\
 &= 0.5(478.0 \cdot 7.5412 \cdot 1 + 2 \cdot 15.0 \cdot 7.5412 \cdot (1 - 1.0)) \\
 &= 18.024 \text{ cm}^2
 \end{aligned}$$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:

$$\begin{aligned}
 &= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1) \\
 &= 478.0(1.0 \cdot 15.0 - 1.0 \cdot 7.541) - 2 \cdot 15.0 \\
 &\quad (1.0 \cdot 15.0 - 1.0 \cdot 7.5412) \cdot (1 - 1.0) \\
 &= 35.653 \text{ cm}^2
 \end{aligned}$$

Area Available in Nozzle Wall Projecting Outward [A2]:

$$\begin{aligned}
 &= (2 \cdot Tlwp) \cdot (tn - trn) \cdot fr2 \\
 &= (2 \cdot 37.5) \cdot (15.0 - 1.23) \cdot 1.0 \\
 &= 10.327 \text{ cm}^2
 \end{aligned}$$

Area Available in Welds [A41 + A42 + A43]:

$$= (Wo^2 - Ar \text{ Lost}) \cdot Fr3 + ((Wi - can / 0.707)^2 - Ar \text{ Lost}) \cdot fr2 + Trapfr4$$



$$= (1.) \cdot 1.0 + (0.0) \cdot 1.0 + 39.0^2 \cdot 1.0$$

$$= 1.390 \text{ cm}^2$$

Area Available in Element [A5]:

$$= (\min(D_p, D_L) - (\text{Nozzle OD})) \cdot (\min(t_p, t_{lp}, t_e)) \cdot f_{r4}$$

$$= (950.0 - 508.0) \cdot 12.0 \cdot 1.0$$

$$= 53.040 \text{ cm}^2$$

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta = 4.3139 mm.
Wall Thickness per UG16(b),	tr16b = 5.5000 mm.
Wall Thickness, shell/head, internal pressure	trb1 = 7.9552 mm.
Wall Thickness	tb1 = max(trb1, tr16b) = 7.9552 mm.
Wall Thickness, shell/head, external pressure	trb2 = 3.7146 mm.
Wall Thickness	tb2 = max(trb2, tr16b) = 5.5000 mm.
Wall Thickness per table UG-45	tb3 = 11.3312 mm.

Determine Nozzle Thickness candidate [tb]:

$$= \min[tb3, \max(tb1, tb2)]$$

$$= \min[11.331, \max(7.9552, 5.5)]$$

$$= 7.9552 \text{ mm.}$$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

$$= \max(ta, tb)$$

$$= \max(4.3139, 7.9552)$$

$$= 7.9552 \text{ mm.}$$

Available Nozzle Neck Thickness = 18.0000 mm. -> OK

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained	:	21.7,	Allowable	:	138.0 N./mm ²	Passed
Expansion	:	0.0,	Allowable	:	323.3 N./mm ²	Passed
Occasional	:	5.5,	Allowable	:	183.5 N./mm ²	Passed
Shear	:	12.4,	Allowable	:	96.6 N./mm ²	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, min(Curve:B, Curve:A)

Govrn. thk, tg = 18.0, tr = 1.314, c = 3.0 mm., E* = 1.0
 Thickness Ratio = tr * E*/(tg - c) = 0.088, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve A	9 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C

Nozzle Neck to Pad Weld for the Nozzle, Curve: B

Govrn. thk, tg = 12.0, c = 3.0 mm., E* = 1.0
 Thickness Ratio = tr * E*/(tg - c) = 0.33, Temp. Reduction = 78 °C
 Attachment governing, Attachment stress = Shell stress per Code

Min Metal Temp. w/o impact per UCS-66, Curve B	-23 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Nozzle Neck to Pad Weld for Reinforcement pad, Curve: B

Govrn. thk, tg = 12.0, c = 3.0 mm., E* = 1.0
 Thickness Ratio = tr * E*/(tg - c) = 0.33, Temp. Reduction = 78 °C
 Attachment governing, Attachment stress = Shell stress per Code

Min Metal Temp. w/o impact per UCS-66, Curve B	-23 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C



Shell to Pad Weld Junction at Pad OD, Curve: B

Govrn. thk, $t_g = 12.0$, $c = 3.0$ mm., $E^* = 1.0$
 Thickness Ratio = $tr * E^* / (t_g - c) = 0.33$, Temp. Reduction = 78 °C
 Attachment governing, Attachment stress = Shell stress per Code

Min Metal Temp. w/o impact per UCS-66, Curve B	-23 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Nozzle-Shell/Head Weld (UCS-66(a)(1)(b)), Curve: B

Govrn. thk, $t_g = 18.0$, $tr = 4.955$, $c = 3.0$ mm., $E^* = 1.0$
 Thickness Ratio = $tr * E^* / (t_g - c) = 0.33$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-11 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Gov. MDMT of the Nozzle	: -104 °C
Gov. MDMT of the Reinforcement Pad	: -104 °C
Gov. MDMT of the nozzle to shell joint welded assembly	: -104 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c)	-18 °C
Flange MDMT with Temp reduction per UCS-66(b)(1)(-b)	-104 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
 Design Pressure/Ambient Rating = $7.15/51.00 = 0.140$

Weld Size Calculations, Description: N1-16"

Intermediate Calc. for nozzle/shell Welds	T_{min}	12.0000 mm.
Intermediate Calc. for pad/shell Welds	T_{minPad}	12.0000 mm.

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	$6.0000 = \text{Min per Code}$	$7.0700 = 0.7 * W_o \text{ mm.}$
Pad Weld	$6.0000 = 0.5 * T_{minPad}$	$5.6560 = 0.7 * W_p \text{ mm.}$

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:
 $= \max(0, (A - A1 + 2 \cdot t_n \cdot fr1 \cdot (E1 \cdot t - tr)) \cdot Sv)$
 $= \max(0, (18.0235 - 35.6529 + 2 \cdot 15.0 \cdot 1.0 \cdot (1.0 \cdot 15.0 - 7.5412)) \cdot 138)$
 $= \max(0, -21658.58) \text{ Kg}$

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:
 $= (A2 + A5 + A4 - (W_i - Can / .707)^2 \cdot fr2) \cdot Sv$
 $= (10.3271 + 53.04 + 1.39 - 0.0 \cdot 1.0) \cdot 138$
 $= 91123.44 \text{ Kg}$

Weld Load [W2]:
 $= (A2 + A3 + A4 + (2 \cdot t_n \cdot t \cdot fr1)) \cdot Sv$
 $= (10.3271 + 0.0 + 1. + (4.5)) \cdot 138$
 $= 22271.24 \text{ Kg}$

Weld Load [W3]:
 $= (A2 + A3 + A4 + A5 + (2 \cdot t_n \cdot t \cdot fr1)) \cdot S$
 $= (10.3271 + 0.0 + 1.39 + 53.04 + (4.5)) \cdot 138$
 $= 97455.65 \text{ Kg}$

Strength of Connection Elements for Failure Path Analysis



Shear, Outward Nozzle Weld [Sonw]:

$$= (\pi/2) \cdot D_{lo} \cdot W_o \cdot 0.49 \cdot S_{nw}$$

$$= (3.1416/2.0) \cdot 508.0 \cdot 10.0 \cdot 0.49 \cdot 138$$

$$= 55020. \text{ Kg}$$

Shear, Pad Element Weld [Spew]:

$$= (\pi/2) \cdot D_P \cdot W_P \cdot 0.49 \cdot S_{EW}$$

$$= (3.1416/2.0) \cdot 950.0 \cdot 8.0 \cdot 0.49 \cdot 138$$

$$= 82314. \text{ Kg}$$

Shear, Nozzle Wall [Snw]:

$$= (\pi \cdot (D_{lr} + D_{lo})/4) \cdot (Thk - Can) \cdot 0.7 \cdot S_n$$

$$= (3.1416 \cdot 246.5) \cdot (18.0 - 3.0) \cdot 0.7 \cdot 138$$

$$= 114419. \text{ Kg}$$

Tension, Pad Groove Weld [Tpgw]:

$$= (\pi/2) \cdot D_{lo} \cdot W_{gpn} \cdot 0.74 \cdot S_{eg}$$

$$= (3.1416/2) \cdot 508.0 \cdot 12.0 \cdot 0.74 \cdot 138$$

$$= 99710. \text{ Kg}$$

Tension, Shell Groove Weld [Tngw]:

$$= (\pi/2) \cdot D_{lo} \cdot W_{gvi} \cdot 0.74 \cdot S_{ng}$$

$$= (3.1416/2.0) \cdot 508.0 \cdot (10.0 - 0.0) \cdot 0.74 \cdot 138$$

$$= 83092. \text{ Kg}$$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:

$$= (S_{PEW} + S_{NW}) = (82314 + 114419) = 196733 \text{ Kg}$$

Failure Path [Path 2-2]:

$$= (S_{onw} + T_{pgw} + T_{ngw} + S_{inw})$$

$$= (55020 + 99710 + 83092 + 0) = 237822 \text{ Kg}$$

Failure Path [Path 3-3]:

$$= (S_{pew} + T_{ngw} + S_{inw})$$

$$= (82314 + 83092 + 0) = 165405 \text{ Kg}$$

Summary of Failure Path Calculations:

Path 1-1 = 196732 Kg, must exceed W = 0 Kg	or W1 = 91123 Kg
Path 2-2 = 237822 Kg, must exceed W = 0 Kg	or W2 = 22271 Kg
Path 3-3 = 165405 Kg, must exceed W = 0 Kg	or W3 = 97455 Kg

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 13 bars

Note: The MAWP of this junction was limited by the parent Shell/Head.

Nozzle is O.K. for the External Pressure: 1 bars

The Drop for this Nozzle is : 34.5854 mm.

The Cut Length for this Nozzle is, Drop + Ho + H + T : 302.5854 mm.

Percent Elongation Calculations:

% Elongation per Table UG-79-1 ($50 \cdot t_{nom} / R_f \cdot (1 - R_f / R_o)$) 3.673 %

Input Echo, WRC297 Item 1, Description: N1-16" :

Diameter Basis for Cylindrical Shell	ID	
Shell Internal Corrosion Allowance	3.0000	mm.
Shell External Corrosion Allowance	0.0000	mm.
Shell Diameter	1900.000	mm.
Shell Thickness	18.0000	mm.
Shell Stress Concentration Factor	1.000	
Vessel Material	SA-516 70	
Vessel Cold S.I. Allowable	Smc	138.00 N./mm ²
Vessel Hot S.I. Allowable	Smh	138.00 N./mm ²



See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Diameter Basis for Nozzle	OD	
Nozzle Corrosion Allowance	3.0000	mm.
Nozzle Diameter	508.000	mm.
Nozzle Thickness	18.0000	mm.
Nozzle Stress Concentration Factor	1.000	
Nozzle Material	SA-516 70	
Nozzle Cold S.I. Allowable	SNmc	138.00 N./mm ²
Nozzle Hot S.I. Allowable	SNmh	138.00 N./mm ²

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Diameter of Reinforcing Pad	950.000	mm.
Thickness of Reinforcing Pad	12.0000	mm.

Note:

*External Forces and Moments in WRC 107/537 Convention:
These loads are assumed to be SUSained loads.*

Design Internal Pressure	Dp	7.15	bars
Radial Load	P	-3824.19	Kgf
Circumferential Shear	Vc	2549.46	Kgf
Longitudinal Shear	Vl	2549.46	Kgf
Circumferential Moment	Mc	3568.05	Kg-m.
Longitudinal Moment	Ml	3568.05	Kg-m.
Torsional Moment	Mt	5097.22	Kg-m.

Include Axial Pressure Thrust	No
Include Pressure Stress Indices per Div. 2	No
Local Loads applied at end of Nozzle/Attachment	No

Stress Computations at the Edge of the Nozzle:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:
 $= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{(R_{\text{mean}}(t - ca))}$
 $= 508.0 + 2 \cdot 1.65 \cdot \sqrt{(966.5(18.0 - 3.0))}$
 $= 905.338 \text{ mm.}$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter	(D) =	1933.000	mm.
Nozzle Outside Diameter	(d) =	508.000	mm.
Vessel Thickness used	(T) =	27.000	mm.
Nozzle Thickness used	(t) =	15.000	mm.
T / t	=	1.800	
d / t	=	33.867	

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 2.224$$

Nr/P	=	0.074
Mr/P	=	0.089
M0/P	=	0.045
N0/P	=	0.143
MrD/Mc	=	0.195
NrDL/Mc	=	0.100
M0d/Mc	=	0.113
N0DL/Mc	=	0.176
MrD/Ml	=	0.101
NrDL/Ml	=	0.059
M0D/Ml	=	0.051
N0DL/Ml	=	0.257

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
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(Stresses Normal to longitudinal plane)	Top Outside	Top Inside	Bottom Outside	Bottom Inside
Outplane Membrane (P)	7	7	7	7
Outplane Bending (P)	13	-13	13	-13
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	-24	-24	24	24
Outplane Bending (ML)	-28	28	28	-28
Normal Pressure Stress	24	25	24	25
Outplane Stress Summary	-7	23	99	14

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stresses parallel to longitudinal plane)	Top Outside	Top Inside	Bottom Outside	Bottom Inside
Inplane Membrane (P)	3	3	3	3
Inplane Bending (P)	27	-27	27	-27
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	-5	-5	5	5
Inplane Bending (ML)	-57	57	57	-57
Inplane Pressure Stress	12	12	12	12
Inplane Stress Summary	-19	40	106	-62

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Shear stress normal to longitudinal plane)	Top Outside	Top Inside	Bottom Outside	Bottom Inside
Outplane Shear (Vc)	1	1	-1	-1
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	4	4	4	4
Shear Stress Summary	5	5	3	3

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stress Intensities)	Top Outside	Top Inside	Bottom Outside	Bottom Inside
Two * Max Shear Stress	21	42	107	77

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stresses Normal to circumferential plane)	Left Outside	Left Inside	Right Outside	Right Inside
Outplane Membrane (P)	7	7	7	7
Outplane Bending (P)	13	-13	13	-13
Outplane Membrane (Mc)	-16	-16	16	16
Outplane Bending (Mc)	-63	63	63	-63
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	12	12	12	12
Outplane Stress Summary	-46	53	114	-41

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stresses parallel to circumferential plane)	Left Outside	Left Inside	Right Outside	Right Inside
Inplane Membrane (P)	3	3	3	3
Inplane Bending (P)	27	-27	27	-27
Inplane Membrane (Mc)	-9	-9	9	9



Inplane Bending (Mc)	-110	110	110	-110
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	24	25	24	25

Inplane Stress Summary	-63	102	175	-98
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Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Shear stress normal to	Left	Left	Right	Right
circumferential plane)	Outside	Inside	Outside	Inside

Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	-1	-1	1	1
Torsional Shear (Mt)	4	4	4	4

Shear Stress Summary	3	3	5	5
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Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stress Intensities	Left	Left	Right	Right
	Outside	Inside	Outside	Inside

Two * Max Shear Stress	64	103	176	99
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Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stresses in the	Top	Top	Bottom	Bottom
hoop direction)	Outside	Inside	Outside	Inside

Hoop Membrane (P)	7	7	7	7
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	-24	-24	24	24
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	13	14	13	14

Hoop Stress Summary	-3	-2	44	45
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Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stresses Normal to	Top	Top	Bottom	Bottom
pipe cross-section)	Outside	Inside	Outside	Inside

Axial Membrane (P)	1	1	1	1
Axial Bending (P)	51	-51	51	-51
Axial Membrane (Mc)	0	0	0	0
Axial Bending (Mc)	0	0	0	0
Axial Membrane (ML)	-12	-12	12	12
Axial Bending (ML)	-131	131	131	-131
Axial Pressure Stress	6	6	6	6

Axial Stress Summary	-84	75	204	-162
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Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Shear stress)	Outside	Inside	Outside	Inside

Shear due to (Vc)	2	2	-2	-2
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	8	8	8	8

Shear Stress Summary	10	10	6	6
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Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stress Intensities	Outside	Inside	Outside	Inside



Two * Max Shear Stress 85 81 204 208

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE
(Stresses in the
hoop direction)

	Cu	Cl	Du	Dl
	Left	Left	Right	Right
	Outside	Inside	Outside	Inside

Hoop Membrane (P)	7	7	7	7
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	-16	-16	16	16
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	13	14	13	14

Hoop Stress Summary 3 4 37 38

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE
(Stresses Normal to
pipe cross-section)

	Cu	Cl	Du	Dl
	Left	Left	Right	Right
	Outside	Inside	Outside	Inside

Axial Membrane (P)	1	1	1	1
Axial Bending (P)	51	-51	51	-51
Axial Membrane (Mc)	-12	-12	12	12
Axial Bending (Mc)	-266	266	266	-266
Axial Membrane (ML)	0	0	0	0
Axial Bending (ML)	0	0	0	0
Axial Pressure Stress	6	6	6	6

Axial Stress Summary -218 210 338 -296

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE
(Shear stress)

	Cu	Cl	Du	Dl
	Outside	Inside	Outside	Inside

Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	-2	-2	2	2
Shear due to Torsion	8	8	8	8

Shear Stress Summary 6 6 10 10

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE
(Stress Intensities)

	Cu	Cl	Du	Dl
	Outside	Inside	Outside	Inside

Two * Max Shear Stress 223 210 338 335

Stress Computations at the Edge of the Reinforcing Pad:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{R_{\text{mean}}(t - ca)}$$

$$= 508.0 + 2 \cdot 1.65 \cdot \sqrt{960.5(18.0 - 3.0)}$$

$$= 904.103 \text{ mm.}$$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter (D) = 1921.000 mm.

Nozzle Outside Diameter (d) = 950.000 mm.

Vessel Thickness used (T) = 15.000 mm.

Nozzle Thickness used (t) = 236.000 mm.

T / t = 0.064

d / t = 10.000

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 5.596$$

Note: Re-pad thickness is added to nozzle thickness to simulate the Re-pad.



Nr/P	=	0.045
Mr/P	=	0.071
M0/P	=	0.019
N0/P	=	0.024
MrD/Mc	=	0.238
NrDL/Mc	=	0.098
M0d/Mc	=	0.070
N0DL/Mc	=	0.047
MrD/M1	=	0.044
NrDL/M1	=	0.043
M0D/M1	=	0.014
N0DL/M1	=	0.022

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Membrane (P)	4	4	4	4
Outplane Bending (P)	19	-19	19	-19
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	-3	-3	3	3
Outplane Bending (ML)	-13	13	13	-13
Normal Pressure Stress	45	45	45	45
Outplane Stress Summary	51	40	85	21

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses parallel to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Inplane Membrane (P)	7	7	7	7
Inplane Bending (P)	70	-70	70	-70
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	-7	-7	7	7
Inplane Bending (ML)	-43	43	43	-43
Inplane Pressure Stress	22	22	22	22
Inplane Stress Summary	50	-4	151	-76

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Shear stress normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Shear (Vc)	1	1	-1	-1
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	2	2	2	2
Shear Stress Summary	3	3	1	1

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stress Intensities)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Two * Max Shear Stress	54	45	151	97

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses Normal to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Outplane Membrane (P)	4	4	4	4
Outplane Bending (P)	19	-19	19	-19
Outplane Membrane (Mc)	-7	-7	7	7



Outplane Bending (Mc)	-68	68	68	-68
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	22	22	22	22

Outplane Stress Summary	-30	68	121	-53
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Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stresses parallel to	Left	Left	Right	Right
circumferential plane)	Outside	Inside	Outside	Inside

Inplane Membrane (P)	7	7	7	7
Inplane Bending (P)	70	-70	70	-70
Inplane Membrane (Mc)	-15	-15	15	15
Inplane Bending (Mc)	-234	234	234	-234
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	45	45	45	45

Inplane Stress Summary	-126	200	373	-235
------------------------	------	-----	-----	------

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Shear stress normal to	Left	Left	Right	Right
circumferential plane)	Outside	Inside	Outside	Inside

Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	-1	-1	1	1
Torsional Shear (Mt)	2	2	2	2

Shear Stress Summary	1	1	3	3
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Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stress Intensities	Left	Left	Right	Right
	Outside	Inside	Outside	Inside

Two * Max Shear Stress	126	200	373	235
------------------------	-----	-----	-----	-----

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:**Vessel Stress Summation at Vessel-Nozzle Junction (N./mm²):**

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	24	25	24	25	24	25	24	25
Circ. Pl (SUS)	-17	-17	31	31	-6	-6	12	12
Circ. Q (SUS)	-15	15	41	-41	-83	83	137	-137
Long. Pm (SUS)	12	12	12	12	12	12	12	12
Long. Pl (SUS)	-2	-2	8	8	-9	-9	23	23
Long. Q (SUS)	-30	30	84	-84	-50	50	76	-76
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	1	1	-1	-1	-1	-1	1	1
Shear Q (SUS)	4	4	4	4	4	4	4	4
Pm (SUS)	24.0	25.0	24.0	25.0	24.0	25.0	24.0	25.0
Pm+Pl (SUS)	10.3	10.4	55.0	56.0	18.1	19.1	36.6	37.4
Pm+Pl+Q (Total)	21.8	41.4	105.0	79.2	65.5	102.2	173.4	100.4

Vessel Stress Summation Comparison (N./mm²):



Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	25.00	138.00	Passed
Pm+Pl (SUS)	56.03	207.00	Passed
Pm+Pl+Q (TOTAL)	173.40	413.99	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Reinforcing Pad Edge (N./mm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	45	45	45	45	45	45	45	45
Circ. Pl (SUS)	1	1	7	7	-8	-8	22	22
Circ. Q (SUS)	6	-6	32	-32	-164	164	304	-304
Long. Pm (SUS)	22	22	22	22	22	22	22	22
Long. Pl (SUS)	0	0	14	14	-3	-3	11	11
Long. Q (SUS)	27	-27	113	-113	-49	49	87	-87
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	1	1	-1	-1	-1	-1	1	1
Shear Q (SUS)	2	2	2	2	2	2	2	2
Pm (SUS)	45.0	45.0	45.0	45.0	45.0	45.0	45.0	45.0
Pm+Pl (SUS)	46.0	46.0	52.1	52.1	37.1	37.1	67.0	67.0
Pm+Pl+Q (Total)	53.9	45.4	149.0	97.0	127.0	201.0	371.0	237.0

Vessel Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	45.00	138.00	Passed
Pm+Pl (SUS)	67.03	207.00	Passed
Pm+Pl+Q (TOTAL)	371.04	413.99	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Nozzle Stress Summation at Vessel-Nozzle Junction (N./mm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	13	14	13	14	13	14	13	14
Circ. Pl (SUS)	-17	-17	31	31	-9	-9	23	23
Circ. Q (SUS)	0	0	0	0	0	0	0	0
Long. Pm (SUS)	6	6	6	6	6	6	6	6
Long. Pl (SUS)	-11	-11	13	13	-11	-11	13	13
Long. Q (SUS)	-80	80	182	-182	-215	215	317	-317
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	2	2	-2	-2	-2	-2	2	2
Shear Q (SUS)	8	8	8	8	8	8	8	8
Pm (SUS)	13.0	14.0	13.0	14.0	13.0	14.0	13.0	14.0
Pm+Pl (SUS)	6.6	6.2	44.2	45.2	9.8	10.8	36.2	37.2



Pm+Pl+Q (Total)	86.2	80.5	201.2	208.3	224.3	210.2	336.3	335.6
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Nozzle Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	14.00	138.00	Passed
Pm+Pl (SUS)	45.15	207.00	Passed
Pm+Pl+Q (TOTAL)	336.33	413.99	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

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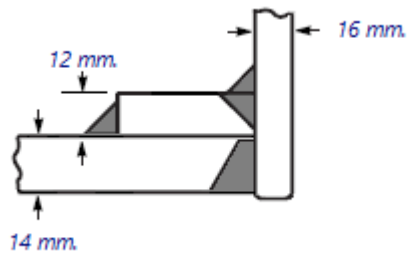
**Input, Nozzle Desc: N6-6"****From: 50**

Pressure for Reinforcement Calculations	P	7.153	bars
Temperature for Internal Pressure	Temp	230	°C
Design External Pressure	Pext	1.03	bars
Temperature for External Pressure	Tempex	198	°C
Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	138.00	N./mm2
Parent Allowable Stress At Ambient	Sva	138.00	N./mm2
Inside Diameter of Cylindrical Shell	D	1900.00	mm.
Design Length of Section	L	3008.3333	mm.
Shell Finished (Minimum) Thickness	t	14.0000	mm.
Shell Internal Corrosion Allowance	cas	3.0000	mm.
Shell External Corrosion Allowance	co	0.0000	mm.
Distance from Bottom/Left Tangent		4450.00	mm.
User Entered Minimum Design Metal Temperature		-1.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-106 B	
Material UNS Number		K03006	
Material Specification/Type		Smls. pipe	
Allowable Stress at Temperature	Sn	118.00	N./mm2
Allowable Stress At Ambient	Sna	118.00	N./mm2
Diameter Basis (for tr calc only)		Outside	
Layout Angle		270.00	deg
Diameter		6.0000	in.
Size and Thickness Basis		Minimum	
Nominal Thickness		160	
Flange Material		SA-105	
Flange Type		Weld Neck Flange	
Corrosion Allowance	can	3.0000	mm.
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	204.7870	mm.
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm.
Groove weld depth between Nozzle and Vessel	Wgnv	12.0000	mm.
Inside Projection	h	0.0000	mm.
Weld leg size, Inside Element to Shell	Wi	0.0000	mm.
Pad Material		SA-516 70	
Pad Allowable Stress at Temperature	Sp	138.00	N./mm2
Pad Allowable Stress At Ambient	Spa	138.00	N./mm2
Diameter of Pad along vessel surface	Dp	318.2750	mm.
Thickness of Pad	te	12.0000	mm.
Weld leg size between Pad and Shell	Wp	10.0000	mm.
Groove weld depth between Pad and Nozzle	Wgpn	12.0000	mm.
Reinforcing Pad Width		75.0000	mm.
Flange Class		300	
Flange Grade		GR 1.1	

The Pressure Design option was MAWP + static head (to the nozzle).

**Reinforcement CALCULATION, Description: N6-6"**

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Outside Diameter Used in Calculation	6.625 in.
Actual Thickness Used in Calculation	0.628 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 7.15 \cdot 953.0 / (138 \cdot 1.0 - 0.6 \cdot 7.15)$
 $= 4.9552 \text{ mm.}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R_o / (S_n \cdot E + 0.4 \cdot P)$ per Appendix 1-1 (a)(1)
 $= 7.15 \cdot 84.1375 / (118 \cdot 1.0 + 0.4 \cdot 7.15)$
 $= 0.5088 \text{ mm.}$

Required Nozzle thickness under External Pressure per UG-28 : 0.5930 mm.

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	284.7198 mm.
Parallel to Vessel Wall, opening length	d	142.3599 mm.
Normal to Vessel Wall (Thickness Limit), pad side Tlwp		27.5000 mm.

The Pad diameter is greater than the Diameter Limit. The excess will not be considered.

Weld Strength Reduction Factor [fr1]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 118.0 / 138.0)$
 $= 0.855$

Weld Strength Reduction Factor [fr2]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 118.0 / 138.0)$
 $= 0.855$

Weld Strength Reduction Factor [fr4]:
 $= \min(1, S_p / S_v)$
 $= \min(1, 138.0 / 138.0)$
 $= 1.000$

Weld Strength Reduction Factor [fr3]:
 $= \min(fr2, fr4)$
 $= \min(0.855, 1.0)$
 $= 0.855$

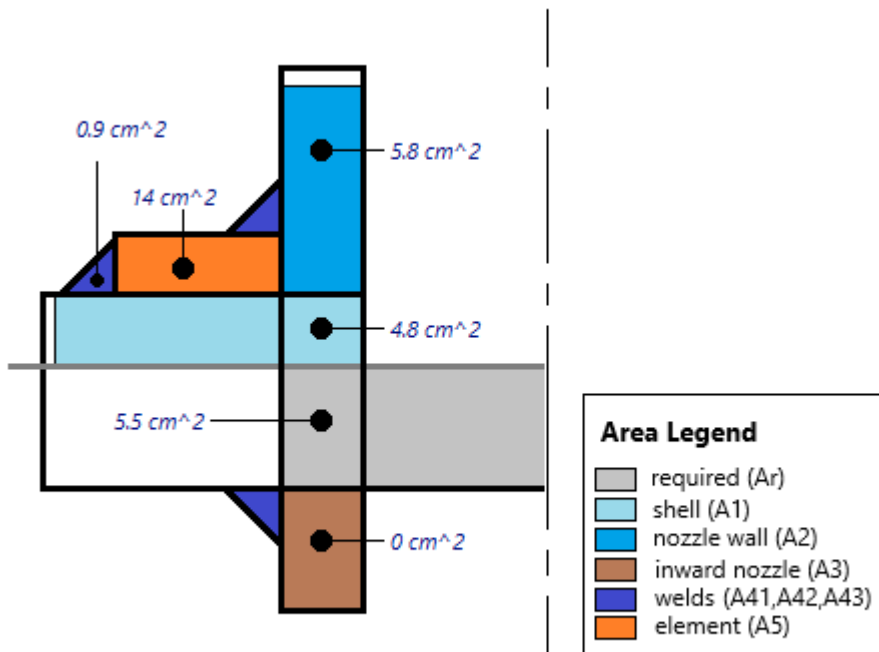


Figure UG 37.1 Nozzle Areas

Results of Nozzle Reinforcement Area Calculations: (cm²)

AREA AVAILABLE, A1 to A5		MAWP	External	Mapnc
Area Required	Ar	7.240	5.509	NA
Area in Shell	A1	8.378	4.794	NA
Area in Nozzle Wall	A2	5.855	5.815	NA
Area in Inward Nozzle	A3	0.000	0.000	NA
Area in Welds	A41+A42+A43	0.855	0.855	NA
Area in Element	A5	13.973	13.973	NA
TOTAL AREA AVAILABLE	Atot	29.061	25.437	NA

The External Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations

90.00 degs.

The area available without a pad is Sufficient.

The area available with the given pad is Sufficient.

Area Required [A]:

$$= 0.5(d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F(1 - fr1)) \text{ per UG-37(d)}$$

$$= 0.5(142.3599 \cdot 7.5412 \cdot 1 + 2 \cdot 12.9576 \cdot 7.5412 \cdot 1(1 - 0.86))$$

$$= 5.509 \text{ cm}^2$$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:

$$= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$$

$$= 142.36(1.0 \cdot 11.0 - 1.0 \cdot 7.541) - 2 \cdot 12.958$$

$$(1.0 \cdot 11.0 - 1.0 \cdot 7.5412) \cdot (1 - 0.855)$$

$$= 4.794 \text{ cm}^2$$

Area Available in Nozzle Wall Projecting Outward [A2]:

$$= (2 \cdot Tlwp) \cdot (tn - trn) \cdot fr2$$

$$= (2 \cdot 27.5) \cdot (12.96 - 0.59) \cdot 0.8551$$

$$= 5.815 \text{ cm}^2$$

Area Available in Welds [A41 + A42 + A43]:

$$= Wo^2 \cdot fr3 + (Wi - can/0.707)^2 \cdot fr2 + Wp^2 \cdot fr4$$



$$= 10.0^2 \cdot 0.86 + (0.0)^2 \cdot 0.86 + 0.0^2 \cdot 1.0$$

$$= 0.855 \text{ cm}^2$$

Area Available in Element [A5]:

$$= (\min(D_p, DL) - (\text{Nozzle OD})) \cdot (\min(t_p, T_{lwp}, t_e)) \cdot f_{r4}$$

$$= (284.7198 - 168.275) \cdot 12.0 \cdot 1.0$$

$$= 13.973 \text{ cm}^2$$

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta = 3.5930 mm.
Wall Thickness per UG16(b),	tr16b = 5.5000 mm.
Wall Thickness, shell/head, internal pressure	trb1 = 7.9552 mm.
Wall Thickness	tb1 = max(trb1, tr16b) = 7.9552 mm.
Wall Thickness, shell/head, external pressure	trb2 = 3.7146 mm.
Wall Thickness	tb2 = max(trb2, tr16b) = 5.5000 mm.
Wall Thickness per table UG-45	tb3 = 9.2200 mm.

Determine Nozzle Thickness candidate [tb]:

$$= \min[tb3, \max(tb1, tb2)]$$

$$= \min[9.22, \max(7.9552, 5.5)]$$

$$= 7.9552 \text{ mm.}$$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

$$= \max(ta, tb)$$

$$= \max(3.593, 7.9552)$$

$$= 7.9552 \text{ mm.}$$

Available Nozzle Neck Thickness = 15.9576 mm. -> OK

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained	: 19.5,	Allowable	: 118.0 N./mm ²	Passed
Expansion	: 0.0,	Allowable	: 275.5 N./mm ²	Passed
Occasional	: 1.8,	Allowable	: 156.9 N./mm ²	Passed
Shear	: 14.6,	Allowable	: 82.6 N./mm ²	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, min(Curve:B, Curve:A)

Govrn. thk, tg = 15.958, tr = 0.509, c = 3.0 mm., E* = 1.0
 Thickness Ratio = tr * E*/(tg - c) = 0.039, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve A	6 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C

Nozzle Neck to Pad Weld for the Nozzle, Curve: B

Govrn. thk, tg = 12.0, c = 3.0 mm., E* = 1.0
 Thickness Ratio = tr * E*/(tg - c) = 0.45, Temp. Reduction = 40 °C
 Attachment governing, Attachment stress = Shell stress per Code

Min Metal Temp. w/o impact per UCS-66, Curve B	-23 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-48 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Nozzle Neck to Pad Weld for Reinforcement pad, Curve: B

Govrn. thk, tg = 12.0, c = 3.0 mm., E* = 1.0
 Thickness Ratio = tr * E*/(tg - c) = 0.45, Temp. Reduction = 40 °C
 Attachment governing, Attachment stress = Shell stress per Code

Min Metal Temp. w/o impact per UCS-66, Curve B	-23 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-48 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C



Shell to Pad Weld Junction at Pad OD, Curve: B

Govrn. thk, $t_g = 12.0$, $c = 3.0$ mm., $E^* = 1.0$
 Thickness Ratio = $tr * E^* / (t_g - c) = 0.45$, Temp. Reduction = 40 °C
 Attachment governing, Attachment stress = Shell stress per Code

Min Metal Temp. w/o impact per UCS-66, Curve B	-23 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-48 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Nozzle-Shell/Head Weld (UCS-66(a)(1)(b)), Curve: B

Govrn. thk, $t_g = 14.0$, $tr = 4.955$, $c = 3.0$ mm., $E^* = 1.0$
 Thickness Ratio = $tr * E^* / (t_g - c) = 0.45$, Temp. Reduction = 40 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-18 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-48 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Gov. MDMT of the Nozzle	:	-48 °C
Gov. MDMT of the Reinforcement Pad	:	-48 °C
Gov. MDMT of the nozzle to shell joint welded assembly	:	-48 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c)	-18 °C
Flange MDMT with Temp reduction per UCS-66(b)(1)(-b)	-104 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
 Design Pressure/Ambient Rating = $7.15/51.00 = 0.140$

Weld Size Calculations, Description: N6-6"

Intermediate Calc. for nozzle/shell Welds	T_{min}	12.0000 mm.
Intermediate Calc. for pad/shell Welds	T_{minPad}	11.0000 mm.

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	$6.0000 = \text{Min per Code}$	$7.0700 = 0.7 * W_o \text{ mm.}$
Pad Weld	$5.5000 = 0.5 * T_{minPad}$	$7.0700 = 0.7 * W_p \text{ mm.}$

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:
 $= \max(0, (A-A1+2 \cdot t_n \cdot fr1 \cdot (E1 \cdot t-tr)) \cdot Sv)$
 $= \max(0, (5.5095 - 4.794 + 2 \cdot 12.9576 \cdot 0.8551 \cdot (1.0 \cdot 11.0 - 7.5412)) \cdot 138)$
 $= 2085.28 \text{ Kg}$

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:
 $= (A2+A5+A4-(W_i-Can/.707)^2 \cdot fr2) \cdot Sv$
 $= (5.8149 + 13.9734 + 0.8551 - 0.0 \cdot 0.86) \cdot 138$
 $= 29048.46 \text{ Kg}$

Weld Load [W2]:
 $= (A2 + A3 + A4 + (2 \cdot t_n \cdot t \cdot fr1)) \cdot Sv$
 $= (5.8149 + 0.0 + 0.8551 + (2.4375)) \cdot 138$
 $= 12815.70 \text{ Kg}$

Weld Load [W3]:
 $= (A2+A3+A4+A5+(2 \cdot t_n \cdot t \cdot fr1)) \cdot S$
 $= (5.8149 + 0.0 + 0.8551 + 13.9734 + (2.4375)) \cdot 138$
 $= 32478.43 \text{ Kg}$

Strength of Connection Elements for Failure Path Analysis



Shear, Outward Nozzle Weld [Sonw]:

$$= (\pi/2) \cdot D_{lo} \cdot W_o \cdot 0.49 \cdot S_{nw}$$

$$= (3.1416/2.0) \cdot 168.275 \cdot 10.0 \cdot 0.49 \cdot 118$$

$$= 15584. \text{ Kg}$$

Shear, Pad Element Weld [Spew]:

$$= (\pi/2) \cdot D_P \cdot W_P \cdot 0.49 \cdot S_{EW}$$

$$= (3.1416/2.0) \cdot 318.275 \cdot 10.0 \cdot 0.49 \cdot 138$$

$$= 34472. \text{ Kg}$$

Shear, Nozzle Wall [Snw]:

$$= (\pi \cdot (D_{lr} + D_{lo})/4) \cdot (Thk - Can) \cdot 0.7 \cdot S_n$$

$$= (3.1416 \cdot 77.6587) \cdot (15.9576 - 3.0) \cdot 0.7 \cdot 118$$

$$= 26626. \text{ Kg}$$

Tension, Pad Groove Weld [Tpgw]:

$$= (\pi/2) \cdot D_{lo} \cdot W_{gpn} \cdot 0.74 \cdot S_{eg}$$

$$= (3.1416/2) \cdot 168.275 \cdot 12.0 \cdot 0.74 \cdot 138$$

$$= 33029. \text{ Kg}$$

Tension, Shell Groove Weld [Tngw]:

$$= (\pi/2) \cdot D_{lo} \cdot (T - cas) \cdot 0.74 \cdot S_{ng}$$

$$= (3.1416/2.0) \cdot 168.275 \cdot (14.0 - 3.0) \cdot 0.74 \cdot 138$$

$$= 30277. \text{ Kg}$$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:

$$= (S_{PEW} + S_{NW}) = (34472 + 26626) = 61098 \text{ Kg}$$

Failure Path [Path 2-2]:

$$= (S_{onw} + T_{pgw} + T_{ngw} + S_{inw})$$

$$= (15584 + 33029 + 30277 + 0) = 78890 \text{ Kg}$$

Failure Path [Path 3-3]:

$$= (S_{pew} + T_{ngw} + S_{inw})$$

$$= (34472 + 30277 + 0) = 64748 \text{ Kg}$$

Summary of Failure Path Calculations:

Path 1-1 = 61097	Kgf, must exceed W = 2085 Kg	or W1 = 29048 Kg
Path 2-2 = 78889	Kgf, must exceed W = 2085 Kg	or W2 = 12815 Kg
Path 3-3 = 64748	Kgf, must exceed W = 2085 Kg	or W3 = 32478 Kg

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 13 bars

Note: The MAWP of this junction was limited by the parent Shell/Head.

Nozzle is O.K. for the External Pressure: 1 bars

The Drop for this Nozzle is : 3.7332 mm.

The Cut Length for this Nozzle is, Drop + Ho + H + T : 222.5202 mm.

Input Echo, WRC297 Item 1, Description: N6-6" :

Diameter Basis for Cylindrical Shell	ID	
Shell Internal Corrosion Allowance	3.0000	mm.
Shell External Corrosion Allowance	0.0000	mm.
Shell Diameter	1900.000	mm.
Shell Thickness	14.0000	mm.
Shell Stress Concentration Factor	1.000	
Vessel Material	SA-516 70	
Vessel Cold S.I. Allowable	Smc	138.00 N./mm ²
Vessel Hot S.I. Allowable	Smh	138.00 N./mm ²

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.



Diameter Basis for Nozzle		OD	
Nozzle Corrosion Allowance		3.0000	mm.
Nozzle Diameter		168.275	mm.
Nozzle Thickness		15.9576	mm.
Nozzle Stress Concentration Factor		1.000	
Nozzle Material		SA-106 B	
Nozzle Cold S.I. Allowable	SNmc	118.00	N./mm ²
Nozzle Hot S.I. Allowable	SNmh	118.00	N./mm ²

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Diameter of Reinforcing Pad	318.275	mm.
Thickness of Reinforcing Pad	12.0000	mm.

Note:

External Forces and Moments in WRC 107/537 Convention:

These loads are assumed to be SUSTAINED loads.

Design Internal Pressure	Dp	7.15	bars
Radial Load	P	-1150.00	Kgf
Circumferential Shear	Vc	764.84	Kgf
Longitudinal Shear	Vl	764.84	Kgf
Circumferential Moment	Mc	321.11	Kg-m.
Longitudinal Moment	Ml	321.11	Kg-m.
Torsional Moment	Mt	458.73	Kg-m.

Include Axial Pressure Thrust	No
Include Pressure Stress Indices per Div. 2	No
Local Loads applied at end of Nozzle/Attachment	No

Stress Computations at the Edge of the Nozzle:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{(R_{\text{mean}}(t - ca))}$$

$$= 168.275 + 2 \cdot 1.65 \cdot \sqrt{(964.5(14.0 - 3.0))}$$

$$= 508.183 \text{ mm.}$$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter	(D) =	1929.000	mm.
Nozzle Outside Diameter	(d) =	168.275	mm.
Vessel Thickness used	(T) =	23.000	mm.
Nozzle Thickness used	(t) =	12.958	mm.
T / t	=	1.775	
d / t	=	12.987	

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 0.799$$

Nr/P	=	0.143
Mr/P	=	0.160
M0/P	=	0.082
N0/P	=	0.157
MrD/Mc	=	0.235
NrDL/Mc	=	0.171
M0d/Mc	=	0.148
N0DL/Mc	=	0.108
MrD/Ml	=	0.191
NrDL/Ml	=	0.146
M0D/Ml	=	0.119
N0DL/Ml	=	0.124

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stresses Normal to	Top	Top	Bottom	Bottom
longitudinal plane)	Outside	Inside	Outside	Inside



Outplane Membrane (P)	3	3	3	3
Outplane Bending (P)	10	-10	10	-10
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	-4	-4	4	4
Outplane Bending (ML)	-25	25	25	-25
Normal Pressure Stress	29	30	29	30

Outplane Stress Summary	13	43	72	1
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Vessel Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stresses parallel to	Top	Top	Bottom	Bottom
longitudinal plane)	Outside	Inside	Outside	Inside

Inplane Membrane (P)	3	3	3	3
Inplane Bending (P)	20	-20	20	-20
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	-5	-5	5	5
Inplane Bending (ML)	-40	40	40	-40
Inplane Pressure Stress	14	14	14	14

Inplane Stress Summary	-7	32	83	-38
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Vessel Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Shear stress normal to	Top	Top	Bottom	Bottom
longitudinal plane)	Outside	Inside	Outside	Inside

Outplane Shear (Vc)	1	1	-1	-1
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	4	4	4	4

Shear Stress Summary	5	5	3	3
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Vessel Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stress Intensities	Top	Top	Bottom	Bottom
	Outside	Inside	Outside	Inside

Two * Max Shear Stress	23	46	84	40
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Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stresses Normal to	Left	Left	Right	Right
circumferential plane)	Outside	Inside	Outside	Inside

Outplane Membrane (P)	3	3	3	3
Outplane Bending (P)	10	-10	10	-10
Outplane Membrane (Mc)	-3	-3	3	3
Outplane Bending (Mc)	-31	31	31	-31
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	14	14	14	14

Outplane Stress Summary	-6	35	63	-20
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Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stresses parallel to	Left	Left	Right	Right
circumferential plane)	Outside	Inside	Outside	Inside

Inplane Membrane (P)	3	3	3	3
Inplane Bending (P)	20	-20	20	-20
Inplane Membrane (Mc)	-6	-6	6	6
Inplane Bending (Mc)	-49	49	49	-49
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0



Inplane Pressure Stress	29	30	29	30
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Inplane Stress Summary	-3	56	108	-31
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Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Shear stress normal to	Left	Left	Right	Right
circumferential plane)	Outside	Inside	Outside	Inside

Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	-1	-1	1	1
Torsional Shear (Mt)	4	4	4	4

Shear Stress Summary	3	3	5	5
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Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stress Intensities	Left	Left	Right	Right
	Outside	Inside	Outside	Inside

Two * Max Shear Stress	8	56	109	33
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Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	B1
(Stresses in the	Top	Top	Bottom	Bottom
hoop direction)	Outside	Inside	Outside	Inside

Hoop Membrane (P)	3	3	3	3
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	-4	-4	4	4
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	4	5	4	5

Hoop Stress Summary	3	4	12	12
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Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	B1
(Stresses Normal to	Top	Top	Bottom	Bottom
pipe cross-section)	Outside	Inside	Outside	Inside

Axial Membrane (P)	1	1	1	1
Axial Bending (P)	35	-35	35	-35
Axial Membrane (Mc)	0	0	0	0
Axial Bending (Mc)	0	0	0	0
Axial Membrane (ML)	-13	-13	13	13
Axial Bending (ML)	-78	78	78	-78
Axial Pressure Stress	2	2	2	2

Axial Stress Summary	-52	33	132	-96
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Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	B1
(Shear stress)	Outside	Inside	Outside	Inside

Shear due to (Vc)	2	2	-2	-2
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	7	7	7	7

Shear Stress Summary	9	9	5	5
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Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	B1
(Stress Intensities	Outside	Inside	Outside	Inside

Two * Max Shear Stress	59	36	132	110
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Nozzle Stresses (N./mm²):



CIRCUMFERENTIAL PLANE (Stresses in the hoop direction)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Hoop Membrane (P)	3	3	3	3
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	-3	-3	3	3
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	4	5	4	5
Hoop Stress Summary	3	4	11	12

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses Normal to pipe cross-section)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Axial Membrane (P)	1	1	1	1
Axial Bending (P)	35	-35	35	-35
Axial Membrane (Mc)	-13	-13	13	13
Axial Bending (Mc)	-99	99	99	-99
Axial Membrane (ML)	0	0	0	0
Axial Bending (ML)	0	0	0	0
Axial Pressure Stress	2	2	2	2
Axial Stress Summary	-73	54	153	-117

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Shear stress)	Cu Outside	Cl Inside	Du Outside	Dl Inside
Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	-2	-2	2	2
Shear due to Torsion	7	7	7	7
Shear Stress Summary	5	5	9	9

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stress Intensities)	Cu Outside	Cl Inside	Du Outside	Dl Inside
Two * Max Shear Stress	78	54	153	131

Stress Computations at the Edge of the Reinforcing Pad:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{R_{\text{mean}}(t - ca)}$$

$$= 168.275 + 2 \cdot 1.65 \cdot \sqrt{958.5(14.0 - 3.0)}$$

$$= 507.124 \text{ mm.}$$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter	(D) =	1917.000 mm.
Nozzle Outside Diameter	(d) =	318.275 mm.
Vessel Thickness used	(T) =	11.000 mm.
Nozzle Thickness used	(t) =	87.958 mm.
T / t	=	0.125
d / t	=	10.000

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 2.192$$

Note: Re-pad thickness is added to nozzle thickness to simulate the Re-pad.

Nr/P	=	0.077
Mr/P	=	0.137
M0/P	=	0.040



NØ/P = 0.056
 MrD/Mc = 0.286
 NrDL/Mc = 0.107
 MØd/Mc = 0.087
 NØDL/Mc = 0.066
 MrD/Ml = 0.103
 NrDL/Ml = 0.101
 MØD/Ml = 0.031
 NØDL/Ml = 0.075

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Membrane (P)	5	5	5	5
Outplane Bending (P)	22	-22	22	-22
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	-6	-6	6	6
Outplane Bending (ML)	-15	15	15	-15
Normal Pressure Stress	61	62	61	62
Outplane Stress Summary	68	54	110	35

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses parallel to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Inplane Membrane (P)	7	7	7	7
Inplane Bending (P)	76	-76	76	-76
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	-8	-8	8	8
Inplane Bending (ML)	-50	50	50	-50
Inplane Pressure Stress	30	30	30	30
Inplane Stress Summary	55	3	173	-80

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Shear stress normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Shear (Vc)	1	1	-1	-1
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	2	2	2	2
Shear Stress Summary	3	3	1	1

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stress Intensities)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Two * Max Shear Stress	69	54	173	116

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses Normal to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Outplane Membrane (P)	5	5	5	5
Outplane Bending (P)	22	-22	22	-22
Outplane Membrane (Mc)	-5	-5	5	5
Outplane Bending (Mc)	-42	42	42	-42
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0



Normal Pressure Stress	30	30	30	30
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Outplane Stress Summary	10	50	106	-23
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Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses parallel to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
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Inplane Membrane (P)	7	7	7	7
Inplane Bending (P)	76	-76	76	-76
Inplane Membrane (Mc)	-8	-8	8	8
Inplane Bending (Mc)	-140	140	140	-140
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	61	62	61	62

Inplane Stress Summary	-3	124	294	-138
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Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Shear stress normal to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
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Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	-1	-1	1	1
Torsional Shear (Mt)	2	2	2	2

Shear Stress Summary	1	1	3	3
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Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stress Intensities)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
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Two * Max Shear Stress	14	124	294	138
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WRC 297 Stress Summations per ASME Sec. VIII Div. 2:**Vessel Stress Summation at Vessel-Nozzle Junction (N./mm²):**

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	29	30	29	30	29	30	29	30
Circ. Pl (SUS)	-1	-1	7	7	-3	-3	9	9
Circ. Q (SUS)	-15	15	35	-35	-29	29	69	-69
Long. Pm (SUS)	14	14	14	14	14	14	14	14
Long. Pl (SUS)	-2	-2	8	8	0	0	6	6
Long. Q (SUS)	-20	20	60	-60	-21	21	41	-41
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	1	1	-1	-1	-1	-1	1	1
Shear Q (SUS)	4	4	4	4	4	4	4	4
Pm (SUS)	29.0	30.0	29.0	30.0	29.0	30.0	29.0	30.0
Pm+Pl (SUS)	28.1	29.1	36.1	37.1	26.1	27.1	38.1	39.1
Pm+Pl+Q (Total)	23.3	45.8	82.8	40.4	8.6	56.4	107.5	32.2

Vessel Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
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Pm (SUS)	30.00	138.00	Passed
Pm+Pl (SUS)	39.05	207.00	Passed
Pm+Pl+Q (TOTAL)	107.54	413.99	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Reinforcing Pad Edge (N./mm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	61	62	61	62	61	62	61	62
Circ. Pl (SUS)	-1	-1	11	11	-1	-1	15	15
Circ. Q (SUS)	7	-7	37	-37	-64	64	216	-216
Long. Pm (SUS)	30	30	30	30	30	30	30	30
Long. Pl (SUS)	-1	-1	15	15	0	0	10	10
Long. Q (SUS)	26	-26	126	-126	-20	20	64	-64
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	1	1	-1	-1	-1	-1	1	1
Shear Q (SUS)	2	2	2	2	2	2	2	2
Pm (SUS)	61.0	62.0	61.0	62.0	61.0	62.0	61.0	62.0
Pm+Pl (SUS)	60.0	61.0	72.0	73.0	60.0	61.0	76.0	77.0
Pm+Pl+Q (Total)	67.7	54.2	171.0	117.0	14.1	125.0	292.0	139.1

Vessel Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	62.00	138.00	Passed
Pm+Pl (SUS)	77.03	207.00	Passed
Pm+Pl+Q (TOTAL)	292.05	413.99	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Nozzle Stress Summation at Vessel-Nozzle Junction (N./mm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	4	5	4	5	4	5	4	5
Circ. Pl (SUS)	-1	-1	7	7	0	0	6	6
Circ. Q (SUS)	0	0	0	0	0	0	0	0
Long. Pm (SUS)	2	2	2	2	2	2	2	2
Long. Pl (SUS)	-12	-12	14	14	-12	-12	14	14
Long. Q (SUS)	-43	43	113	-113	-64	64	134	-134
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	2	2	-2	-2	-2	-2	2	2
Shear Q (SUS)	7	7	7	7	7	7	7	7
Pm (SUS)	4.0	5.0	4.0	5.0	4.0	5.0	4.0	5.0
Pm+Pl (SUS)	13.6	14.6	16.7	16.8	14.6	15.5	16.6	16.7
Pm+Pl+Q (Total)	58.8	35.6	129.2	109.5	78.6	54.5	150.6	130.2

**Nozzle Stress Summation Comparison (N./mm²):**

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	5.00	118.00	Passed
Pm+Pl (SUS)	16.83	177.00	Passed
Pm+Pl+Q (TOTAL)	150.58	353.99	Passed

*Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.*

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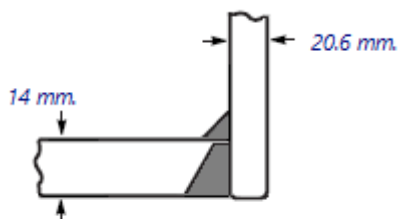
**Input, Nozzle Desc: N4-3"****From: 50**

Pressure for Reinforcement Calculations	P	7.153	bars
Temperature for Internal Pressure	Temp	230	°C
Design External Pressure	Pext	1.03	bars
Temperature for External Pressure	Tempex	198	°C
Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	138.00	N./mm2
Parent Allowable Stress At Ambient	Sva	138.00	N./mm2
Inside Diameter of Cylindrical Shell	D	1900.00	mm.
Design Length of Section	L	3008.3333	mm.
Shell Finished (Minimum) Thickness	t	14.0000	mm.
Shell Internal Corrosion Allowance	cas	3.0000	mm.
Shell External Corrosion Allowance	co	0.0000	mm.
Distance from Bottom/Left Tangent		4050.00	mm.
User Entered Minimum Design Metal Temperature		-1.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-106 B	
Material UNS Number		K03006	
Material Specification/Type		Smls. pipe	
Allowable Stress at Temperature	Sn	118.00	N./mm2
Allowable Stress At Ambient	Sna	118.00	N./mm2
Diameter Basis (for tr calc only)		Inside	
Layout Angle		90.00	deg
Diameter		3.0000	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	20.5740	mm.
Flange Material		SA-106 B	
Flange Type		Long Weld Neck	
Corrosion Allowance	can	3.0000	mm.
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	200.0250	mm.
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm.
Groove weld depth between Nozzle and Vessel	Wgnv	9.0000	mm.
Inside Projection	h	0.0000	mm.
Weld leg size, Inside Element to Shell	Wi	0.0000	mm.
Flange Class		300	
Flange Grade		GR 1.1	

The Pressure Design option was MAWP + static head (to the nozzle).

**Reinforcement CALCULATION, Description: N4-3"**



ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation 3.000 in.
 Actual Thickness Used in Calculation 0.810 in.

Messages regarding the Nozzle Input Data:

For this nozzle, the actual wall thickness was specified. In these cases, the weight of the attached nozzle flange cannot be determined. It is required to manually compute the weight of the assembly and enter it into the nozzle dialog's overriding weight field.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 7.15 \cdot 953.0 / (138 \cdot 1.0 - 0.6 \cdot 7.15)$
 $= 4.9552$ mm.

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R / (S_n \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 7.15 \cdot 41.1 / (118 \cdot 1.0 - 0.6 \cdot 7.15)$
 $= 0.2501$ mm.

Required Nozzle thickness under External Pressure per UG-28 : 0.4762 mm.

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	164.4000	mm.
Parallel to Vessel Wall, opening length	d	82.2000	mm.
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	27.5000	mm.

*Taking a UG-36(c)(3)(a) exemption for nozzle: N4-3".
 This calculation is valid for nozzles that meet all the requirements of paragraph UG-36. Please check the Code carefully, especially for nozzles that are not isolated or do not meet Code spacing requirements. To force the computation of areas for small nozzles go to Tools->Configuration and check the box to force the UG-37 small nozzle area calculation.*

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta	= 3.4762	mm.
Wall Thickness per UG16(b),	tr16b	= 5.5000	mm.
Wall Thickness, shell/head, internal pressure	trb1	= 7.9552	mm.
Wall Thickness	tb1 = max(trb1, tr16b)	= 7.9552	mm.
Wall Thickness, shell/head, external pressure	trb2	= 3.7146	mm.
Wall Thickness	tb2 = max(trb2, tr16b)	= 5.5000	mm.
Wall Thickness per table UG-45	tb3	= 8.7300	mm.

Determine Nozzle Thickness candidate [tb]:
 $= \min[tb3, \max(tb1, tb2)]$
 $= \min[8.73, \max(7.9552, 5.5)]$
 $= 7.9552$ mm.

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
 $= \max(ta, tb)$
 $= \max(3.4762, 7.9552)$
 $= 7.9552$ mm.

Available Nozzle Neck Thickness = 20.5740 mm. -> OK

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained	:	12.3,	Allowable	:	118.0	N./mm2	Passed
Expansion	:	0.0,	Allowable	:	282.7	N./mm2	Passed
Occasional	:	0.7,	Allowable	:	156.9	N./mm2	Passed
Shear	:	9.4,	Allowable	:	82.6	N./mm2	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

**Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:****Nozzle-Shell/Head Weld (UCS-66(a)(b)), Curve: B**

Govrn. thk, tg = 14.0, tr = 4.955, c = 3.0 mm., $E^* = 1.0$
 Thickness Ratio = $tr * E^* / (tg - c) = 0.45$, Temp. Reduction = 40 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-18 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-48 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Gov. MDMT of the nozzle to shell joint welded assembly : -48 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c)	-18 °C
Flange MDMT with Temp reduction per UCS-66(b)(1)(-b)	-104 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
 Design Pressure/Ambient Rating = $7.15/51.00 = 0.140$

Weld Size Calculations, Description: N4-3"

Intermediate Calc. for nozzle/shell Welds Tmin 11.0000 mm.

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * Wo mm.

Skipping the nozzle attachment weld strength calculations. Per UW-15(b)(2) the nozzles exempted by UG-36(c)(3)(a) (small nozzles) do not require a weld strength check.

The Drop for this Nozzle is : 1.8136 mm.
 The Cut Length for this Nozzle is, Drop + Ho + H + T : 215.8387 mm.

Input Echo, WRC297 Item 1, Description: N4-3" :

Diameter Basis for Cylindrical Shell	ID	
Shell Internal Corrosion Allowance	3.0000	mm.
Shell External Corrosion Allowance	0.0000	mm.
Shell Diameter	1900.000	mm.
Shell Thickness	14.0000	mm.
Shell Stress Concentration Factor	1.000	
Vessel Material	SA-516 70	
Vessel Cold S.I. Allowable	Smc	138.00 N./mm ²
Vessel Hot S.I. Allowable	Smh	138.00 N./mm ²

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Diameter Basis for Nozzle	ID	
Nozzle Corrosion Allowance	3.0000	mm.
Nozzle Diameter	76.200	mm.
Nozzle Thickness	20.5740	mm.
Nozzle Stress Concentration Factor	1.000	
Nozzle Material	SA-106 B	
Nozzle Cold S.I. Allowable	SNmc	118.00 N./mm ²
Nozzle Hot S.I. Allowable	SNmh	118.00 N./mm ²

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Note:

External Forces and Moments in WRC 107/537 Convention:
 These loads are assumed to be SUSstained loads.

Design Internal Pressure	Dp	7.15	bars
Radial Load	P	611.87	Kgf
Circumferential Shear	Vc	407.91	Kgf



Longitudinal Shear	Vl	407.91	Kgf
Circumferential Moment	Mc	91.34	Kg-m.
Longitudinal Moment	Ml	91.34	Kg-m.
Torsional Moment	Mt	130.48	Kg-m.

Include Axial Pressure Thrust	No
Include Pressure Stress Indices per Div. 2	No
Local Loads applied at end of Nozzle/Attachment	No

*Warning - The ratio, Dn/Tn (5.704) is < 10
check the limitations of WRC bulletin 297.*

Stress Computations at the Edge of the Nozzle:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{(R_{\text{mean}}(t - c_a))}$$

$$= 117.348 + 2 \cdot 1.65 \cdot \sqrt{(958.5(14.0 - 3.0))}$$

$$= 456.197 \text{ mm.}$$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter	(D) =	1917.000 mm.
Nozzle Outside Diameter	(d) =	117.348 mm.
Vessel Thickness used	(T) =	11.000 mm.
Nozzle Thickness used	(t) =	17.574 mm.
T / t	=	0.626
d / t	=	10.000

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 0.808$$

Nr/P	=	0.122
Mr/P	=	0.244
M0/P	=	0.074
N0/P	=	0.142
MrD/Mc	=	0.303
NrDL/Mc	=	0.102
M0d/Mc	=	0.101
N0DL/Mc	=	0.003
MrD/Ml	=	0.221
NrDL/Ml	=	0.098
M0D/Ml	=	0.076
N0DL/Ml	=	0.098

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Membrane (P)	-7	-7	-7	-7
Outplane Bending (P)	-22	22	-22	22
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (Ml)	-6	-6	6	6
Outplane Bending (Ml)	-28	28	28	-28
Normal Pressure Stress	61	62	61	62
Outplane Stress Summary	-2	99	67	54

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses parallel to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Inplane Membrane (P)	-6	-6	-6	-6
Inplane Bending (P)	-72	72	-72	72
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (Ml)	-6	-6	6	6
Inplane Bending (Ml)	-83	83	83	-83



Inplane Pressure Stress	30	30	30	30
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Inplane Stress Summary	-137	174	41	19
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Vessel Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Shear stress normal to longitudinal plane)	Top Outside	Top Inside	Bottom Outside	Bottom Inside

Outplane Shear (Vc)	1	1	-1	-1
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	5	5	5	5

Shear Stress Summary	7	7	3	3
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Vessel Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stress Intensities)	Top Outside	Top Inside	Bottom Outside	Bottom Inside

Two * Max Shear Stress	138	175	67	55
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Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stresses Normal to circumferential plane)	Left Outside	Left Inside	Right Outside	Right Inside

Outplane Membrane (P)	-7	-7	-7	-7
Outplane Bending (P)	-22	22	-22	22
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	-38	38	38	-38
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	30	30	30	30

Outplane Stress Summary	-36	83	40	7
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Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stresses parallel to circumferential plane)	Left Outside	Left Inside	Right Outside	Right Inside

Inplane Membrane (P)	-6	-6	-6	-6
Inplane Bending (P)	-72	72	-72	72
Inplane Membrane (Mc)	-6	-6	6	6
Inplane Bending (Mc)	-114	114	114	-114
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	61	62	61	62

Inplane Stress Summary	-138	237	104	20
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Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Shear stress normal to circumferential plane)	Left Outside	Left Inside	Right Outside	Right Inside

Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	-1	-1	1	1
Torsional Shear (Mt)	5	5	5	5

Shear Stress Summary	3	3	7	7
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Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stress Intensities)	Left Outside	Left Inside	Right Outside	Right Inside

Two * Max Shear Stress	138	237	104	23
------------------------	-----	-----	-----	----

**Nozzle Stresses (N./mm²):**

LONGITUDINAL PLANE (Stresses in the hoop direction)	Au	Al	Bu	Bl
	Top	Top	Bottom	Bottom
	Outside	Inside	Outside	Inside
Hoop Membrane (P)	-7	-7	-7	-7
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	-6	-6	6	6
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	1	2	1	2
Hoop Stress Summary	-11	-10	0	1

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses Normal to pipe cross-section)	Au	Al	Bu	Bl
	Top	Top	Bottom	Bottom
	Outside	Inside	Outside	Inside
Axial Membrane (P)	-1	-1	-1	-1
Axial Bending (P)	-21	21	-21	21
Axial Membrane (Mc)	0	0	0	0
Axial Bending (Mc)	0	0	0	0
Axial Membrane (ML)	-7	-7	7	7
Axial Bending (ML)	-25	25	25	-25
Axial Pressure Stress	0	0	0	0
Axial Stress Summary	-54	39	11	3

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Shear stress)	Au	Al	Bu	Bl
	Outside	Inside	Outside	Inside
	Outside	Inside	Outside	Inside
Shear due to (Vc)	1	1	-1	-1
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	3	3	3	3
Shear Stress Summary	4	4	2	2

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Stress Intensities)	Au	Al	Bu	Bl
	Outside	Inside	Outside	Inside
	Outside	Inside	Outside	Inside
Two * Max Shear Stress	55	50	11	4

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses in the hoop direction)	Cu	Cl	Du	Dl
	Left	Left	Right	Right
	Outside	Inside	Outside	Inside
Hoop Membrane (P)	-7	-7	-7	-7
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	1	2	1	2
Hoop Stress Summary	-5	-4	-5	-4

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses Normal to pipe cross-section)	Cu	Cl	Du	Dl
	Left	Left	Right	Right
	Outside	Inside	Outside	Inside
Axial Membrane (P)	-1	-1	-1	-1
Axial Bending (P)	-21	21	-21	21



Axial Membrane (Mc)	-7	-7	7	7
Axial Bending (Mc)	-37	37	37	-37
Axial Membrane (ML)	0	0	0	0
Axial Bending (ML)	0	0	0	0
Axial Pressure Stress	0	0	0	0

Axial Stress Summary	-66	51	23	-8
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Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Shear stress)	Cu	Cl	Du	Dl
	Outside	Inside	Outside	Inside
Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	-1	-1	1	1
Shear due to Torsion	3	3	3	3

Shear Stress Summary	2	2	4	4
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Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stress Intensities)	Cu	Cl	Du	Dl
	Outside	Inside	Outside	Inside
Two * Max Shear Stress	66	56	29	11

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:**Vessel Stress Summation at Vessel-Nozzle Junction (N./mm²):**

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	61	62	61	62	61	62	61	62
Circ. Pl (SUS)	-13	-13	-1	-1	-12	-12	0	0
Circ. Q (SUS)	-50	50	6	-6	-186	186	42	-42
Long. Pm (SUS)	30	30	30	30	30	30	30	30
Long. Pl (SUS)	-12	-12	0	0	-7	-7	-7	-7
Long. Q (SUS)	-155	155	11	-11	-60	60	16	-16
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	1	1	-1	-1	-1	-1	1	1
Shear Q (SUS)	5	5	5	5	5	5	5	5
Pm (SUS)	61.0	62.0	61.0	62.0	61.0	62.0	61.0	62.0
Pm+Pl (SUS)	48.0	49.0	60.0	61.0	49.0	50.0	61.0	62.0
Pm+Pl+Q (Total)	137.3	173.5	66.6	55.4	137.2	236.1	103.6	22.3

Vessel Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	62.00	138.00	Passed
Pm+Pl (SUS)	62.03	207.00	Passed
Pm+Pl+Q (TOTAL)	236.10	413.99	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:**Nozzle Stress Summation at Vessel-Nozzle Junction (N./mm²):**

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl



Circ. Pm (SUS)	1	2	1	2	1	2	1	2
Circ. Pl (SUS)	-13	-13	-1	-1	-7	-7	-7	-7
Circ. Q (SUS)	0	0	0	0	0	0	0	0
Long. Pm (SUS)	0	0	0	0	0	0	0	0
Long. Pl (SUS)	-8	-8	6	6	-8	-8	6	6
Long. Q (SUS)	-46	46	4	-4	-58	58	16	-16
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	1	1	-1	-1	-1	-1	1	1
Shear Q (SUS)	3	3	3	3	3	3	3	3
Pm (SUS)	1.0	2.0	1.0	2.0	1.0	2.0	1.0	2.0
Pm+Pl (SUS)	12.2	11.3	6.3	6.2	8.4	8.3	12.2	11.2
Pm+Pl+Q (Total)	54.4	49.6	10.8	4.1	66.1	55.1	29.1	12.2

Nozzle Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	2.00	118.00	Passed
Pm+Pl (SUS)	12.24	177.00	Passed
Pm+Pl+Q (TOTAL)	66.07	353.99	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

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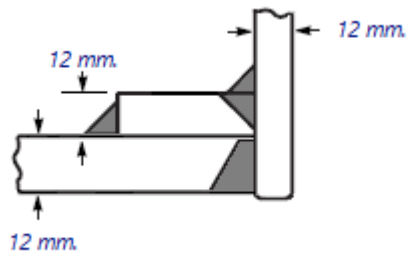
**Input, Nozzle Desc: N3-20"****From: 60**

Pressure for Reinforcement Calculations	P	7.153	bars
Temperature for Internal Pressure	Temp	230	°C
Design External Pressure	Pext	1.03	bars
Temperature for External Pressure	Tempex	198	°C
Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	138.00	N./mm2
Parent Allowable Stress At Ambient	Sva	138.00	N./mm2
Inside Diameter of Elliptical Head	D	1900.00	mm.
Aspect Ratio of Elliptical Head	Ar	2.00	
Head Finished (Minimum) Thickness	t	12.0000	mm.
Head Internal Corrosion Allowance	cas	3.0000	mm.
Head External Corrosion Allowance	co	0.0000	mm.
Distance from Head Centerline	L1	0.0000	mm.
User Entered Minimum Design Metal Temperature		-1.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-516 70	
Material UNS Number		K02700	
Material Specification/Type		Plate	
Allowable Stress at Temperature	Sn	138.00	N./mm2
Allowable Stress At Ambient	Sna	138.00	N./mm2
Diameter Basis (for tr calc only)		Outside	
Layout Angle		0.00	deg
Diameter		20.0000	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	12.0000	mm.
Flange Material		SA-105	
Flange Type		Weld Neck Flange	
Corrosion Allowance	can	3.0000	mm.
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	150.0000	mm.
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm.
Groove weld depth between Nozzle and Vessel	Wgnv	9.0000	mm.
Inside Projection	h	0.0000	mm.
Weld leg size, Inside Element to Shell	Wi	0.0000	mm.
Pad Material		SA-516 70	
Pad Allowable Stress at Temperature	Sp	138.00	N./mm2
Pad Allowable Stress At Ambient	Spa	138.00	N./mm2
Diameter of Pad along vessel surface	Dp	622.0000	mm.
Thickness of Pad	te	12.0000	mm.
Weld leg size between Pad and Shell	Wp	8.0000	mm.
Groove weld depth between Pad and Nozzle	Wgpn	12.0000	mm.
Reinforcing Pad Width		57.0000	mm.
Flange Class		150	
Flange Grade		GR 1.1	

The Pressure Design option was MAWP + static head (to the nozzle).

**Reinforcement CALCULATION, Description: N3-20"**

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Outside Diameter Used in Calculation	20.000 in.
Actual Thickness Used in Calculation	0.472 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Elliptical Head, tr [Int. Press]
 $= P \cdot D \cdot K1 / (2 \cdot Sv \cdot E - 0.2 \cdot P)$ per Appendix 1-4(c)
 $= 7.15 \cdot 1906.0 \cdot 0.8973 / (2 \cdot 138.0 \cdot 1.0 - 0.2 \cdot 7.15)$
 $= 4.4348 \text{ mm.}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot Ro / (Sn \cdot E + 0.4 \cdot P)$ per Appendix 1-1 (a)(1)
 $= 7.15 \cdot 254.0 / (138 \cdot 1.0 + 0.4 \cdot 7.15)$
 $= 1.3139 \text{ mm.}$

Required Nozzle thickness under External Pressure per UG-28 : 0.9923 mm.

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	980.0001 mm.
Parallel to Vessel Wall, opening length	d	490.0000 mm.
Normal to Vessel Wall (Thickness Limit), pad side Tlwp		22.5000 mm.

Weld Strength Reduction Factor [fr1]:
 $= \min(1, Sn/Sv)$
 $= \min(1, 138.0/138.0)$
 $= 1.000$

Weld Strength Reduction Factor [fr2]:
 $= \min(1, Sn/Sv)$
 $= \min(1, 138.0/138.0)$
 $= 1.000$

Weld Strength Reduction Factor [fr4]:
 $= \min(1, Sp/Sv)$
 $= \min(1, 138.0/138.0)$
 $= 1.000$

Weld Strength Reduction Factor [fr3]:
 $= \min(fr2, fr4)$
 $= \min(1.0, 1.0)$
 $= 1.000$

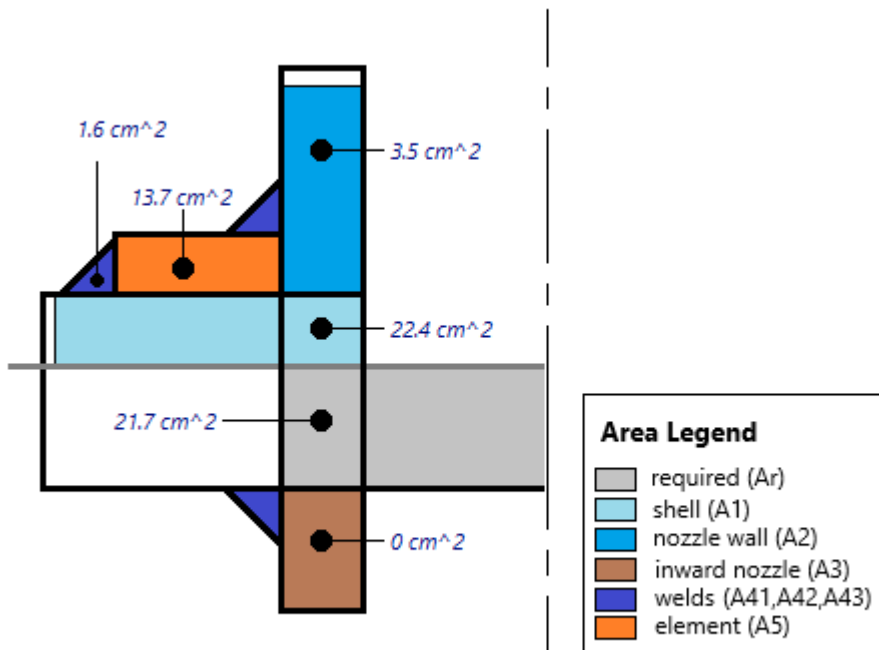


Figure UG 37.1 Nozzle Areas

Results of Nozzle Reinforcement Area Calculations: (cm²)

AREA AVAILABLE, A1 to A5		MAWP	External	Mapnc
Area Required	Ar	21.731	12.395	NA
Area in Shell	A1	22.369	19.309	NA
Area in Nozzle Wall	A2	3.459	3.603	NA
Area in Inward Nozzle	A3	0.000	0.000	NA
Area in Welds	A41+A42+A43	1.640	1.640	NA
Area in Element	A5	13.680	13.680	NA
TOTAL AREA AVAILABLE	Atot	41.148	38.233	NA

The MAWP Case Governs the Analysis.

Nozzle Angle Used in Area Calculations

90.00 degs.

The area available without a pad is Sufficient.

The area available with the given pad is Sufficient.

Area Required [A]:

$$\begin{aligned}
 &= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1)) \text{ UG-37(c)} \\
 &= (490.0 \cdot 4.4348 \cdot 1.0 + 2 \cdot 9.0 \cdot 4.4348 \cdot 1.0 \cdot (1 - 1.0)) \\
 &= 21.731 \text{ cm}^2
 \end{aligned}$$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:

$$\begin{aligned}
 &= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1) \\
 &= 490.0(1.0 \cdot 9.0 - 1.0 \cdot 4.435) - 2 \cdot 9.0 \\
 &\quad (1.0 \cdot 9.0 - 1.0 \cdot 4.4348) \cdot (1 - 1.0) \\
 &= 22.369 \text{ cm}^2
 \end{aligned}$$

Area Available in Nozzle Wall Projecting Outward [A2]:

$$\begin{aligned}
 &= (2 \cdot Tlwp) \cdot (tn - trn) \cdot fr2 \\
 &= (2 \cdot 22.5) \cdot (9.0 - 1.31) \cdot 1.0 \\
 &= 3.459 \text{ cm}^2
 \end{aligned}$$

Area Available in Welds [A41 + A42 + A43]:

$$= Wo^2 \cdot fr3 + (Wi - can/0.707)^2 \cdot fr2 + Wp^2 \cdot fr4$$



$$= 10.0^2 \cdot 1.0 + (0.0)^2 \cdot 1.0 + 8.0^2 \cdot 1.0$$

$$= 1.640 \text{ cm}^2$$

Area Available in Element [A5]:

$$= (\min(D_p, D_L) - (\text{Nozzle OD})) \cdot (\min(t_p, t_{lwp}, t_e)) \cdot f_{r4}$$

$$= (622.0 - 508.0) \cdot 12.0 \cdot 1.0$$

$$= 13.680 \text{ cm}^2$$

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta = 4.3139 mm.
Wall Thickness per UG16(b),	tr16b = 5.5000 mm.
Wall Thickness, shell/head, internal pressure	trb1 = 7.9217 mm.
Wall Thickness	tb1 = max(trb1, tr16b) = 7.9217 mm.
Wall Thickness, shell/head, external pressure	trb2 = 3.7113 mm.
Wall Thickness	tb2 = max(trb2, tr16b) = 5.5000 mm.
Wall Thickness per table UG-45	tb3 = 11.3312 mm.

Determine Nozzle Thickness candidate [tb]:

$$= \min[tb3, \max(tb1, tb2)]$$

$$= \min[11.331, \max(7.9217, 5.5)]$$

$$= 7.9217 \text{ mm.}$$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

$$= \max(ta, tb)$$

$$= \max(4.3139, 7.9217)$$

$$= 7.9217 \text{ mm.}$$

Available Nozzle Neck Thickness = 12.0000 mm. -> OK

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained	:	35.5,	Allowable	:	138.0 N./mm ²	Passed
Expansion	:	0.0,	Allowable	:	309.5 N./mm ²	Passed
Occasional	:	9.6,	Allowable	:	183.5 N./mm ²	Passed
Shear	:	19.8,	Allowable	:	96.6 N./mm ²	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, min(Curve:B, Curve:A)

Govern. thk, tg = 12.0, tr = 1.314, c = 3.0 mm., E* = 1.0
 Thickness Ratio = tr * E*/(tg - c) = 0.146, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve A	-3 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Nozzle Neck to Pad Weld for the Nozzle, Curve: B

Govern. thk, tg = 12.0, tr = 1.314, c = 3.0 mm., E* = 1.0
 Thickness Ratio = tr * E*/(tg - c) = 0.146, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-23 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Nozzle Neck to Pad Weld for Reinforcement pad, Curve: B

Govern. thk, tg = 12.0, tr = 1.314, c = 3.0 mm., E* = 1.0
 Thickness Ratio = tr * E*/(tg - c) = 0.146, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-23 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

**Shell to Pad Weld Junction at Pad OD, Curve: B**

Govrn. thk, tg = 12.0, tr = 4.435, c = 3.0 mm., $E^* = 1.0$
 Thickness Ratio = $tr * E^* / (tg - c) = 0.493$, Temp. Reduction = 33 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-23 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-48 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Nozzle-Shell/Head Weld (UCS-66(a)(1)(b)), Curve: B

Govrn. thk, tg = 12.0, tr = 4.435, c = 3.0 mm., $E^* = 1.0$
 Thickness Ratio = $tr * E^* / (tg - c) = 0.493$, Temp. Reduction = 33 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-23 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-48 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Gov. MDMT of the Nozzle	:	-48 °C
Gov. MDMT of the Reinforcement Pad	:	-48 °C
Gov. MDMT of the nozzle to shell joint welded assembly	:	-48 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c)	-18 °C
Flange MDMT with Temp reduction per UCS-66(b)(1)(-b)	-48 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
 Design Pressure/Ambient Rating = $7.15/19.60 = 0.365$

Weld Size Calculations, Description: N3-20"

Intermediate Calc. for nozzle/shell Welds	Tmin	9.0000 mm.
Intermediate Calc. for pad/shell Welds	TminPad	11.0000 mm.

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	$6.0000 = \text{Min per Code}$	$7.0700 = 0.7 * W_o \text{ mm.}$
Pad Weld	$5.5000 = 0.5 * T_{minPad}$	$5.6560 = 0.7 * W_p \text{ mm.}$

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:
 $= \max(0, (A-A1+2 \cdot t_n \cdot f_{r1} \cdot (E1 \cdot t - tr)) \cdot S_v)$
 $= \max(0, (21.7306 - 22.3695 + 2 \cdot 9.0 \cdot 1.0 \cdot (1.0 \cdot 9.0 - 4.4348)) \cdot 138)$
 $= 257.28 \text{ Kgf}$

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:
 $= (A2+A5+A4-(W_i \cdot \text{Can}/.707)^2 \cdot f_{r2}) \cdot S_v$
 $= (3.4588 + 13.68 + 1.64 - 0.0 \cdot 1.0) \cdot 138$
 $= 26424.67 \text{ Kgf}$

Weld Load [W2]:
 $= (A2 + A3 + A4 + (2 \cdot t_n \cdot t \cdot f_{r1})) \cdot S_v$
 $= (3.4588 + 0.0 + 1. + (1.62)) \cdot 138$
 $= 8553.77 \text{ Kgf}$

Weld Load [W3]:
 $= (A2+A3+A4+A5+(2 \cdot t_n \cdot t \cdot f_{r1})) \cdot S$
 $= (3.4588 + 0.0 + 1.64 + 13.68 + (1.62)) \cdot 138$
 $= 28704.27 \text{ Kgf}$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:
 $= (\pi/2) \cdot D_{lo} \cdot W_o \cdot 0.49 \cdot S_{nw}$



$$= (3.1416/2.0) \cdot 508.0 \cdot 10.0 \cdot 0.49 \cdot 138$$

$$= 55020. \text{ Kgf}$$

Shear, Pad Element Weld [Spew]:

$$= (\pi/2) \cdot DP \cdot WP \cdot 0.49 \cdot SEW$$

$$= (3.1416/2.0) \cdot 622.0 \cdot 8.0 \cdot 0.49 \cdot 138$$

$$= 53894. \text{ Kgf}$$

Shear, Nozzle Wall [Snw]:

$$= (\pi \cdot (Dlr + Dlo) / 4) \cdot (Thk - Can) \cdot 0.7 \cdot Sn$$

$$= (3.1416 \cdot 249.5) \cdot (12.0 - 3.0) \cdot 0.7 \cdot 138$$

$$= 69487. \text{ Kgf}$$

Tension, Pad Groove Weld [Tpgw]:

$$= (\pi/2) \cdot Dlo \cdot Wgpn \cdot 0.74 \cdot Seg$$

$$= (3.1416/2) \cdot 508.0 \cdot 12.0 \cdot 0.74 \cdot 138$$

$$= 99710. \text{ Kgf}$$

Tension, Shell Groove Weld [Tngw]:

$$= (\pi/2) \cdot Dlo \cdot (T - cas) \cdot 0.74 \cdot Sng$$

$$= (3.1416/2.0) \cdot 508.0 \cdot (12.0 - 3.0) \cdot 0.74 \cdot 138$$

$$= 74783. \text{ Kgf}$$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:

$$= (SPEW + SNW) = (53894 + 69487) = 123381 \text{ Kgf}$$

Failure Path [Path 2-2]:

$$= (Sonw + Tpgw + Tngw + Sinw)$$

$$= (55020 + 99710 + 74783 + 0) = 229513 \text{ Kgf}$$

Failure Path [Path 3-3]:

$$= (Spew + Tngw + Sinw)$$

$$= (53894 + 74783 + 0) = 128676 \text{ Kgf}$$

Summary of Failure Path Calculations:

Path 1-1 = 123380 Kgf, must exceed W = 257 Kgf	or W1 = 26424 Kgf
Path 2-2 = 229512 Kgf, must exceed W = 257 Kgf	or W2 = 8553 Kgf
Path 3-3 = 128676 Kgf, must exceed W = 257 Kgf	or W3 = 28704 Kgf

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 10 bars

Nozzle is O.K. for the External Pressure: 1 bars

The Drop for this Nozzle is : 18.8624 mm.

The Cut Length for this Nozzle is, Drop + Ho + H + T : 180.8625 mm.

Percent Elongation Calculations:% Elongation per Table UG-79-1 ($50 \cdot t_{nom} / R_f \cdot (1 - R_f / R_o)$) 2.419 %**Input Echo, WRC107/537 Item 1, Description: N3-20" :**

Diameter Basis for Vessel	Vbasis	ID	
Cylindrical or Spherical Vessel	Cylsph	Spherical	
Internal Corrosion Allowance	Cas	3.0000	mm.
Vessel Diameter	Dv	3420.000	mm.
Vessel Thickness	Tv	12.000	mm.
Design Temperature	T1	230.0	C
Vessel Material		SA-516 70	
Vessel UNS Number		K02700	
Vessel Cold S.I. Allowable	Smc	138.00	N./mm2
Vessel Hot S.I. Allowable	Smh	138.00	N./mm2

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.



Attachment Type	Type	Round
WRC107 Attachment Classification	Holsol	Hollow
Diameter Basis for Nozzle	Nbasis	OD
Corrosion Allowance for Nozzle	Can	3.0000 mm.
Nozzle Diameter	Dn	508.000 mm.
Nozzle Thickness	Tn	12.000 mm.
Nozzle Material		SA-516 70
Nozzle UNS Number		K02700
Nozzle Cold S.I. Allowable	SNmc	138.00 N./mm2
Nozzle Hot S.I. Allowable	SNmh	138.00 N./mm2
Thickness of Reinforcing Pad	Tpad	12.000 mm.
Diameter of Reinforcing Pad	Dpad	622.000 mm.
Design Internal Pressure	Dp	7.153 bars
Include Pressure Thrust		No

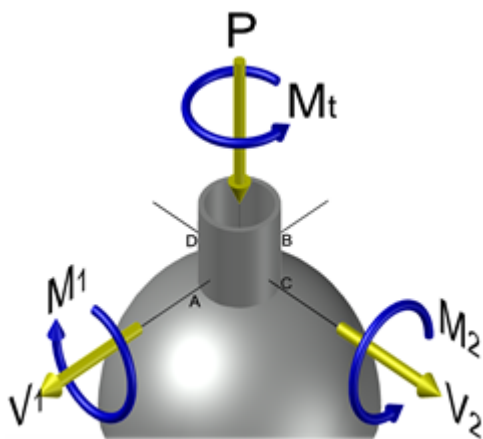
External Forces and Moments in WRC 107/537 Convention:

Radial Load (SUS)	P	-3824.2	Kgf
Longitudinal Shear (SUS)	(V1) V1	2549.5	Kgf
Circumferential Shear (SUS)	(Vc) V2	2549.5	Kgf
Circumferential Moment (SUS)	(Mc) M1	3567.9	Kg-m.
Longitudinal Moment (SUS)	(Ml) M2	3567.9	Kg-m.
Torsional Moment (SUS)	Mt	5097.0	Kg-m.

Use Interactive Control	No
WRC107 Version	Version March 1979

Include Pressure Stress Indices per Div. 2	No
Compute Pressure Stress per WRC-368	No
Local Loads applied at end of Nozzle/Attachment	No

WRC Bulletin 537 provides equations for the dimensionless curves found in bulletin 107. As noted in the foreword to bulletin 537, "537 is equivalent to WRC 107". Where 107 is printed in the results below, "537" can be interchanged with "107".



WRC 107/537 has dimensional limitations that can be found in the bulletin's Foreword and which can also be inferred by review of the non-dimensional curve limits.

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$\begin{aligned}
 &= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{(R_{\text{mean}}(t - ca))} \\
 &= 508.0 + 2 \cdot 1.65 \cdot \sqrt{(1717.5 (12.0 - 3.0))} \\
 &= 918.283 \text{ mm.}
 \end{aligned}$$

WRC 107 Stress Calculation for SUSTained loads:

Radial Load	P	-3824.2	Kgf
Circumferential Shear	(VC) V2	2549.5	Kgf



Longitudinal Shear	(VL) V1	2549.5	Kgf
Circumferential Moment	(MC) M1	3567.9	Kg-m.
Longitudinal Moment	(ML) M2	3567.9	Kg-m.
Torsional Moment	MT	5097.0	Kg-m.

Dimensionless Param: U = 1.34 TAU = 27.72 RHO = 4.00 (2.33)

Dimensionless Loads for Spherical Shells at Attachment Junction:

Curves read for 1979	Figure	Value	Location
N(x) * T / P	SP 7	0.02060	(A,B,C,D)
M(x) / P	SP 7	0.00626	(A,B,C,D)
N(x) * T * SQRT(Rm * T) / MC	SM 7	0.02639	(A,B,C,D)
M(x) * SQRT(Rm * T) / MC	SM 7	0.00938	(A,B,C,D)
N(x) * T * SQRT(Rm * T) / ML	SM 7	0.02639	(A,B,C,D)
M(x) * SQRT(Rm * T) / ML	SM 7	0.00938	(A,B,C,D)
N(y) * T / P	SP 7	0.14984	(A,B,C,D)
M(y) / P	SP 7	0.01467	(A,B,C,D)
N(y) * T * SQRT(Rm * T) / MC	SM 7	0.16830	(A,B,C,D)
M(y) * SQRT(Rm * T) / MC	SM 7	0.02697	(A,B,C,D)
N(y) * T * SQRT(Rm * T) / ML	SM 7	0.16830	(A,B,C,D)
M(y) * SQRT(Rm * T) / ML	SM 7	0.02697	(A,B,C,D)

Stress Concentration Factors: Kn = 1.00, Kb = 1.00

Stresses in the Vessel at the Attachment Junction (N./mm²)

Type of Stress	Load	Stress Intensity Values at							
		Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Rad.	Memb. P	1.8	1.8	1.8	1.8	1.8	1.8	1.8	1.8
Rad.	Bend. P	3.2	-3.2	3.2	-3.2	3.2	-3.2	3.2	-3.2
Rad.	Memb. MC	0.0	0.0	0.0	0.0	-11.0	-11.0	11.0	11.0
Rad.	Bend. MC	0.0	0.0	0.0	0.0	-23.5	23.5	23.5	-23.5
Rad.	Memb. ML	-11.0	-11.0	11.0	11.0	0.0	0.0	0.0	0.0
Rad.	Bend. ML	-23.5	23.5	23.5	-23.5	0.0	0.0	0.0	0.0
Tot.	Rad. Str.	-29.5	11.0	39.4	-13.9	-29.5	11.0	39.4	-13.9
Tang.	Memb. P	12.7	12.7	12.7	12.7	12.7	12.7	12.7	12.7
Tang.	Bend. P	7.5	-7.5	7.5	-7.5	7.5	-7.5	7.5	-7.5
Tang.	Memb. MC	0.0	0.0	0.0	0.0	-70.2	-70.2	70.2	70.2
Tang.	Bend. MC	0.0	0.0	0.0	0.0	-67.5	67.5	67.5	-67.5
Tang.	Memb. ML	-70.2	-70.2	70.2	70.2	0.0	0.0	0.0	0.0
Tang.	Bend. ML	-67.5	67.5	67.5	-67.5	0.0	0.0	0.0	0.0
Tot.	Tang. Str.	-117.4	2.6	157.9	8.0	-117.4	2.6	157.9	8.0
	Shear VC	1.5	1.5	-1.5	-1.5	0.0	0.0	0.0	0.0
	Shear VL	0.0	0.0	0.0	0.0	-1.5	-1.5	1.5	1.5
	Shear MT	5.9	5.9	5.9	5.9	5.9	5.9	5.9	5.9
	Tot. Shear	7.4	7.4	4.4	4.4	4.4	4.4	7.4	7.4
	Str. Int.	118.1	17.0	158.1	23.6	117.7	12.9	158.4	26.4

Unitless Prm: U = 2.50 TAU = 0.00 (34.06) RHO = 0.00 (1.00)

Dimensionless Loads for Spherical Shells at Pad edge:

Curves read for 1979	Figure	Value	Location
N(x) * T / P	SR 2	0.02175	(A,B,C,D)
M(x) / P	SR 2	0.00990	(A,B,C,D)
N(x) * T * SQRT(Rm * T) / MC	SR 3	0.01691	(A,B,C,D)
M(x) * SQRT(Rm * T) / MC	SR 3	0.01053	(A,B,C,D)



$N(x) * T * \sqrt{R_m * T} / ML$ SR 3 0.01691 (A,B,C,D)
 $M(x) * \sqrt{R_m * T} / ML$ SR 3 0.01053 (A,B,C,D)

 $N(y) * T / P$ SR 2 0.00650 (A,B,C,D)
 $M(y) / P$ SR 2 0.00300 (A,B,C,D)
 $N(y) * T * \sqrt{R_m * T} / MC$ SR 3 0.00507 (A,B,C,D)
 $M(y) * \sqrt{R_m * T} / MC$ SR 3 0.00317 (A,B,C,D)
 $N(y) * T * \sqrt{R_m * T} / ML$ SR 3 0.00507 (A,B,C,D)
 $M(y) * \sqrt{R_m * T} / ML$ SR 3 0.00317 (A,B,C,D)

Stress Concentration Factors: $K_n = 1.00$, $K_b = 1.00$

Stresses in the Vessel at the Edge of Reinforcing Pad (N./mm2)

Type of Stress	Load	Stress Intensity Values at							
		Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Rad. Memb. P		10.1	10.1	10.1	10.1	10.1	10.1	10.1	10.1
Rad. Bend. P		27.5	-27.5	27.5	-27.5	27.5	-27.5	27.5	-27.5
Rad. Memb. MC		0.0	0.0	0.0	0.0	-58.8	-58.8	58.8	58.8
Rad. Bend. MC		0.0	0.0	0.0	0.0	-219.5	219.5	219.5	-219.5
Rad. Memb. ML		-58.8	-58.8	58.8	58.8	0.0	0.0	0.0	0.0
Rad. Bend. ML		-219.5	219.5	219.5	-219.5	0.0	0.0	0.0	0.0
Tot. Rad. Str.		-240.6	143.2	315.8	-178.1	-240.6	143.2	315.8	-178.1
Tang. Memb. P		3.0	3.0	3.0	3.0	3.0	3.0	3.0	3.0
Tang. Bend. P		8.3	-8.3	8.3	-8.3	8.3	-8.3	8.3	-8.3
Tang. Memb. MC		0.0	0.0	0.0	0.0	-17.6	-17.6	17.6	17.6
Tang. Bend. MC		0.0	0.0	0.0	0.0	-66.2	66.2	66.2	-66.2
Tang. Memb. ML		-17.6	-17.6	17.6	17.6	0.0	0.0	0.0	0.0
Tang. Bend. ML		-66.2	66.2	66.2	-66.2	0.0	0.0	0.0	0.0
Tot. Tang. Str.		-72.4	43.3	95.1	-53.9	-72.4	43.3	95.1	-53.9
Shear VC		2.8	2.8	-2.8	-2.8	0.0	0.0	0.0	0.0
Shear VL		0.0	0.0	0.0	0.0	-2.8	-2.8	2.8	2.8
Shear MT		9.1	9.1	9.1	9.1	9.1	9.1	9.1	9.1
Tot. Shear		12.0	12.0	6.3	6.3	6.3	6.3	12.0	12.0
Str. Int.		241.5	144.7	316.0	178.4	240.9	143.6	316.4	179.3

WRC 107/537 Stress Summations:

Vessel Stress Summation at Attachment Junction (N./mm2)

Type of Stress	Load	Stress Intensity Values at							
		Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Rad. Pm (SUS)		29.2	29.2	29.2	29.2	29.2	29.2	29.2	29.2
Rad. Pl (SUS)		-9.3	-9.3	12.8	12.8	-9.3	-9.3	12.8	12.8
Rad. Q (SUS)		-20.3	20.3	26.7	-26.7	-20.3	20.3	26.7	-26.7
Long. Pm (SUS)		29.2	29.2	29.2	29.2	29.2	29.2	29.2	29.2
Long. Pl (SUS)		-57.4	-57.4	82.9	82.9	-57.4	-57.4	82.9	82.9
Long. Q (SUS)		-60.0	60.0	75.0	-75.0	-60.0	60.0	75.0	-75.0
Shear Pm (SUS)		0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Shear Pl (SUS)		1.5	1.5	-1.5	-1.5	-1.5	-1.5	1.5	1.5
Shear Q (SUS)		5.9	5.9	5.9	5.9	5.9	5.9	5.9	5.9
Pm (SUS)		29.2	29.2	29.2	29.2	29.2	29.2	29.2	29.2
Pm+Pl (SUS)		48.3	48.3	112.1	112.1	48.3	48.3	112.1	112.1



Pm+Pl+Q (Total)	89.1	44.5	187.2	38.0	88.5	42.1	187.5	39.4
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Vessel Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	29.18	138.00	Passed
Pm+Pl (SUS)	112.14	207.00	Passed
Pm+Pl+Q (TOTAL)	187.54	413.99	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 107/537 Stress Summations:**Vessel Stress Summation at Reinforcing Pad Edge (N./mm²)**

Type of Stress	Load	Stress Intensity Values at							
		Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Rad.	Pm (SUS)	68.1	68.1	68.1	68.1	68.1	68.1	68.1	68.1
Rad.	Pl (SUS)	-48.7	-48.7	68.8	68.8	-48.7	-48.7	68.8	68.8
Rad.	Q (SUS)	-191.9	191.9	247.0	-247.0	-191.9	191.9	247.0	-247.0
Long.	Pm (SUS)	68.1	68.1	68.1	68.1	68.1	68.1	68.1	68.1
Long.	Pl (SUS)	-14.6	-14.6	20.6	20.6	-14.6	-14.6	20.6	20.6
Long.	Q (SUS)	-57.8	57.8	74.5	-74.5	-57.8	57.8	74.5	-74.5
Shear	Pm (SUS)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Shear	Pl (SUS)	2.8	2.8	-2.8	-2.8	-2.8	-2.8	2.8	2.8
Shear	Q (SUS)	9.1	9.1	9.1	9.1	9.1	9.1	9.1	9.1
Pm (SUS)		68.1	68.1	68.1	68.1	68.1	68.1	68.1	68.1
Pm+Pl (SUS)		53.7	53.7	137.1	137.1	53.7	53.7	137.1	137.1
Pm+Pl+Q (Total)		173.4	212.7	384.0	124.9	172.8	211.7	384.5	126.5

Vessel Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	68.07	138.00	Passed
Pm+Pl (SUS)	137.08	207.00	Passed
Pm+Pl+Q (TOTAL)	384.51	413.99	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

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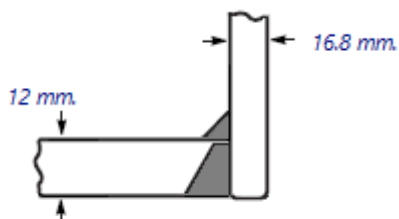
**Input, Nozzle Desc: K1A-2"****From: 60**

Pressure for Reinforcement Calculations	P	7.153	bars
Temperature for Internal Pressure	Temp	230	°C
Design External Pressure	Pext	1.03	bars
Temperature for External Pressure	Tempex	198	°C
Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	138.00	N./mm ²
Parent Allowable Stress At Ambient	Sva	138.00	N./mm ²
Inside Diameter of Elliptical Head	D	1900.00	mm.
Aspect Ratio of Elliptical Head	Ar	2.00	
Head Finished (Minimum) Thickness	t	12.0000	mm.
Head Internal Corrosion Allowance	cas	3.0000	mm.
Head External Corrosion Allowance	co	0.0000	mm.
Distance from Head Centerline	L1	600.0000	mm.
User Entered Minimum Design Metal Temperature		-1.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-105	
Material UNS Number		K03504	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	136.80	N./mm ²
Allowable Stress At Ambient	Sna	138.00	N./mm ²
Diameter Basis (for tr calc only)		Inside	
Layout Angle		90.00	deg
Diameter		2.0000	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	16.7640	mm.
Flange Material		SA-105	
Flange Type		Long Weld Neck	
Corrosion Allowance	can	3.0000	mm.
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	206.3750	mm.
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm.
Groove weld depth between Nozzle and Vessel	Wgnv	9.0000	mm.
Inside Projection	h	0.0000	mm.
Weld leg size, Inside Element to Shell	Wi	0.0000	mm.
Flange Class		300	
Flange Grade		GR 1.1	

The Pressure Design option was MAWP + static head (to the nozzle).



Note : Checking Nozzle in the Meridional direction.

**Reinforcement CALCULATION, Description: K1A-2"**

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation	2.000 in.
Actual Thickness Used in Calculation	0.660 in.

Messages regarding the Nozzle Input Data:

For this nozzle, the actual wall thickness was specified. In these cases, the weight of the attached nozzle flange cannot be determined. It is required to manually compute the weight of the assembly and enter it into the nozzle dialog's overriding weight field.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Elliptical Head, tr [Int. Press]
 $= P \cdot D \cdot K1 / (2 \cdot S \cdot E - 0.2 \cdot P)$ per Appendix 1-4(c)
 $= 7.15 \cdot 1906.0 \cdot 0.8973 / (2 \cdot 138.0 \cdot 1.0 - 0.2 \cdot 7.15)$
 $= 4.4348 \text{ mm.}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R / (S \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 7.15 \cdot 28.4 / (137 \cdot 1.0 - 0.6 \cdot 7.15)$
 $= 0.1490 \text{ mm.}$

Required Nozzle thickness under External Pressure per UG-28 : 0.3975 mm.

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	122.6047 mm.
Parallel to Vessel Wall, opening length	d	61.3023 mm.
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	22.5000 mm.

*Taking a UG-36(c)(3)(a) exemption for nozzle: K1A-2".
 This calculation is valid for nozzles that meet all the requirements of paragraph UG-36. Please check the Code carefully, especially for nozzles that are not isolated or do not meet Code spacing requirements. To force the computation of areas for small nozzles go to Tools->Configuration and check the box to force the UG-37 small nozzle area calculation.*

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:**Nozzle-Shell/Head Weld (UCS-66(a)(1)(b)), min(Curve:A, Curve:B)**

Govrn. thk, tg = 12.0, tr = 4.435, c = 3.0 mm., E* = 1.0
 Thickness Ratio = $tr \cdot E^* / (tg - c) = 0.493$, Temp. Reduction = 33 °C

Min Metal Temp. w/o impact per UCS-66, Curve A	-3 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-36 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Gov. MDMT of the nozzle to shell joint welded assembly : -36 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c)	-18 °C
Flange MDMT with Temp reduction per UCS-66(b)(1)(-b)	-104 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
 Design Pressure/Ambient Rating = $7.15 / 51.00 = 0.140$

Weld Size Calculations, Description: K1A-2"

Intermediate Calc. for nozzle/shell Welds Tmin 11.0000 mm.

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * Wo mm.

Skipping the nozzle attachment weld strength calculations. Per UW-15(b)(2) the nozzles

exempted by UG-36(c)(3)(a) (small nozzles) do not require a weld strength check.

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 11 bars

Note: The MAWP of this junction was limited by the parent Shell/Head.

Note : Checking Nozzle in the Latitudinal direction.

Reinforcement CALCULATION, Description: K1A-2"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation 2.000 in.
 Actual Thickness Used in Calculation 0.660 in.

Messages regarding the Nozzle Input Data:

For this nozzle, the actual wall thickness was specified. In these cases, the weight of the attached nozzle flange cannot be determined. It is required to manually compute the weight of the assembly and enter it into the nozzle dialog's overriding weight field.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Elliptical Head, tr [Int. Press]
 $= P \cdot D \cdot K1 / (2 \cdot S_v \cdot E - 0.2 \cdot P)$ per Appendix 1-4(c)
 $= 7.15 \cdot 1906.0 \cdot 0.8973 / (2 \cdot 138.0 \cdot 1.0 - 0.2 \cdot 7.15)$
 $= 4.4348$ mm.

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R / (S_n \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 7.15 \cdot 28.4 / (137 \cdot 1.0 - 0.6 \cdot 7.15)$
 $= 0.1490$ mm.

Required Nozzle thickness under External Pressure per UG-28 : 0.3975 mm.

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	113.6000	mm.
Parallel to Vessel Wall, opening length	d	56.8000	mm.
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	22.5000	mm.

*Taking a UG-36(c)(3)(a) exemption for nozzle: K1A-2".
 This calculation is valid for nozzles that meet all the requirements of paragraph UG-36. Please check the Code carefully, especially for nozzles that are not isolated or do not meet Code spacing requirements. To force the computation of areas for small nozzles go to Tools->Configuration and check the box to force the UG-37 small nozzle area calculation.*

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta	= 3.3975	mm.
Wall Thickness per UG16(b),	tr16b	= 5.5000	mm.
Wall Thickness, shell/head, internal pressure	trb1	= 7.9217	mm.
Wall Thickness	tb1 = max(trb1, tr16b)	= 7.9217	mm.
Wall Thickness, shell/head, external pressure	trb2	= 3.7113	mm.
Wall Thickness	tb2 = max(trb2, tr16b)	= 5.5000	mm.
Wall Thickness per table UG-45	tb3	= 7.8000	mm.

Determine Nozzle Thickness candidate [tb]:
 $= \min[tb3, \max(tb1, tb2)]$
 $= \min[7.8, \max(7.9217, 5.5)]$
 $= 7.8000$ mm.

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

$= \max(ta, tb)$
 $= \max(3.3975, 7.8)$
 $= 7.8000$ mm.

Available Nozzle Neck Thickness = 16.7640 mm. -> OK

Weld Size Calculations, Description: K1A-2"

Intermediate Calc. for nozzle/shell Welds Tmin 11.0000 mm.

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * Wo mm.

Skipping the nozzle attachment weld strength calculations. Per UW-15(b)(2) the nozzles exempted by UG-36(c)(3)(a) (small nozzles) do not require a weld strength check.

The Drop for this Nozzle is : 16.3327 mm.

The Cut Length for this Nozzle is, Drop + Ho + H + T : 235.5065 mm.

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PV Elite® Pressure Vessel Design Notes

Project Information	
Project Number	
Client	
Date	
Designer	
Design Code	
Additional Standards	

Design Conditions		
Internal Design Pressure:		
External Design Pressure:		
Design Int. Temperature:		
Design Ext. Temperature:		
Corrosion Allowance:		

Vessel Geometry		
Vessel Type		
Diameter		
Tangent-to-Tangent Length		
Head Type		
Number of Segments		

Material Specifications	
Shell:	
Head:	
Nozzle:	

Nozzle/Opening Details	ID/OD	Size	Rating	Matl.	Reinf.

Miscellaneous		
Support Type		
Number of Supports		
Support Material		
Seismic Code		
Wind Code		
Insulation		
Pressure Test		
Radiography		
Fatigue Analysis		

Analysis Notes: